

Clinical Reasoning Manual

NBEO Part II PAM / TMOD. An emergency gate, condition cards, and a full 175-item simulation with a generated, auditable key.

CONTENTS

2	P1	How to use this manual	20	P6	Law, ethics, public health
3	P1	What was verified, and what was not	21	P7	Competing pairs
4	P1	Key integrity	24	P8	Session 2, thirty-five cases
5	P2	The exam, to scale	25	P8	Case index and score sheet
6	P2	Item types	62	P9	Teaching keys, case by case
7	P2	Pacing and the case algorithm	99	P10	Answer key, Session 2
8	P3	Emergency gate	100	P10	Eleven days
11	P4	Condition cards	101	P10	Rapid recall deck
15	P5	Pharmacology	103	P10	Sources
18	P6	Optics, formulas, drills			

175	35	3.5	104
ITEMS IN THIS VOLUME	CASES, OF 3 TO 6 ITEMS	HOURS, ONE SITTING	PAGES, NO FILLER

WHAT CHANGED IN THIS EDITION

The previous key was 82% option A, and 95% in Session 2: guessing A scored 272 of 333 items with no stem read. Letters are now generated to a distribution and audited on page 4, options are length-matched, and every rationale explains the tempting wrong answer as well as the right one.

HOW THE TWO VOLUMES DIVIDE

Each volume is complete: the same reference apparatus, then one session. Cases come first and the teaching keys follow in Part 9, so a session can be sat cold. Volume One carries the other 175 items, on the same letter distribution, so the two scores are comparable.

Six passes

Reading this manual is not studying it. Retrieval under time pressure is what transfers to the exam room.

PASS	WORK	PROOF IT WORKED
1 • Systems	Learn the algorithm, the emergency gate, the condition cards, the drug anchors.	You can state recognition, danger, next test, treatment, contraindication, follow-up, and failure trigger without looking.
2 • Retrieval	Closed-book recall at roughly 1, 3, 7, 14, and 30 days, always from a changed prompt.	Successful recalls are logged. Hours are not.
3 • Session 1	175 items, 210 minutes, uninterrupted, confidence marked on every item.	Scored by item type as well as total, with pacing and error categories written down.
4 • Repair	High-confidence misses first. Each becomes one if-then rule.	Every false rule has a replacement you can recite.
5 • Session 2	Second 175 items under identical conditions.	Weak item types improve; late-session accuracy holds.
6 • Taper	Red flags, drugs, formulas, image algorithms, logistics.	No new content in the final 48 hours. Sleep protected.

Error log: seven fields, one line each

Concept	Short disease or decision label.
Answer contrast	What you chose against what was correct.
Error type	Knowledge, discrimination, image reading, management sequence, calculation, or misreading the stem.
False rule	The exact belief that produced the miss, written as a sentence.
New rule	One executable if-then replacement.
Next recall	Tomorrow for a high-confidence miss, three days for low-confidence, seven days after a success.
Transfer test	A changed patient or the competing diagnosis.

WHY SCORE BY ITEM TYPE

The blueprint weights treatment and management far above basic science, and TMOD is decided separately. A 74% total can hide a failing treatment score. Every item here is tagged, so scoring by tag takes two minutes and tells you what to repair.

INTERLEAVE, DO NOT BLOCK

Study competing pairs together: episcleritis against scleritis, optic neuritis against ischaemic neuropathy, AMD against central serous, HSV against zoster, preseptal against orbital. The exam tests the discriminator, never the textbook description.

What is checked, and what is not

A guide that labels its own uncertainty is more useful than one that asserts it was verified. Every factual claim carries one of three states.

Verification table

CLAIM	STATE	BASIS
350 items, presented as cases of 3-6 items	VERIFIED	NBEO exam page, fetched 15 Aug 2026
Two sessions of 175 items, 3.5 hours each	VERIFIED	NBEO exam page
Approximately 120 TMOD items, separate breakout decision	VERIFIED	NBEO exam page
Delivered at Pearson centres; tutorial being updated	VERIFIED	NBEO exam page
Per-domain item ranges, 17 domains	UNCONFIRMED	Transcribed from prior edition. Content Matrix PDF resolves but was not machine-read
Item-type percentage bands	UNCONFIRMED	As above
Deep link to the Pearson tutorial	REMOVED	URL did not resolve; reach it from the exam page

The defect this edition exists to fix

ANSWER KEY	SESSION 1	SESSION 2	BOTH
Single-answer items	161	172	333
Correct answer was A	67.1%	95.3%	81.7%
Longest run of one letter	30	140	-
Letters never correct	E	D, E	E

WHAT THAT MEANT IN PRACTICE

Answering A to every single-answer item scored 272/333, about 82%, without reading one stem. Session 2 alone scored 95%. The two sessions were also biased by different amounts, so a Session 1 score could not be compared with a Session 2 score, which broke the guide's own repair-and-confirm workflow.

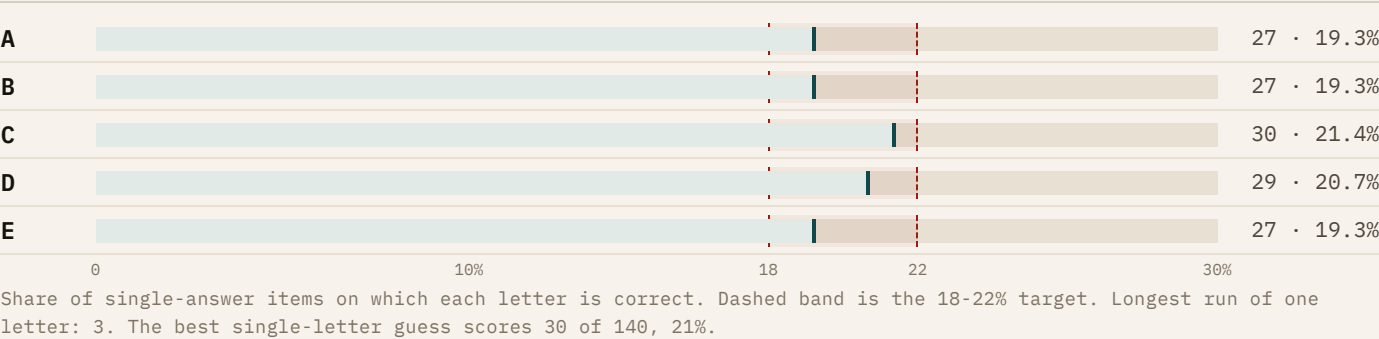
REBUILD RULE, AUDITABLE

Each letter A to E is correct on 18-22% of single-answer items in each session, no letter repeats more than three times consecutively, all five letters appear in both sessions, and option length is matched within each item. The build fails if any of that breaks. Re-run `tools/build.py --audit` after any edit to the bank.

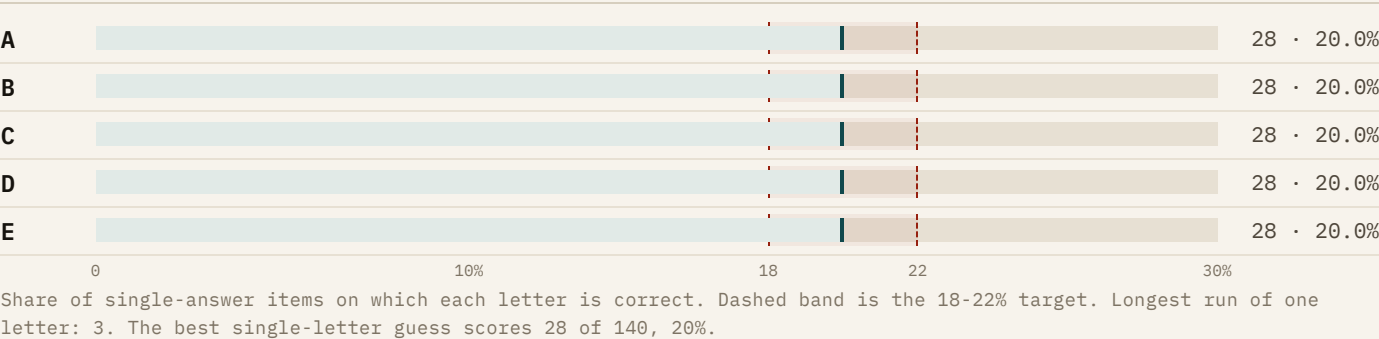
The key, audited

A biased key trains the wrong skill and then flatters you for it. This page is the evidence that the bank you are about to sit does not.

Session 1, 140 single-answer items



Session 2, 140 single-answer items



How the letters were chosen

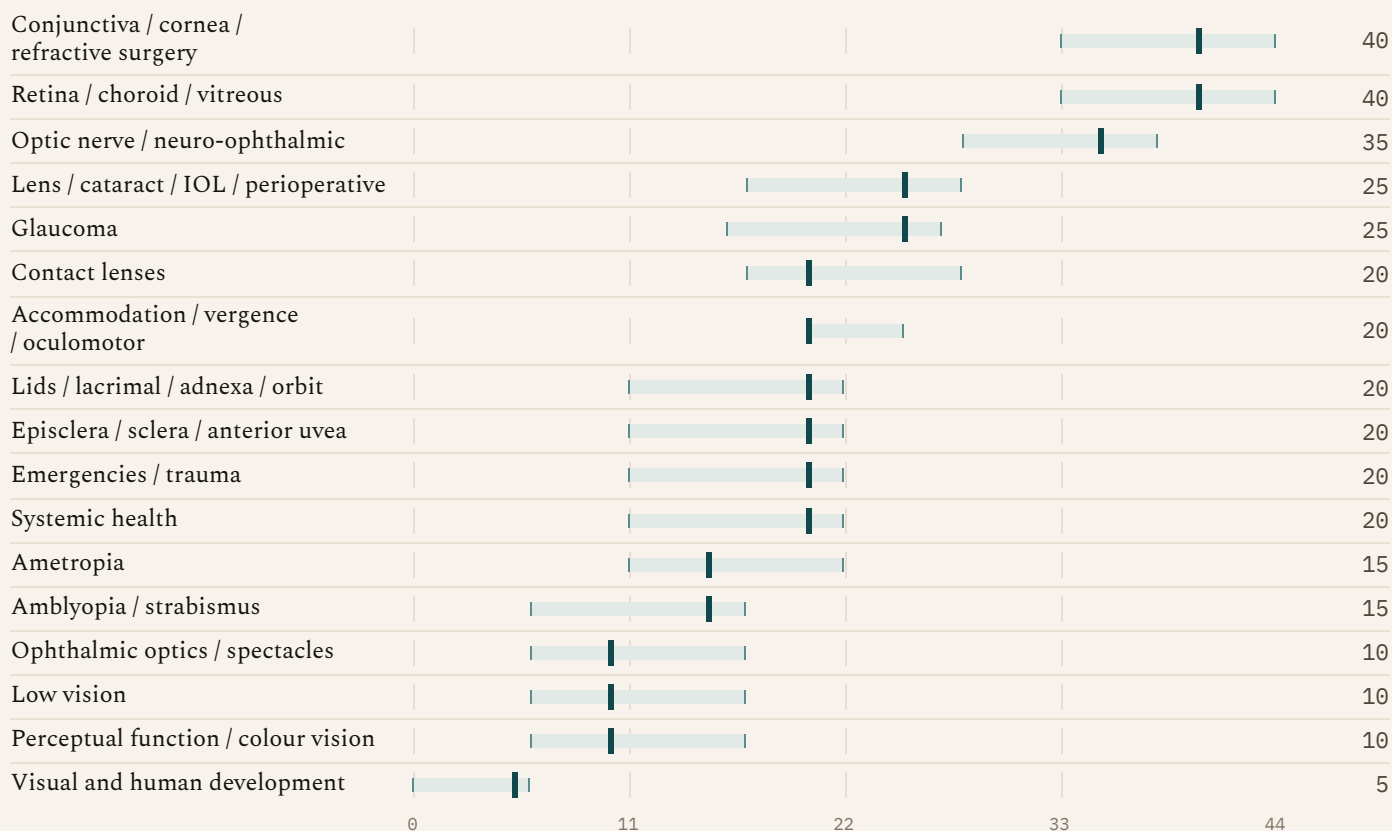
Not by the author	Item writers mark the correct option in place and never see a letter. Letters are assigned afterwards by a generator, so a habit of putting the right answer first cannot survive into print.
Even by construction	Each session's single-answer items are dealt an even multiset of A to E, so every letter lands on 19 to 21% before any adjustment.
No streaks	The sequence is rejected and redealt until no letter is correct more than three times in a row.
Order preserved	Options are rotated, not shuffled, so graded or paired options stay adjacent and a numeric ladder stays in order.
Length controlled	Within each item, the longest option is at most about twice the shortest, and across a session the correct option is the longest no more often than a third of the time.
Checked on build	The manual will not build if any of those conditions fails. The numbers above are printed from the same run that produced these pages.

TEST IT YOURSELF

Mark A for every single-answer item in either session and score it against the answer key in Part 10. You should land near a fifth. The previous edition scored 82% for exactly that, and 95% in Session 2, which is set out on page 3.

Know the target

Band shows the published range. Bar shows this manual's 350-item allocation. Scale runs 0 to 44 items.



READ IT THIS WAY

Cornea and retina alone can carry 88 items. Revision split evenly across 17 domains is misallocated by a factor of eight at the extremes. Development items usually arrive inside paediatric cases.

Case conventions worth memorising

BVA	Best achievable or best-corrected acuity. If it is reduced, pinhole improvement has already been accounted for, so do not spend an item hunting refractive causes.
Near acuity	Assume 16 inches unless the case says otherwise.
Fields	Side by side: OD right, OS left. Stacked: OD above OS.
Images	Fundus, OCT, fields, angiography, ultrasound, radiology, anterior segment. Describe before diagnosing.

245

ITEMS · NORMAL HEALTH, DISEASE, TRAUMA

105

ITEMS · REFRACTIVE, SENSORY, OCULOMOTOR

Where the marks are

Four item types, weighted very unevenly. Every item here is tagged, and the score sheets total by tag.

Item-type weighting and this manual's allocation

ITEM TYPE	BAND	HERE	WHAT IT ACTUALLY ASKS
Treatment and management	40-57%	155 · 44%	Choice, mechanism, dose, schedule, duration, follow-up, education, escalation.
Diagnosis	26-34%	119 · 34%	Most likely diagnosis, the supporting or missing datum, the next test.
Basic science, applied	11-17%	58 · 17%	Anatomy, physiology, microbiology, optics, pharmacology, tied to this patient.
Law, ethics, public health	4-10%	18 · 5%	Duty, consent, refusal, records, delegation, telehealth, screening arithmetic.

Bands transcribed, marked unconfirmed on page 3. The right-hand column is the real allocation across 350 items and 17 domains, counted at build time. 70 items are all-or-none, a format rather than a type.

Two item formats, two disciplines

Single answerONE OF FIVE	Select all that applyALL OR NONE
READ The question before the options. Decide what you think, then look. COMPARE Options in pairs. The pair differing on one variable shows what is tested. COMMIT Once. Changed management answers usually move to the more cautious wrong option.	COUNT Underline the requested number before reading the options. JUDGE Each option alone, true or false for this patient. PREFER Patient-specific over merely true. SCORE Every correct choice and no incorrect choice. A partial set earns zero.
TRAP Length is not a signal in this manual. Correct options are the longest in roughly a fifth of items, which is chance.	TRAP Findings that prove the diagnosis are not the same as findings that measure organ function. Read which the item wants.

What a complete management answer contains

Immediate action	What happens in the next minutes, before any prescription.
Agent and route	Drug or procedure, form, concentration, site of action.
Dose and duration	Loading and maintenance, frequency, taper, stop point.
Contraindication	The host factor in the stem that changes the choice.
Follow-up interval	A number, not as needed. Often the only difference between two options.
Measurable response	What you will remeasure to know it worked.
Escalation trigger	What failure looks like, and what you do then.

HOW THIS IS EXAMINED

Wrong options are right actions in the wrong order, right drugs at the wrong interval, or plans missing the contraindication the stem planted. An option with no interval and no failure trigger is incomplete.

Pacing, and the seven-step algorithm

175 items in 210 minutes is 72 seconds per item. The cost of running late is not the last case, it is the rereading you do in the middle.

Budget by case size

CASE	BUDGET	MOVE	SITUATION	DECISION
3 items	4 min	Read once, answer the direct items, flag one uncertainty.	Two options survive and you have spent 90 seconds	Choose the one that acts sooner on the more dangerous possibility, flag, move.
4 items	5 min	Commit before option-hunting for a rescue.	The stem contains a number you have not used	Reread. Numbers are planted to be used: pulse, weight, dose, pressure, age.
5 items	6 min	The typical unit. Do not reread the stem for each item.	An image is described but not shown	Answer from the described findings. Do not invent what is not stated.
6 items	7 min	Diagnosis, threat, test, treatment, follow-up, then the science.	You do not know the disease	Answer the management item anyway. Threat level is often decidable without the name.
Checkpoints: item 50 by 60 minutes, item 100 by 120, item 150 by 180. That leaves 30 minutes for the last 25 items and the flags. Behind schedule means cutting rereading, not standards.			15 minutes left, six flags	Spend it on flagged management items, not on flagged basic science.

Seven-step case algorithm

1 Frame	Age, laterality, tempo, pain, vision loss, systemic context, medications, risk behaviour.
2 Threat gate	Could delay cost life, the eye, or the fellow eye? Run the eighteen patterns in Part 3 before anything else.
3 Localise	Media, cornea and anterior segment, retina, optic nerve, post-chiasmal pathway, neuromuscular junction, orbit, binocular motor system.
4 Build three	Most likely, most dangerous plausible, best mimic. The mimic is usually one of the five options.
5 Discriminate	The one finding or test that most changes probability or management. Not a panel.
6 Treat in order	Stabilise, protect tissue and vision, treat the cause, prevent recurrence and fellow-eye injury.
7 Close the loop	Interval, measurable response, adverse effects, red flags, referral urgency, and what failure triggers.

INTERLEAVE, DO NOT BLOCK

Study competing pairs together: episcleritis against scleritis, optic neuritis against ischaemic neuropathy, AMD against central serous, HSV against zoster, preseptal against orbital, pupillary block against ciliochoroidal effusion. Part 7 sets them out side by side, because the exam tests the discriminator and never the textbook description.

Six patterns that cannot wait

Recognise, act, and know the specific thing that must not be done. The prohibition is examined as often as the treatment.

Chemical injury SURFACE	Open globe TRAUMA
RECOGNISE Exposure history. Do not wait for acuity or pH before acting.	RECOGNISE Full-thickness wound, Seidel, peaked pupil, uveal prolapse, low IOP with a mechanism.
DO FIRST Irrigate immediately and copiously. Evert lids, sweep fornices for retained particles, continue until pH is neutral and stays neutral after a pause. Then grade limbal ischaemia and epithelial loss.	DO FIRST Rigid shield, nil by mouth, antiemetic and analgesia, tetanus and systemic antibiotics per protocol, CT orbits when appropriate, immediate surgical referral.
NEVER Neutralise alkali with acid. Delay irrigation to find equipment.	NEVER Tonometry, gonioscopy, ultrasound, or any pressure on the globe.
Orbital cellulitis ORBIT	Microbial keratitis CORNEA
RECOGNISE Lid swelling plus any of: painful or restricted motility, proptosis, reduced acuity or colour, RAPD, raised IOP, fever, severe chemosis.	RECOGNISE Contact lens wear, epithelial defect over a stromal infiltrate, chamber reaction, pain, central location.
DO FIRST Emergency department, contrast CT of orbits and sinuses, intravenous antibiotics, ophthalmology and ENT. Drain a subperiosteal abscess when vision or the apex is threatened.	DO FIRST Measure and photograph. Culture central, large, deep, atypical, post-surgical, or unresponsive ulcers. Intensive topical antimicrobial, review inside 24 hours.
NEVER Call it preseptal until orbital signs have been actively excluded.	NEVER Send a central ulcer home on conjunctivitis dosing, or patch it.
Acute angle closure GLAUCOMA	Giant cell arteritis SYSTEMIC
RECOGNISE Pain, halos, nausea, corneal oedema, shallow chamber, mid-dilated fixed pupil, very high IOP.	RECOGNISE Over 50 with new headache, jaw claudication, scalp tenderness, polymyalgia or constitutional symptoms, chalky pallid disc oedema.
DO FIRST Aqueous suppressants the patient can safely take, systemic acetazolamide if appropriate, control inflammation and nausea, then definitive iridotomy. Assess the fellow eye.	DO FIRST ESR, CRP, and platelets, and start high-dose systemic corticosteroid immediately. Temporal artery imaging or biopsy follows treatment.
NEVER Lead with pilocarpine at extreme pressure: an ischaemic sphincter will not respond.	NEVER Let a normal ESR reassure you, or wait for biopsy. The fellow eye is what is at stake.

CONTINUED ON PAGE 9

Retinal artery occlusion, tear and detachment, endophthalmitis, papilloedema, acute third-nerve palsy and painful Horner syndrome, then six more on page 10 that arrive looking mild.

And six more

Retinal artery occlusion VASCULAR	Tear and detachment RETINA
RECOGNISE Sudden painless profound monocular loss, dense RAPD, retinal whitening or cherry-red spot.	RECOGNISE New flashes, shower of floaters, pigment in the vitreous, vitreous haemorrhage, curtain or field loss.
DO FIRST Treat as acute stroke: activate the emergency stroke pathway now, with carotid and cardiac source evaluation and a GCA screen by age.	DO FIRST Dilated peripheral examination with indentation, B-scan if the view is blocked, urgent retina referral. Macula-on detachment is the most time-critical.
NEVER Spend the treatment window on ocular massage or other office manoeuvres.	NEVER Accept a normal central OCT as evidence against a peripheral break.
Endophthalmitis POSTOPERATIVE	Papilloedema NEURO
RECOGNISE Pain and severe acuity loss after surgery or injection, hypopyon, fibrin, vitritis, lost red reflex.	RECOGNISE Bilateral disc oedema with headache, transient obscurations, pulsatile tinnitus, diplopia, possible sixth-nerve palsy.
DO FIRST Same-hour retina-level care: intraocular sampling then intravitreal antibiotics, with vitrectomy by severity.	DO FIRST Urgent brain imaging including venous evaluation, then lumbar puncture with opening pressure when imaging makes it safe.
NEVER Treat it as routine postoperative inflammation, or wait for culture before treating.	NEVER Puncture before imaging, or dismiss true oedema as crowded discs without evidence.
Third-nerve palsy NEURO-VASCULAR	Painful Horner syndrome NEURO-VASCULAR
RECOGNISE Ptosis with the eye down and out; pupil involvement, pain, or an incomplete palsy raises compression.	RECOGNISE Acute ptosis and miosis, anisocoria greater in darkness, dilation lag, with head or neck pain.
DO FIRST Emergency aneurysm-focused CTA or MRA and neurovascular referral. Modern practice images acute isolated palsies.	DO FIRST Emergency head and neck vascular imaging for carotid dissection.
NEVER Treat pupil sparing as proof against aneurysm in an acute palsy.	NEVER Delay imaging to complete pharmacologic localisation.

TREATMENT GRAMMAR, EVERY TIME

Immediate action, drug or procedure, dose and form and schedule and duration, contraindication, patient education, follow-up interval, the measurable response, and the escalation trigger. An answer that names a drug but no interval or failure trigger is incomplete, and incomplete options are how these items are written.

Six that are missed because they look mild

These are the ones that arrive looking like something routine. The recognition line is the whole point.

Acute retinal necrosis VIRAL RETINA RECOGNISE Peripheral confluent whitening, occlusive arteritis, dense vitritis, rapid circumferential spread, often with pain and a red eye. DO FIRST Same-day retina referral. Systemic antiviral therapy starts on clinical suspicion, with intravitreal antiviral cover for posterior pole threat. NEVER Treat as idiopathic intermediate uveitis with steroid alone. Steroid without antiviral cover accelerates it.	Sympathetic ophthalmia UVEA RECOGNISE Bilateral granulomatous panuveitis weeks to years after penetrating injury or intraocular surgery to the other eye. The fellow eye is the eye at risk. DO FIRST Urgent high-dose systemic corticosteroid with immunomodulatory planning, and specialist co-management. NEVER Dismiss mild fellow-eye inflammation after trauma as sympathetic irritation.
Ciliochoroidal effusion angle closure GLAUCOMA RECOGNISE Bilateral acute myopic shift with uniformly shallow chambers, days after starting a sulfa-derived drug such as topiramate. DO FIRST Stop the drug with the prescriber, suppress aqueous, cycloplege to deepen the chamber, treat pressure medically. NEVER Treat it as pupillary block. Iridotomy does not help and miotics make the anterior rotation worse.	Non-accidental injury PAEDIATRIC RECOGNISE Multi-layered retinal haemorrhages extending to the periphery, retinoschisis, in an infant with a history that does not fit the findings. DO FIRST Admit through the paediatric safeguarding pathway the same day, document findings and images verbatim, notify per jurisdiction. NEVER Arrange outpatient follow-up and let the child leave, or record an opinion about who caused it.
Retinoblastoma PAEDIATRIC TUMOUR RECOGNISE Leukocoria, new strabismus, or unexplained poor fixation under three years. Any white reflex is abnormal until proven otherwise. DO FIRST Same-week referral to ocular oncology for examination under anaesthesia, with B-scan and MRI. Examine the parents and siblings. NEVER Biopsy an intraocular mass in a child, and never book routine review for leukocoria.	Pituitary apoplexy CHIASM RECOGNISE Sudden severe headache with bitemporal field loss, ophthalmoplegia, and reduced consciousness or hypotension. DO FIRST Emergency department now: urgent MRI, endocrine support for adrenal insufficiency, neurosurgical assessment. NEVER Book an outpatient field to confirm a chiasmal defect in an acutely unwell patient.

THE GATE IN ONE SENTENCE

Before you name the disease, decide whether delay costs life, the eye, or the fellow eye, because that decision is answerable from the stem even when the diagnosis is not.

Ocular surface and adnexa

Recall each card in one order: decisive pattern, confirming test, first action, trap. Then name the most dangerous mimic.

Preseptal cellulitis LIDS / ORBIT PATTERN Lid erythema and oedema with normal acuity, pupils, colour, motility, and globe position. CONFIRM Clinical. Image when the examination is limited, severe, atypical, or orbital spread is possible. ACT Oral antibiotic covering skin and sinus organisms, review inside 24-48 hours, written orbital red flags. TRAP In a febrile child with sinusitis this is a diagnosis of exclusion. Motility and colour are the gate.	Dacryocystitis LACRIMAL PATTERN Tender erythematous swelling below the medial canthal tendon, with reflux or tearing. CONFIRM Clinical, culture the drainage, assess for orbital extension. ACT Systemic antibiotics, warm compresses, drainage of an abscess, definitive lacrimal surgery once settled. TRAP Do not probe or irrigate through acutely infected tissue.
Recurrent chalazion LIDS PATTERN Firm lesion returning to the same site, madarosis, yellow tarsal thickening, unilateral chronic blepharoconjunctivitis. CONFIRM Full-thickness biopsy with conjunctival mapping, not a third curettage. ACT Urgent oculoplastic referral. Evert both lids, examine conjunctiva and regional nodes. TRAP Sebaceous carcinoma masquerades as chalazion and as blepharitis, and spreads pagetoid.	Thyroid eye disease ORBIT PATTERN Lid retraction, proptosis, exposure, restrictive diplopia, apical crowding. CONFIRM Thyroid function, colour and pupil testing, fields, exophthalmometry; imaging when atypical or neuropathy is suspected. ACT Surface protection, smoking cessation, urgent specialist care for optic neuropathy or corneal breakdown. TRAP Activity and severity are separate axes. Muscle belly enlargement with tendon sparing is the signature.
Adenoviral keratoconjunctivitis CONJUNCTIVA PATTERN Follicles, watery discharge, preauricular node, high contagion; subepithelial infiltrates may follow. CONFIRM Clinical; rapid antigen testing helps in selected settings. ACT Supportive care and strict hygiene. Corticosteroid only for visually significant infiltrates, monitored. TRAP Contagious period and instrument disinfection matter as much as the drop.	Episcleritis vs scleritis EPISCLERA / SCLERA PATTERN Sectoral mobile vessels that blanch with phenylephrine, against deep violaceous non-blanching injection with boring pain and globe tenderness. CONFIRM Phenylephrine, palpation, B-scan for posterior disease, phenotype-directed systemic workup. ACT Episcleritis: lubricant, cool compress, NSAID. Scleritis: systemic anti-inflammatory or immunomodulatory therapy, urgent specialist. TRAP Topical steroid alone is inadequate in scleritis and dangerous if the cause is infectious.

Cornea and refractive surgery

Recall each card in one order: decisive pattern, confirming test, first action, trap. Then name the most dangerous mimic.

Bacterial keratitis CORNEA	HSV keratitis CORNEA
PATTERN Epithelial defect over a dense stromal infiltrate, pain, chamber reaction, lens wear or trauma. CONFIRM Measure and photograph. Scrape for culture when central, large, deep, atypical, post-surgical or unresponsive. ACT Intensive topical antimicrobial with a loading phase, no lens wear, review inside 24 hours.	PATTERN Dendrite with terminal bulbs and reduced corneal sensation for epithelial disease; stromal haze or endothelial oedema with keratic precipitates for immune disease. CONFIRM Clinical, with sensation testing. PCR when atypical. ACT Oral or topical antiviral for epithelial disease. Add steroid only for immune stromal or endothelial disease, always under antiviral cover.
TRAP Never patch, and never treat a central ulcer on conjunctivitis dosing.	TRAP Steroid over an active dendrite converts a treatable ulcer into a geographic one.
Acanthamoeba keratitis CORNEA	Keratoconus ECTASIA
PATTERN Pain far out of proportion to signs, water exposure or poor lens hygiene, radial keratoneuritis, poor response to antibacterials and antivirals. CONFIRM Culture on non-nutrient agar with bacterial overlay, PCR, or confocal. Repeat if suspicion persists. ACT Prolonged intensive biguanide therapy, often with a diamidine, cornea specialist involved from the start.	PATTERN Rising irregular astigmatism, scissoring reflex, Vogt striae, Fleischer ring, inferior steepening and thinning on tomography. CONFIRM Serial tomography for progression: thinnest point, maximum keratometry, posterior elevation. ACT Cross-linking for documented progression, specialty lenses for vision. Treat the eye rubbing and the atopy.
TRAP A ring infiltrate is late and not required for the diagnosis. Steroid worsens it.	TRAP Cross-linking stabilises, it does not correct. Progression is a tomographic judgement, not a refraction change.
Post-LASIK DLK REFRACTIVE SURGERY	Neurotrophic keratopathy SURFACE
PATTERN Diffuse sterile interface haze in the first days, no epithelial defect, no chamber reaction, minimal pain. CONFIRM Clinical, staged by extent and central involvement. ACT Intensive topical steroid, close review. Flap lift and irrigation for stage three or four.	PATTERN Persistent epithelial defect with rolled edges, minimal pain, reduced sensation, after zoster, HSV, surgery or a fifth-nerve lesion. CONFIRM Corneal sensation testing before anaesthetic, staged by epithelial and stromal loss. ACT Preservative-free lubrication, punctal occlusion, autologous serum or nerve growth factor therapy, tarsorrhaphy or bandage lens for progression.
TRAP Focal infiltrate, pain, discharge or chamber reaction is infectious keratitis, where steroid is the wrong answer.	TRAP The absence of pain is why it presents late and perforates. Sensation is the test that names it.

Retina, choroid and vitreous

Neovascular AMD MACULA PATTERN New distortion or a central scotoma over days to weeks, subretinal or intraretinal fluid, haemorrhage, pigment epithelial elevation. CONFIRM OCT for fluid and its compartment, angiography or OCT angiography when the lesion type is unclear. ACT Urgent retina referral for anti-VEGF therapy, with a loading course and interval-based review. Amsler self-monitoring and fellow-eye counselling. TRAP Good acuity is a reason to treat quickly, not a reason to wait. Drusen do not explain new fluid.	Central serous chorioretinopathy MACULA PATTERN Younger patient, hyperopic shift, mild distortion, a shallow serous detachment with a thick choroid, steroid exposure or high stress. CONFIRM OCT for subretinal fluid and choroidal thickness. Angiography for chronic or atypical disease. ACT Remove exogenous corticosteroid where possible, observe acute episodes, treat persistent disease with photodynamic therapy or a mineralocorticoid antagonist per specialist. TRAP Anti-VEGF is not first-line here, and the corticosteroid in the stem is the answer to the management item.
Proliferative diabetic retinopathy RETINA PATTERN New vessels at the disc or elsewhere, vitreous or preretinal haemorrhage, with or without centre-involving oedema. CONFIRM Dilated examination, OCT for the macula, wide-field imaging or angiography for the periphery. ACT Panretinal photocoagulation or anti-VEGF per specialist, treat centre-involving oedema, and drive systemic control: glycaemia, blood pressure, lipids, renal review. TRAP Anti-VEGF without a reliable follow-up plan is unsafe in proliferative disease. Ask who is coming back.	Retinal vein occlusion VASCULAR PATTERN Sector or four-quadrant haemorrhage with dilated tortuous veins, cotton wool spots, macular oedema. CONFIRM OCT for oedema, RAPD and field for ischaemia, blood pressure, glucose, lipids, and a clotting or inflammatory screen in the young. ACT Anti-VEGF for centre-involving oedema, treat neovascular complications, and manage the vascular risk factors that caused it. TRAP An ischaemic central occlusion threatens the angle. Undilated review at 4 to 6 weeks is not enough.
Rhegmatogenous detachment VITREORETINAL PATTERN Flashes, a shower of floaters, a curtain, relative field loss, pigment or haemorrhage in the vitreous. CONFIRM Dilated peripheral examination with indentation. B-scan when the view is obscured. ACT Urgent retina referral. Macula-on disease is the most time-critical thing on this page after an artery occlusion. TRAP A normal central OCT says nothing about the periphery. Positioning advice is not treatment.	Hydroxychloroquine retinopathy TOXIC PATTERN Often asymptomatic early. Parafoveal outer retinal thinning, later a bull's-eye. Asian eyes may show a pericentral pattern. CONFIRM Baseline then annual screening after five years, earlier with risk: dose above roughly 5 mg/kg, renal impairment, tamoxifen. Fields plus OCT, with mfERG or FAF when equivocal. ACT Discuss stopping with the prescriber when toxicity is confirmed. Damage does not reverse. TRAP Fundus appearance is a late sign. Calculate the dose per real body weight and test the pericentral field in Asian patients.

Glaucoma, uvea and the optic nerve

Primary open angle glaucoma GLAUCOMA PATTERN Open angle, structural rim and RNFL loss matching a repeatable field defect, often asymmetric, usually asymptomatic. CONFIRM Gonioscopy, corrected pressure with central thickness in mind, disc photography, OCT, and at least two reliable fields before committing. ACT Set a target pressure from the damage, the baseline and life expectancy. First-line selective laser trabeculoplasty or a prostaglandin analogue, then reassess against the target. TRAP Pressure is a risk factor, not the diagnosis. Treating a number without structure and function is how normal-tension disease is missed.	Angle closure mechanisms GLAUCOMA PATTERN Pupillary block gives a convex iris that flattens after iridotomy. Plateau iris persists after iridotomy. Effusion gives bilateral myopic shift and uniform shallowing. CONFIRM Gonioscopy with indentation, anterior segment OCT or ultrasound, and reassessment after any iridotomy. ACT Iridotomy for block, iridoplasty or long-term miotic for plateau, drug withdrawal with cycloplegia for effusion. TRAP One treatment does not fit the three mechanisms. Naming the mechanism is the item.
Anterior uveitis UVEA PATTERN Pain, photophobia, circumlimbal injection, cells and flare, keratic precipitates, posterior synechiae, pressure high or low. CONFIRM Grade cells and flare, dilate to check the posterior segment, and phenotype before ordering: granulomatous, bilateral, recurrent, or systemic features. ACT Topical steroid at a frequency matched to the grade with a real taper, cycloplegia for pain and synechiae, monitor pressure, treat the cause. TRAP Stopping steroid when the eye looks comfortable causes the rebound. The taper is the treatment plan.	Optic neuritis OPTIC NERVE PATTERN Younger patient, subacute monocular loss over days, pain on eye movement, RAPD, dyschromatopsia, disc normal or swollen. CONFIRM MRI of brain and orbits with contrast, and antibody testing for MOG and aquaporin-4 when the picture is atypical or severe. ACT Neurology referral. Intravenous corticosteroid speeds recovery without changing final acuity, and is used for severe or bilateral loss and for MRI-defined risk. TRAP Oral prednisone at standard dose alone is the wrong answer. Painless loss over 50 with a pallid swollen disc is arteritic, not this.
Ischaemic optic neuropathy OPTIC NERVE PATTERN Sudden painless loss with an altitudinal defect and a swollen disc. Chalky pallid oedema, jaw claudication and constitutional symptoms mean arteritic. CONFIRM ESR, CRP and platelets immediately for any suspicion over 50; temporal artery imaging or biopsy after treatment starts. ACT High-dose systemic corticosteroid at once for arteritic disease. Non-arteritic disease gets vascular risk management and sleep apnoea screening. TRAP A normal ESR does not exclude arteritis. The fellow eye is what the treatment protects.	Compressive and chiasmal disease PATHWAY PATTERN Slowly progressive loss, disc pallor without prior swelling, bitemporal defects respecting the vertical midline, or a junctional defect. CONFIRM Formal fields both eyes, OCT for a band or bow-tie pattern, and MRI with attention to the sella and chiasm. ACT Neurosurgery and endocrinology referral by urgency, with acute presentations treated as apoplexy. TRAP A field defect that respects the vertical midline is never retinal. Pallor with no history of swelling means chronic compression.

Anchor, caveat, monitor

A memorised dose used in the wrong tissue or the wrong host is a wrong answer.

Antimicrobial anchors

INDICATION	ANCHOR	CAVEAT THAT GETS EXAMINED
Bacterial keratitis	Fluoroquinolone for small peripheral ulcers; fortified therapy for central, deep or resistant disease. Load every 5-15 minutes, then hourly.	Culture first when central, large, deep or unresponsive. Do not wait for results to treat.
HSV epithelial keratitis	Aciclovir 400 mg five times daily, or valaciclovir 500 mg three times daily, 7-10 days. Topical ganciclovir 0.15% is an alternative.	No steroid over an active epithelial ulcer. Steroid belongs to immune stromal or endothelial disease, with antiviral cover.
Herpes zoster ophthalmicus	Valaciclovir 1 g three times daily, or aciclovir 800 mg five times daily, 7-10 days, renally adjusted.	Ocular disease lags the rash. A quiet first examination does not end surveillance.
Fungal keratitis	Natamycin 5% hourly initially for filamentous organisms, prolonged course, cornea specialist involved.	Corticosteroid accelerates progression. Vegetative trauma and feathery margins are the clues.
Acanthamoeba keratitis	Intensive biguanide therapy, PHMB or chlorhexidine 0.02%, often with a diamidine, for months.	Pain out of proportion, water exposure. Early disease is mislabelled HSV.
Hyperacute conjunctivitis	Systemic ceftriaxone-based treatment per current CDC guidance, saline lavage, STI testing, partner care.	Copious purulence with corneal risk is an emergency, not a drop problem.

Contraindication anchors

Topical beta-blocker CAUTION AVOID Bronchospastic asthma or COPD, bradycardia, heart block, decompensated heart failure. REDUCE Punctal occlusion and lid closure lower systemic absorption. TRAP A resting pulse in the low 50s is in the stem for a reason.	Alpha-2 agonist CAUTION AVOID Brimonidine in infants and very young children: CNS depression and apnoea. EXPECT Fatigue, dry mouth, late follicular allergy, hypotension. TRAP Check interacting systemic drugs before prescribing.
Miotic CAUTION AVOID Susceptible peripheral retina, active inflammation, ciliochoroidal effusion mechanisms. EXPECT Brow ache, induced myopia, poor night vision. TRAP Useless with an ischaemic sphincter, harmful in topiramate-type effusion.	Topical corticosteroid CAUTION AVOID Active HSV epithelial disease, fungal and Acanthamoeba keratitis, uncontrolled bacterial infection. MONITOR IOP, epithelial healing, cataract, and the taper. TRAP Name tissue, cause, dose, and monitoring, or the answer is incomplete.

Lowering pressure

Pressure-lowering classes

CLASS	MECHANISM	ANCHOR, AND THE CAVEAT THAT GETS EXAMINED
Prostaglandin analogue	Increases uveoscleral outflow	Once nightly, the largest single-agent reduction. Hyperaemia, lash growth, iris and periocular pigmentation, orbital fat atrophy. Caution with active uveitis or a history of herpetic keratitis.
Beta-blocker	Reduces aqueous production	Twice daily, or once for gel-forming. Avoid in bronchospastic disease, bradycardia, heart block and decompensated failure. A resting pulse in the low 50s is in the stem for a reason.
Alpha-2 agonist	Reduces production, some outflow	Two or three times daily. Late follicular allergy is common. Contraindicated in infants and young children: CNS depression and apnoea.
Topical carbonic anhydrase inhibitor	Reduces production	Two or three times daily. Stinging, metallic taste. Caution with sulfonamide reactions and low endothelial reserve.
Systemic acetazolamide	Reduces production strongly	For acute pressure crises. Paraesthesia, malaise, metabolic acidosis, renal stones, hypokalaemia. Avoid in sickle disease and significant renal impairment.
Rho kinase inhibitor	Increases trabecular outflow	Once nightly, additive to other classes. Hyperaemia is very common, with subconjunctival haemorrhage and corneal verticillata.
Cholinergic agonist	Increases trabecular outflow	Useful in plateau iris and pseudophakic eyes. Brow ache, induced myopia, poor night vision. Useless against an ischaemic sphincter and harmful in ciliochoroidal effusion.

CHOOSING THE SECOND AGENT

Add a class with a different mechanism, not a second drug of the same one. Check adherence and instillation technique before escalating, and count the preservative load on a compromised surface. Every added bottle costs adherence, which is the commonest reason a target pressure is missed.

Controlling inflammation

Four steps, and the specific thing each one fails at. Naming the step is usually easier than naming where it stops working, which is what the item asks.

Anti-inflammatory ladder, and where each step fails

Topical NSAID	SURFACE AND MACULA	Topical steroid	ANTERIOR SEGMENT
USE	Postoperative pain and macular oedema prophylaxis, episcleritis, some allergic disease.	USE	Anterior uveitis, immune stromal keratitis with antiviral cover, significant postoperative inflammation.
FAILS	Not adequate for significant intraocular inflammation.	FAILS	Does not reach the posterior segment reliably, and does not control scleritis.
TRAP	Prolonged use on a compromised surface, particularly a neurotrophic or post-refractive cornea, causes melt.	TRAP	Contraindicated over active HSV epithelial disease, fungal and Acanthamoeba keratitis, and uncontrolled bacterial infection. Monitor pressure and the taper.
Periocular and intravitreal steroid	POSTERIOR	Systemic steroid and steroid sparing	HOST
USE	Intermediate and posterior uveitis, refractory macular oedema.	USE	Scleritis, arteritic ischaemic neuropathy, sympathetic ophthalmia, sight-threatening systemic disease.
FAILS	Not for undiagnosed infectious disease, and not a substitute for systemic control in systemic vasculitis.	FAILS	Long-term control. Immunomodulatory therapy is planned from the start, co-managed.
TRAP	Steroid response and cataract are expected, not surprises. Exclude infection first, every time.	TRAP	Glucose, bone, blood pressure, mood, infection risk and vaccination status are all part of the correct answer.

HOST FACTORS THAT CHANGE THE DRUG

Pregnancy and breastfeeding, age at either extreme, renal and hepatic function, asthma and cardiac conduction, sulfonamide reactions, sickle disease, immunosuppression, and the systemic drug list. When a stem gives you a weight, a pulse, a creatinine or a pregnancy, it is the discriminator, not decoration.

Arithmetic, fast

The fastest marks on the exam, and the easiest to lose to a sign error.

Formulas

NAME	FORM	USE AND SIGN DISCIPLINE
Spherical equivalent	sph + cyl/2	Halve the cylinder with its own sign. -6.00 -2.00 x 180 gives -7.00 D.
Transposition	sph + cyl, flip, 90°	New sphere is old sphere plus old cylinder, reverse the cylinder sign, rotate the axis by 90 degrees.
Prentice rule	$\Delta = \text{cm} \times \text{D}$	Decentration in centimetres. Base follows the thick edge for plus lenses, the thin edge for minus.
Vertex conversion	$F2 = F1/(1 - dF1)$	Distance in metres, signs preserved. Matters beyond about ± 4.00 D.
Kestenbaum estimate	1 / acuity fraction	Starting magnification only. Refine by task, working distance, contrast, field, endurance.
Equivalent viewing power	1 / metres	10 cm working distance is 10 D.
Hofstetter	$18.5 - 0.30 \text{ age}$	Average. Minimum is $15 - 0.25 \text{ age}$, maximum $25 - 0.40 \text{ age}$. A screening comparison, not a diagnosis.
Gradient AC/A	$\Delta \text{phoria} / \Delta \text{D}$	State the sign convention before you calculate.
Screening arithmetic	$\text{TP}/(\text{TP}+\text{FN})$	Sensitivity. Specificity is $\text{TN}/(\text{TN}+\text{FP})$. PPV rises with prevalence; NPV falls.

Drills: cover the right column

PROMPT	WORKING AND ANSWER
Transpose +2.00 -3.00 x 180	-1.00 +3.00 x 090
Spherical equivalent of -4.00 -2.00 x 090	-4.00 + (-1.00) = -5.00 D
4 mm decentration through +5.00 D	$0.4 \times 5 = 2\Delta$
-10.00 D spectacle at 12 mm, to the corneal plane	$-10/(1 - (0.012 \times -10)) = -8.93 \text{ D}$
Starting magnification for 20/160	$160/20 = 8\times$
Hofstetter average amplitude at age 20	$18.5 - 6.0 = 12.5 \text{ D}$
Hydroxychloroquine 400 mg daily at 60 kg	6.67 mg/kg, above the 5 mg/kg threshold
Soft toric rotated 10° left of a 180 axis	LARS: order 010
GP steepened 0.50 D, over-refraction -0.50 D, from -3.00 D	SAM: $-3.00 -0.50 -0.50 = -4.00 \text{ D}$
Anisometropia -2.00 OD, -5.00 OS, 10 mm below centre	2Δ and 5Δ base down: 3Δ imbalance

Marks are lost to sign errors in transposition and vertex, to forgetting that Prentice needs centimetres, and to quoting a magnification estimate as a prescription. Write one line of working every time.

Lenses, and the arithmetic that follows

Fitting decisions and screening arithmetic. Both are fast marks, and both are lost to sign errors rather than to ignorance.

Contact lens and refractive decisions that get examined

SITUATION	THE RULE, AND THE ARITHMETIC IF THERE IS ANY
Soft toric rotation	Left add, right subtract, from the lens marking as observed. Rotated 10 degrees left of a 180 axis means order axis 010.
Rigid lens fit change	Steeper add minus, flatter add plus. Steepening 0.50 D from -3.00 D with a -0.50 D over-refraction gives -4.00 D.
Lens-related red eye	Stop wear, exclude infiltrative keratitis, and decide sterile against infectious on pain, epithelial defect, chamber reaction and location.
Vertex distance	Convert beyond about four dioptres. $F2 = F1$ divided by $(1 \text{ minus } dF1)$, distance in metres, signs preserved.
Induced prism	Prentice: prism dioptres equal decentration in centimetres times power. Anisometropia through a reading position is the classic item.
Myopia control	Discuss the options with their evidence, expected effect size and rebound, and document the discussion, including the option of doing nothing.

Screening arithmetic, worked once

QUANTITY	FORMULA	WORKED EXAMPLE: 100 WITH DISEASE, 900 WITHOUT, 80 DETECTED, 90 FALSE POSITIVES
Sensitivity	$TP / (TP + FN)$	$80 / 100 = 80\%$
Specificity	$TN / (TN + FP)$	$810 / 900 = 90\%$
Positive predictive value	$TP / (TP + FP)$	$80 / 170 = 47\%$
Negative predictive value	$TN / (TN + FN)$	$810 / 830 = 98\%$

THE POINT OF THAT TABLE

Sensitivity and specificity are properties of the test. Predictive values are properties of the population. Halve the prevalence and the positive predictive value falls while the test does not change, which is exactly what the item is testing.

The cheapest marks on the paper

Fifteen or so items turn on duty, consent and documentation. They are the least prepared part of the exam, and the answers are almost always the same shape: do the clinically safe thing, say so plainly, write it down, and keep the patient in the system.

Law, ethics and public health

ITEM TYPE	WHAT THE CORRECT ANSWER LOOKS LIKE
Informed refusal	Explain the risk in plain terms, offer the alternative, document the refusal and the specific risks discussed, arrange the shortest safe review, and keep the door open. Do not discharge the patient for refusing.
Capacity and consent	Capacity is decision-specific. Assess understanding, retention, weighing and communication. Involve a surrogate only when capacity is genuinely absent.
Minors	Consent from a parent or guardian, assent from the child, and jurisdiction-specific exceptions. Safeguarding concerns override confidentiality.
Records	Contemporaneous, factual, timed and attributed. Amend by addendum, never overwrite. Record what you were told, what you found, what you advised and what was declined.
Delegation	Delegate tasks, never clinical judgement. The delegating clinician remains responsible for the decision.
Telehealth	Establish that the standard of care can be met remotely. If the decision needs a finding you cannot obtain, arrange in-person examination.
Driving and work	Advise against driving where the standard is not met, document the advice, and follow the local reporting duty exactly.
Notifiable and public health	Report as jurisdiction requires, treat the patient, and address contacts and partners where the disease demands it.

THE SHAPE OF A CORRECT ANSWER

Act on the clinical risk first. Explain in plain language. Document what you found, advised and were refused. Set the shortest safe review interval. Follow the local duty exactly, and never make the patient leave the system as a punishment for disagreeing with you.

Competing pairs

Each pair is two conditions that trade places under time pressure. The decider row is the single finding that separates them, which is the finding the item will turn on.

Pseudomyopia		vs	The mimic
TEMPO	Hours		Weeks
PAIN	Severe		None
PUPIL	RAPD		Normal
DECIDER	Stub decider The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the.		

Toric Rotation		vs	The mimic
TEMPO	Hours		Weeks
PAIN	Severe		None
PUPIL	RAPD		Normal
DECIDER	Stub decider The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the.		

Accommodative Insufficiency		vs	The mimic
TEMPO	Hours		Weeks
PAIN	Severe		None
PUPIL	RAPD		Normal
DECIDER	Stub decider The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the.		

Intermittent Exotropia		vs	The mimic
TEMPO	Hours		Weeks
PAIN	Severe		None
PUPIL	RAPD		Normal
DECIDER	Stub decider The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the.		

Retinoblastoma		vs	Coats disease
FUNDUS	Chalky elevated mass with vitreous seeds		Yellow lipid exudate with retinal telangiectasia
LATERALITY	Unilateral or bilateral, often multifocal		Almost always unilateral, usually male
ULTRASOUND	Calcification within the mass		No calcification, exudate and detachment only
AGE	Median 18 months, most under 2 years		Usually 4 to 8 years at presentation
DECIDER	Calcification with vitreous seeding means retinoblastoma and a same-day oncology referral. Exudate with telangiectasia and no calcium means Coats, urgent but not oncological.		

Competing pairs, continued

Infectious keratitis after LASIK		vs	Diffuse lamellar keratitis
ONSET	Day 2 onwards, worsening hour by hour		Day 1 to 2, then improving on steroid
SYMPTOMS	Increasing pain with photophobia		Mild or no pain, blur dominates
INFILTRATE	Focal and dense, crosses into stroma		Diffuse granular haze within the interface
EPITHELIUM, CHAMBER	Defect present, cells in the chamber		Epithelium intact, chamber quiet
DECIDER	A focal infiltrate with an epithelial break or chamber cells means lift, culture and treat as infection. Diffuse haze with intact epithelium means hourly steroid and same-day review.		

Acute bacterial endophthalmitis		vs	Toxic anterior segment syndrome
ONSET	Day 2 to 5, worsening by the hour		Within 24 hours of surgery, then static
SYMPTOMS	Pain with lid swelling and injection		Little or no pain, blur dominates
CORNEA	Clear or focally hazy		Limbus-to-limbus oedema
VITREOUS	Dense opacities on ultrasound		Clear, the reaction stays anterior
DECIDER	Vitreous involvement with pain means tap and inject within the hour. Limbus-to-limbus oedema on day 1 with a clear vitreous means intensive steroid and a look at the theatre process.		

Acute retinal necrosis		vs	Cytomegalovirus retinitis
HOST	Immunocompetent, often after shingles		Deeply immunosuppressed, low CD4 count
VITREOUS	Dense vitritis with an aching eye		Little or no vitreous reaction, painless
BORDER	Scalloped, spreads circumferentially in days		Granular edge, creeps over weeks
VESSELS	Occlusive arteritis with sheathing		Haemorrhage with perivascular whitening
DECIDER	A well patient with dense vitritis and arteritis needs induction antiviral today. A granular indolent border in an immunosuppressed patient points at cytomegalovirus and its own agents.		

Choroidal melanoma		vs	Choroidal naevus
THICKNESS	Over 2 mm, and growing between records		Under 2 mm and unchanged for years
SURFACE	Orange lipofuscin, no drusen		Drusen, often with a pigmented halo
FLUID	Subretinal fluid at the margin		No fluid, retina flat over it
ULTRASOUND	Low internal reflectivity, excavation		Too flat to measure, no excavation
DECIDER	Thickness over two millimetres, subretinal fluid, orange pigment or documented growth each turn observation into a same-week ocular oncology referral.		

Competing pairs, continued

Internuclear ophthalmoplegia		vs	Ocular myasthenia mimicking it
CONVERGENCE	Full, because it bypasses the fasciculus		Variable, and it fatigues with effort
NYSTAGMUS	Coarse in the abducting eye		Absent, though a lid twitch may be seen
LIDS	Level and symmetrical		Ptosis, worsening through the day
COURSE	Fixed over days, with other lesions		Fluctuates within a day, better after rest
DECIDER	Fatigue, fluctuation and ptosis make it myasthenia and move the test to antibodies. A fixed adduction deficit with abducting nystagmus and full convergence needs MRI of the brainstem.		

Drug-induced ciliochoroidal effusion		vs	Acute primary angle closure
LATERALITY	Both eyes within hours of each other		One eye now, the fellow eye at risk later
CHAMBER	Shallow centrally and peripherally		Shallow peripherally, central depth near normal
REFRACTION	Myopic shift of 2 to 4 D		No shift once the corneal oedema clears
IRIDOTOMY	Angle stays closed afterwards		Angle opens and the attack breaks
DECIDER	A myopic shift with uniform shallowing in both eyes means the ciliary body has rotated forward. Stop the drug, cycloplege and suppress aqueous, and do not cut an iridotomy for it.		

Plateau iris syndrome		vs	Residual pupil block
IRIDOTOMY	Patent, and the angle closes anyway		Small, sealed or absent on retroillumination
IRIS CONTOUR	Flat plane with a sudden peripheral rise		Bowed forward from the pupil outward
CENTRAL DEPTH	Normal for the eye		Shallow, and it deepens once the hole works
INDENTATION	Double hump over the ciliary processes		Angle opens smoothly and stays open
DECIDER	A patent iridotomy with a flat iris and normal central depth means the ciliary body holds the angle shut. The answer is iridoplasty or lens extraction, never a second hole.		

PART 8

Session 2

Thirty-five cases under conditions identical to Volume One. This session is comparable with that one, which is the whole point of rebuilding the key.

WHY IT IS COMPARABLE

Both sessions carry the same letter distribution, the same item-type mix and the same option-length discipline. A four-point improvement here is four points of learning, not an easier key.

WEIGHTING

Session 2 leans harder on postoperative and surgical co-management, neuro-ophthalmic localisation, paediatrics, and the refractive and binocular vision domains that decide TMOD.

SIT IT AFTER THE REPAIR WORK

Volume One scored, the weakest two item-type columns worked through, the error log written as if-then rules. Then this.

AFTERWARDS

Compare item-type columns across the two volumes, not totals. A total that rose while the treatment column fell is a warning, not progress.

Session 2 · case index

Sorted as sat. Dx diagnosis, Tx treatment, Sci basic science, Law law and ethics, Sel select all that apply. Use the item-type column to work a single weakness across the whole session.

CASE	PRESENTATION	DOMAIN	ITEMS	CASE	KEY
S2-01	Stub presentation for pseudomyopia	Cornea and refractive surgery	Dx Tx Sci Dx* Tx	27	63
S2-02	Stub presentation for anisometropia	Retina, choroid, vitreous	Dx Tx Sci Dx* Tx	28	64
S2-03	Stub presentation for vertical imbalance	Optic nerve and neuro-ophthalmic	Dx Tx Sci Dx* Tx	29	65
S2-04	Stub presentation for toric rotation	Glaucoma	Dx Tx Sci Dx* Tx	30	66
S2-05	Stub presentation for flat GP fit	Contact lenses	Dx Tx Sci Dx* Tx	31	67
S2-06	Stub presentation for RP-spectrum low vision	Lids, lacrimal, adnexa, orbit	Dx Tx Sci Dx* Tx	32	68
S2-07	Stub presentation for accommodative insufficiency	Episclera, sclera, anterior uvea	Dx Tx Sci Dx* Tx	33	69
S2-08	Stub presentation for high AC/A esophoria	Emergencies and trauma	Dx Tx Sci Dx* Tx	34	70
S2-09	Stub presentation for divergence insufficiency	Accommodation and vergence	Dx Tx Sci Dx* Tx	35	71
S2-10	Stub presentation for intermittent exotropia	Lens, cataract, IOL, perioperative	Dx Tx Sci Dx* Tx	36	72
S2-11	Three months into treatment the colours went flat	Optic nerve and neuro-ophthalmic	Dx Dx* Tx Tx Sci	37	73
S2-12	The white reflex in the holiday photograph	Retina, choroid, vitreous	Dx Tx Dx* Law Sci	38	74
S2-13	The lump that keeps coming back in the same place	Lids, lacrimal, adnexa, orbit	Dx Dx Tx Dx* Tx	39	75
S2-14	Watering since the first weeks of life	Lids, lacrimal, adnexa, orbit	Dx Dx Tx Tx* Tx	40	76
S2-15	The eye that will not look up after football	Emergencies and trauma	Dx Tx Tx Law* Sci	41	77
S2-16	Haze under the flap on the third day	Cornea and refractive surgery	Dx Dx* Tx Tx Sci	42	78
S2-17	Wet cement, and a cornea that has gone grey	Emergencies and trauma	Tx Tx* Dx Sci Tx	43	79
S2-18	A large epithelial defect that does not hurt	Cornea and refractive surgery	Dx Sci Tx Tx* Tx	44	80
S2-19	Pain and a lost view on the third day after surgery	Lens, cataract, IOL, perioperative	Dx Tx Tx* Tx Sci	45	81
S2-20	A cloudy cornea the morning after routine surgery	Lens, cataract, IOL, perioperative	Dx Dx* Tx Tx Sci	46	82
S2-21	The eye that bleeds a little every few months	Lens, cataract, IOL, perioperative	Dx Tx* Tx Sci Tx	47	83
S2-22	A wobble in the vision nine years after surgery	Lens, cataract, IOL, perioperative	Dx Dx* Tx Tx Sci	48	84
S2-23	A haze in the vision beside an old scar	Retina, choroid, vitreous	Dx Tx* Tx Sci Tx	49	85
S2-24	A crescent of cornea melting at the edge	Cornea and refractive surgery	Dx Tx Tx* Tx Sci	50	86
S2-25	The upper field that went and did not come back	Retina, choroid, vitreous	Dx Dx Tx Tx Law*	51	87
S2-26	An aching eye that lost its field in five days	Retina, choroid, vitreous	Dx Tx* Tx Sci Tx	52	88
S2-27	The dark lesion that grew between two photographs	Retina, choroid, vitreous	Dx Dx* Tx Tx Sci	53	89
S2-28	Two years of headache and a field she never noticed	Optic nerve and neuro-ophthalmic	Dx Dx Tx Tx Sci*	54	90
S2-29	The eye that will not follow its partner across	Accommodation, vergence, oculomotor	Dx Dx Dx* Tx Sci	55	91
S2-30	The same half of the world gone in both eyes	Optic nerve and neuro-ophthalmic	Dx Dx Tx Tx* Law	56	92
S2-31	Vision that dropped like a blind for four minutes	Systemic health	Dx Tx* Tx Sci Law	57	93
S2-32	The eye that took a squash ball twenty years ago	Glaucoma	Dx Dx* Tx Tx Law	58	94
S2-33	Both eyes blurred a week after a new tablet	Glaucoma	Dx Tx* Tx Sci Dx	59	95
S2-34	Pressure that climbed again after a laser hole	Glaucoma	Dx Dx* Tx Tx Tx	60	96
S2-35	The screening result he does not want to act on	Glaucoma	Sci Sci* Law Law Tx	61	97

Session 2 • score by item type

Each cell carries its item type. Mark it right or wrong, then total the five columns on the right.

CASE	1	2	3	4	5	6	DX	TX	SCI	LAW
S2-01	Dx	Tx	Sci	Dx*	Tx					
S2-02	Dx	Tx	Sci	Dx*	Tx					
S2-03	Dx	Tx	Sci	Dx*	Tx					
S2-04	Dx	Tx	Sci	Dx*	Tx					
S2-05	Dx	Tx	Sci	Dx*	Tx					
S2-06	Dx	Tx	Sci	Dx*	Tx					
S2-07	Dx	Tx	Sci	Dx*	Tx					
S2-08	Dx	Tx	Sci	Dx*	Tx					
S2-09	Dx	Tx	Sci	Dx*	Tx					
S2-10	Dx	Tx	Sci	Dx*	Tx					
S2-11	Dx	Dx*	Tx	Tx	Sci					
S2-12	Dx	Tx	Dx*	Law	Sci					
S2-13	Dx	Dx	Tx	Dx*	Tx					
S2-14	Dx	Dx	Tx	Tx*	Tx					
S2-15	Dx	Tx	Tx	Law*	Sci					
S2-16	Dx	Dx*	Tx	Tx	Sci					
S2-17	Tx	Tx*	Dx	Sci	Tx					
S2-18	Dx	Sci	Tx	Tx*	Tx					
S2-19	Dx	Tx	Tx*	Tx	Sci					
S2-20	Dx	Dx*	Tx	Tx	Sci					
S2-21	Dx	Tx*	Tx	Sci	Tx					
S2-22	Dx	Dx*	Tx	Tx	Sci					
S2-23	Dx	Tx*	Tx	Sci	Tx					
S2-24	Dx	Tx	Tx*	Tx	Sci					
S2-25	Dx	Dx	Tx	Tx	Law*					
S2-26	Dx	Tx*	Tx	Sci	Tx					
S2-27	Dx	Dx*	Tx	Tx	Sci					
S2-28	Dx	Dx	Tx	Tx	Sci*					
S2-29	Dx	Dx	Dx*	Tx	Sci					
S2-30	Dx	Dx	Tx	Tx*	Law					
S2-31	Dx	Tx*	Tx	Sci	Law					
S2-32	Dx	Dx*	Tx	Tx	Law					
S2-33	Dx	Tx*	Tx	Sci	Dx					
S2-34	Dx	Dx*	Tx	Tx	Tx					
S2-35	Sci	Sci*	Law	Law	Tx					

Totals

Mark each item R or W. A star means all or none

Then total the right-hand columns by item type

READING YOUR OWN TOTALS

Under 60% in any single column is a study plan, not a bad day. Diagnosis misses send you to the condition cards.

Treatment misses send you to pharmacology and the emergency gate. Basic science misses are the cheapest to fix and the easiest to postpone.

CASE S2-01 • CORNEA AND REFRACTIVE SURGERY • 5 ITEMS

Stub presentation for pseudomyopia

Stub stem for pseudomyopia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

1 Stub question 1 about pseudomyopia The stem carries the host factor, the interval and the number. DIAGNOSIS

- A Stub option 1 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- B Stub option 2 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- C Stub option 3 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- D Stub option 4 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- E Stub option 5 for pseudomyopia The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

2 Stub question 2 about pseudomyopia The stem carries the host factor, the interval and the number. TREATMENT

- A Stub option 1 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- B Stub option 2 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- C Stub option 3 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- D Stub option 4 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- E Stub option 5 for pseudomyopia The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

3 Stub question 3 about pseudomyopia The stem carries the host factor, the interval and the number. BASIC SCIENCE

- A Stub option 2 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- B Stub option 3 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- C Stub option 4 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- D Stub option 5 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- E Stub option 1 for pseudomyopia The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

4 Stub question 4 about pseudomyopia The stem carries the host factor, the interval and the number. DIAGNOSIS • SELECT 3

- A Stub option 5 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- B Stub option 6 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- C Stub option 1 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- D Stub option 2 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- E Stub option 3 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- F Stub option 4 for pseudomyopia The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

5 Stub question 5 about pseudomyopia The stem carries the host factor, the interval and the number. TREATMENT

- A Stub option 4 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- B Stub option 5 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- C Stub option 1 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- D Stub option 2 for pseudomyopia The stem carries the host factor, the interval and the number that decides.
- E Stub option 3 for pseudomyopia The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

CASE S2-02 • RETINA, CHOROID, VITREOUS • 5 ITEMS

Stub presentation for anisometropia

Stub stem for anisometropia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

1 Stub question 1 about anisometropia The stem carries the host factor, the interval and the number. DIAGNOSIS

- A Stub option 2 for anisometropia The stem carries the host factor, the interval and the number that decides.
- B Stub option 3 for anisometropia The stem carries the host factor, the interval and the number that decides.
- C Stub option 4 for anisometropia The stem carries the host factor, the interval and the number that decides.
- D Stub option 5 for anisometropia The stem carries the host factor, the interval and the number that decides.
- E Stub option 1 for anisometropia The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

2 Stub question 2 about anisometropia The stem carries the host factor, the interval and the number. TREATMENT

- A Stub option 3 for anisometropia The stem carries the host factor, the interval and the number that decides.
- B Stub option 4 for anisometropia The stem carries the host factor, the interval and the number that decides.
- C Stub option 5 for anisometropia The stem carries the host factor, the interval and the number that decides.
- D Stub option 1 for anisometropia The stem carries the host factor, the interval and the number that decides.
- E Stub option 2 for anisometropia The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

3 Stub question 3 about anisometropia The stem carries the host factor, the interval and the number. BASIC SCIENCE

- A Stub option 5 for anisometropia The stem carries the host factor, the interval and the number that decides.
- B Stub option 1 for anisometropia The stem carries the host factor, the interval and the number that decides.
- C Stub option 2 for anisometropia The stem carries the host factor, the interval and the number that decides.
- D Stub option 3 for anisometropia The stem carries the host factor, the interval and the number that decides.
- E Stub option 4 for anisometropia The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

4 Stub question 4 about anisometropia The stem carries the host factor, the interval and the number. DIAGNOSIS • SELECT 3

- A Stub option 6 for anisometropia The stem carries the host factor, the interval and the number that decides.
- B Stub option 1 for anisometropia The stem carries the host factor, the interval and the number that decides.
- C Stub option 2 for anisometropia The stem carries the host factor, the interval and the number that decides.
- D Stub option 3 for anisometropia The stem carries the host factor, the interval and the number that decides.
- E Stub option 4 for anisometropia The stem carries the host factor, the interval and the number that decides.
- F Stub option 5 for anisometropia The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

5 Stub question 5 about anisometropia The stem carries the host factor, the interval and the number. TREATMENT

- A Stub option 4 for anisometropia The stem carries the host factor, the interval and the number that decides.
- B Stub option 5 for anisometropia The stem carries the host factor, the interval and the number that decides.
- C Stub option 1 for anisometropia The stem carries the host factor, the interval and the number that decides.
- D Stub option 2 for anisometropia The stem carries the host factor, the interval and the number that decides.
- E Stub option 3 for anisometropia The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

CASE S2-03 • OPTIC NERVE AND NEURO-OPHTHALMIC • 5 ITEMS

Stub presentation for vertical imbalance

Stub stem for vertical imbalance The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the.

1 Stub question 1 about vertical imbalance The stem carries the host factor, the interval and the.

DIAGNOSIS

- A Stub option 1 for vertical imbalance The stem carries the host factor, the interval and the number that.
- B Stub option 2 for vertical imbalance The stem carries the host factor, the interval and the number that.
- C Stub option 3 for vertical imbalance The stem carries the host factor, the interval and the number that.
- D Stub option 4 for vertical imbalance The stem carries the host factor, the interval and the number that.
- E Stub option 5 for vertical imbalance The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

2 Stub question 2 about vertical imbalance The stem carries the host factor, the interval and the.

TREATMENT

- A Stub option 2 for vertical imbalance The stem carries the host factor, the interval and the number that.
- B Stub option 3 for vertical imbalance The stem carries the host factor, the interval and the number that.
- C Stub option 4 for vertical imbalance The stem carries the host factor, the interval and the number that.
- D Stub option 5 for vertical imbalance The stem carries the host factor, the interval and the number that.
- E Stub option 1 for vertical imbalance The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

3 Stub question 3 about vertical imbalance The stem carries the host factor, the interval and the.

BASIC SCIENCE

- A Stub option 4 for vertical imbalance The stem carries the host factor, the interval and the number that.
- B Stub option 5 for vertical imbalance The stem carries the host factor, the interval and the number that.
- C Stub option 1 for vertical imbalance The stem carries the host factor, the interval and the number that.
- D Stub option 2 for vertical imbalance The stem carries the host factor, the interval and the number that.
- E Stub option 3 for vertical imbalance The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

4 Stub question 4 about vertical imbalance The stem carries the host factor, the interval and the.

DIAGNOSIS • SELECT 3

- A Stub option 1 for vertical imbalance The stem carries the host factor, the interval and the number that.
- B Stub option 2 for vertical imbalance The stem carries the host factor, the interval and the number that.
- C Stub option 3 for vertical imbalance The stem carries the host factor, the interval and the number that.
- D Stub option 4 for vertical imbalance The stem carries the host factor, the interval and the number that.
- E Stub option 5 for vertical imbalance The stem carries the host factor, the interval and the number that.
- F Stub option 6 for vertical imbalance The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

5 Stub question 5 about vertical imbalance The stem carries the host factor, the interval and the.

TREATMENT

- A Stub option 2 for vertical imbalance The stem carries the host factor, the interval and the number that.
- B Stub option 3 for vertical imbalance The stem carries the host factor, the interval and the number that.
- C Stub option 4 for vertical imbalance The stem carries the host factor, the interval and the number that.
- D Stub option 5 for vertical imbalance The stem carries the host factor, the interval and the number that.
- E Stub option 1 for vertical imbalance The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

CASE S2-04 • GLAUCOMA • 5 ITEMS

Stub presentation for toric rotation

Stub stem for toric rotation The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

1 Stub question 1 about toric rotation The stem carries the host factor, the interval and the.

DIAGNOSIS

- A Stub option 3 for toric rotation The stem carries the host factor, the interval and the number that decides.
- B Stub option 4 for toric rotation The stem carries the host factor, the interval and the number that decides.
- C Stub option 5 for toric rotation The stem carries the host factor, the interval and the number that decides.
- D Stub option 1 for toric rotation The stem carries the host factor, the interval and the number that decides.
- E Stub option 2 for toric rotation The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

2 Stub question 2 about toric rotation The stem carries the host factor, the interval and the.

TREATMENT

- A Stub option 5 for toric rotation The stem carries the host factor, the interval and the number that decides.
- B Stub option 1 for toric rotation The stem carries the host factor, the interval and the number that decides.
- C Stub option 2 for toric rotation The stem carries the host factor, the interval and the number that decides.
- D Stub option 3 for toric rotation The stem carries the host factor, the interval and the number that decides.
- E Stub option 4 for toric rotation The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

3 Stub question 3 about toric rotation The stem carries the host factor, the interval and the.

BASIC SCIENCE

- A Stub option 1 for toric rotation The stem carries the host factor, the interval and the number that decides.
- B Stub option 2 for toric rotation The stem carries the host factor, the interval and the number that decides.
- C Stub option 3 for toric rotation The stem carries the host factor, the interval and the number that decides.
- D Stub option 4 for toric rotation The stem carries the host factor, the interval and the number that decides.
- E Stub option 5 for toric rotation The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

4 Stub question 4 about toric rotation The stem carries the host factor, the interval and the.

DIAGNOSIS • SELECT 3

- A Stub option 2 for toric rotation The stem carries the host factor, the interval and the number that decides.
- B Stub option 3 for toric rotation The stem carries the host factor, the interval and the number that decides.
- C Stub option 4 for toric rotation The stem carries the host factor, the interval and the number that decides.
- D Stub option 5 for toric rotation The stem carries the host factor, the interval and the number that decides.
- E Stub option 6 for toric rotation The stem carries the host factor, the interval and the number that decides.
- F Stub option 1 for toric rotation The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

5 Stub question 5 about toric rotation The stem carries the host factor, the interval and the.

TREATMENT

- A Stub option 3 for toric rotation The stem carries the host factor, the interval and the number that decides.
- B Stub option 4 for toric rotation The stem carries the host factor, the interval and the number that decides.
- C Stub option 5 for toric rotation The stem carries the host factor, the interval and the number that decides.
- D Stub option 1 for toric rotation The stem carries the host factor, the interval and the number that decides.
- E Stub option 2 for toric rotation The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-05 • CONTACT LENSES • 5 ITEMS

Stub presentation for flat GP fit

Stub stem for flat GP fit The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

1 Stub question 1 about flat GP fit The stem carries the host factor, the interval and the number.

DIAGNOSIS

- A Stub option 4 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- B Stub option 5 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- C Stub option 1 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- D Stub option 2 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- E Stub option 3 for flat GP fit The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

2 Stub question 2 about flat GP fit The stem carries the host factor, the interval and the number.

TREATMENT

- A Stub option 1 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- B Stub option 2 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- C Stub option 3 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- D Stub option 4 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- E Stub option 5 for flat GP fit The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

3 Stub question 3 about flat GP fit The stem carries the host factor, the interval and the number.

BASIC SCIENCE

- A Stub option 5 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- B Stub option 1 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- C Stub option 2 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- D Stub option 3 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- E Stub option 4 for flat GP fit The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

4 Stub question 4 about flat GP fit The stem carries the host factor, the interval and the number.

DIAGNOSIS • SELECT 3

- A Stub option 3 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- B Stub option 4 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- C Stub option 5 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- D Stub option 6 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- E Stub option 1 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- F Stub option 2 for flat GP fit The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

5 Stub question 5 about flat GP fit The stem carries the host factor, the interval and the number.

TREATMENT

- A Stub option 1 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- B Stub option 2 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- C Stub option 3 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- D Stub option 4 for flat GP fit The stem carries the host factor, the interval and the number that decides.
- E Stub option 5 for flat GP fit The stem carries the host factor, the interval and the number that decides.

ANSWER _____ CONFIDENCE L • M • H

CASE S2-06 • LIDS, LACRIMAL, ADNEXA, ORBIT • 5 ITEMS

Stub presentation for RP-spectrum low vision

Stub stem for RP-spectrum low vision The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries.

1 Stub question 1 about RP-spectrum low vision The stem carries the host factor, the interval and.

DIAGNOSIS

- A Stub option 1 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- B Stub option 2 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- C Stub option 3 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- D Stub option 4 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- E Stub option 5 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

2 Stub question 2 about RP-spectrum low vision The stem carries the host factor, the interval and.

TREATMENT

- A Stub option 4 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- B Stub option 5 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- C Stub option 1 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- D Stub option 2 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- E Stub option 3 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

3 Stub question 3 about RP-spectrum low vision The stem carries the host factor, the interval and.

BASIC SCIENCE

- A Stub option 2 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- B Stub option 3 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- C Stub option 4 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- D Stub option 5 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- E Stub option 1 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

4 Stub question 4 about RP-spectrum low vision The stem carries the host factor, the interval and.

DIAGNOSIS • SELECT 3

- A Stub option 4 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- B Stub option 5 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- C Stub option 6 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- D Stub option 1 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- E Stub option 2 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- F Stub option 3 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

5 Stub question 5 about RP-spectrum low vision The stem carries the host factor, the interval and.

TREATMENT

- A Stub option 1 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- B Stub option 2 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- C Stub option 3 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- D Stub option 4 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.
- E Stub option 5 for RP-spectrum low vision The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

CASE S2-07 • EPISCLERA, SCLERA, ANTERIOR UVEA • 5 ITEMS

Stub presentation for accommodative insufficiency

Stub stem for accommodative insufficiency The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem.

1 Stub question 1 about accommodative insufficiency The stem carries the host factor, the interval.

DIAGNOSIS

- A Stub option 5 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- B Stub option 1 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- C Stub option 2 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- D Stub option 3 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- E Stub option 4 for accommodative insufficiency The stem carries the host factor, the interval and the number.

ANSWER _____ CONFIDENCE L • M • H

2 Stub question 2 about accommodative insufficiency The stem carries the host factor, the interval.

TREATMENT

- A Stub option 5 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- B Stub option 1 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- C Stub option 2 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- D Stub option 3 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- E Stub option 4 for accommodative insufficiency The stem carries the host factor, the interval and the number.

ANSWER _____ CONFIDENCE L • M • H

3 Stub question 3 about accommodative insufficiency The stem carries the host factor, the interval.

BASIC SCIENCE

- A Stub option 3 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- B Stub option 4 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- C Stub option 5 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- D Stub option 1 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- E Stub option 2 for accommodative insufficiency The stem carries the host factor, the interval and the number.

ANSWER _____ CONFIDENCE L • M • H

4 Stub question 4 about accommodative insufficiency The stem carries the host factor, the interval.

DIAGNOSIS • SELECT 3

- A Stub option 5 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- B Stub option 6 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- C Stub option 1 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- D Stub option 2 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- E Stub option 3 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- F Stub option 4 for accommodative insufficiency The stem carries the host factor, the interval and the number.

ANSWER _____ CONFIDENCE L • M • H

5 Stub question 5 about accommodative insufficiency The stem carries the host factor, the interval.

TREATMENT

- A Stub option 4 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- B Stub option 5 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- C Stub option 1 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- D Stub option 2 for accommodative insufficiency The stem carries the host factor, the interval and the number.
- E Stub option 3 for accommodative insufficiency The stem carries the host factor, the interval and the number.

ANSWER _____ CONFIDENCE L • M • H

CASE S2-08 • EMERGENCIES AND TRAUMA • 5 ITEMS

Stub presentation for high AC/A esophoria

Stub stem for high AC/A esophoria The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the.

1 Stub question 1 about high AC/A esophoria The stem carries the host factor, the interval and the.

DIAGNOSIS

- A Stub option 4 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- B Stub option 5 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- C Stub option 1 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- D Stub option 2 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- E Stub option 3 for high AC/A esophoria The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

2 Stub question 2 about high AC/A esophoria The stem carries the host factor, the interval and the.

TREATMENT

- A Stub option 3 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- B Stub option 4 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- C Stub option 5 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- D Stub option 1 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- E Stub option 2 for high AC/A esophoria The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

3 Stub question 3 about high AC/A esophoria The stem carries the host factor, the interval and the.

BASIC SCIENCE

- A Stub option 1 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- B Stub option 2 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- C Stub option 3 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- D Stub option 4 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- E Stub option 5 for high AC/A esophoria The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

4 Stub question 4 about high AC/A esophoria The stem carries the host factor, the interval and the.

DIAGNOSIS • SELECT 3

- A Stub option 6 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- B Stub option 1 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- C Stub option 2 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- D Stub option 3 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- E Stub option 4 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- F Stub option 5 for high AC/A esophoria The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

5 Stub question 5 about high AC/A esophoria The stem carries the host factor, the interval and the.

TREATMENT

- A Stub option 1 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- B Stub option 2 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- C Stub option 3 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- D Stub option 4 for high AC/A esophoria The stem carries the host factor, the interval and the number that.
- E Stub option 5 for high AC/A esophoria The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

CASE S2-09 • ACCOMMODATION AND VERGENCE • 5 ITEMS

Stub presentation for divergence insufficiency

Stub stem for divergence insufficiency The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries.

1 Stub question 1 about divergence insufficiency The stem carries the host factor, the interval and.

DIAGNOSIS

- A Stub option 4 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- B Stub option 5 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- C Stub option 1 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- D Stub option 2 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- E Stub option 3 for divergence insufficiency The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

2 Stub question 2 about divergence insufficiency The stem carries the host factor, the interval and.

TREATMENT

- A Stub option 3 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- B Stub option 4 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- C Stub option 5 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- D Stub option 1 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- E Stub option 2 for divergence insufficiency The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

3 Stub question 3 about divergence insufficiency The stem carries the host factor, the interval and.

BASIC SCIENCE

- A Stub option 1 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- B Stub option 2 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- C Stub option 3 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- D Stub option 4 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- E Stub option 5 for divergence insufficiency The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

4 Stub question 4 about divergence insufficiency The stem carries the host factor, the interval and.

DIAGNOSIS • SELECT 3

- A Stub option 1 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- B Stub option 2 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- C Stub option 3 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- D Stub option 4 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- E Stub option 5 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- F Stub option 6 for divergence insufficiency The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

5 Stub question 5 about divergence insufficiency The stem carries the host factor, the interval and.

TREATMENT

- A Stub option 4 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- B Stub option 5 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- C Stub option 1 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- D Stub option 2 for divergence insufficiency The stem carries the host factor, the interval and the number that.
- E Stub option 3 for divergence insufficiency The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

CASE S2-10 • LENS, CATARACT, IOL, PERIOPERATIVE • 5 ITEMS

Stub presentation for intermittent exotropia

Stub stem for intermittent exotropia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries.

1 Stub question 1 about intermittent exotropia The stem carries the host factor, the interval and.

DIAGNOSIS

- A Stub option 1 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- B Stub option 2 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- C Stub option 3 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- D Stub option 4 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- E Stub option 5 for intermittent exotropia The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

2 Stub question 2 about intermittent exotropia The stem carries the host factor, the interval and.

TREATMENT

- A Stub option 2 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- B Stub option 3 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- C Stub option 4 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- D Stub option 5 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- E Stub option 1 for intermittent exotropia The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

3 Stub question 3 about intermittent exotropia The stem carries the host factor, the interval and.

BASIC SCIENCE

- A Stub option 4 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- B Stub option 5 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- C Stub option 1 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- D Stub option 2 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- E Stub option 3 for intermittent exotropia The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

4 Stub question 4 about intermittent exotropia The stem carries the host factor, the interval and.

DIAGNOSIS • SELECT 3

- A Stub option 2 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- B Stub option 3 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- C Stub option 4 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- D Stub option 5 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- E Stub option 6 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- F Stub option 1 for intermittent exotropia The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

5 Stub question 5 about intermittent exotropia The stem carries the host factor, the interval and.

TREATMENT

- A Stub option 1 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- B Stub option 2 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- C Stub option 3 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- D Stub option 4 for intermittent exotropia The stem carries the host factor, the interval and the number that.
- E Stub option 5 for intermittent exotropia The stem carries the host factor, the interval and the number that.

ANSWER _____ CONFIDENCE L • M • H

CASE S2-11 • OPTIC NERVE AND NEURO-OPHTHALMIC • 5 ITEMS

Three months into treatment the colours went flat

A 58-year-old man on isoniazid, rifampicin, pyrazinamide and ethambutol for 3 months reports 6 weeks of dimming vision and washed-out colour. Weight 52 kg, ethambutol 1200 mg daily, eGFR 42. BVA 20/60 OD, 20/70 OS. Pupils equal and brisk, no RAPD. Ishihara 2 of 14 plates each eye. Symmetric central scotomas on both fields. Discs flat with normal colour. IOP 15 both eyes.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Ethambutol toxic optic neuropathy
- B Bilateral non-arteritic ischaemic optic neuropathy
- C Compressive lesion at the optic chiasm
- D Leber hereditary optic neuropathy
- E Isoniazid-related nutritional optic neuropathy

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings place this at the optic nerves as a symmetric process?

DIAGNOSIS • SELECT 3

- A Dense red-green loss, 2 of 14 plates each eye
- B Central scotomas of matched size in both fields
- C No RAPD despite acuity of 20/60 or worse
- D Onset while on four-drug antituberculous therapy
- E Ethambutol at 23 mg/kg with an eGFR of 42
- F Best corrected acuity reduced in both eyes

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate next step?

TREATMENT

- A Contact the TB team today to stop ethambutol
- B Reduce ethambutol to 15 mg/kg and review in 3 months
- C Stop all four antituberculous drugs immediately
- D Add high-dose pyridoxine and continue current doses
- E Start oral prednisolone and repeat fields in 4 weeks

ANSWER _____ CONFIDENCE L • M • H

4 He asks what to expect. Which advice and follow-up plan is correct?

TREATMENT

- A Hydroxocobalamin injections will restore the colour vision
- B Vision may fall for weeks after stopping, review at 6 weeks
- C Vision should return to normal within 2 weeks of stopping the drug
- D Restart ethambutol at a lower dose once acuity improves
- E Discharge now, since stopping the drug settles the matter

ANSWER _____ CONFIDENCE L • M • H

5 Which mechanism best explains the pattern of loss?

BASIC SCIENCE

- A Infarction in the short posterior ciliary artery territory
- B Drug deposition in the retinal pigment epithelium
- C Excitotoxic loss of the magnocellular ganglion cells
- D Metal chelation impairing mitochondrial complex IV
- E Immune demyelination of the retrobulbar optic nerve

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-12 • RETINA, CHOROID, VITREOUS • 5 ITEMS

The white reflex in the holiday photograph

A 14-month-old girl is brought in after a family photograph showed a white reflex in the right eye. Six weeks of intermittent right esotropia. Born at term, no supplemental oxygen, no family history of eye disease. She fixes and follows with the left eye only. No red reflex OD, white pupillary reflex, quiet conjunctiva, clear lens, normal corneal diameters. Dilated view OD: elevated chalky retinal mass with vitreous seeds.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Coats disease with exudative detachment
- B Persistent fetal vasculature
- C Ocular toxocariasis with a granuloma
- D Stage 5 retinopathy of prematurity
- E Retinoblastoma of the right eye

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate next step?

TREATMENT

- A Routine referral to paediatric ophthalmology in 6 weeks
- B Topical steroid for presumed intraocular inflammation
- C Same-day referral to a retinoblastoma service
- D Arrange a B-scan and review the child in 2 weeks
- E Fine needle biopsy of the mass to confirm the diagnosis

ANSWER _____ CONFIDENCE L • M • H

3 Which three steps belong in the specialist assessment of this child?

DIAGNOSIS • SELECT 3

- A RB1 testing for the child and both parents
- B CT of the head to demonstrate calcification
- C Transvitreal biopsy of the mass for histology
- D Bone marrow aspirate before any treatment
- E Examination under anaesthesia with indentation of both eyes
- F MRI of the orbits and the brain

ANSWER _____ CONFIDENCE L • M • H

4 Which action best meets the duty owed to this child?

LAW AND ETHICS

- A Refer today, escalate to safeguarding if she is not brought
- B Give the parents a letter to hand to their family doctor this week
- C Refer today and close the episode once the letter is sent
- D Wait for the parents to choose a date that suits them
- E Contact the safeguarding team before making any referral

ANSWER _____ CONFIDENCE L • M • H

5 Which genetic point should shape the counselling offered?

BASIC SCIENCE

- A Sporadic tumours need one somatic hit at the RB1 locus
- B Around 15% of unilateral cases carry a germline mutation
- C Unilateral disease excludes a germline mutation
- D Germline cases follow autosomal recessive inheritance
- E Pineoblastoma risk arises only with bilateral tumours

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-13 • LIDS, LACRIMAL, ADNEXA, ORBIT • 5 ITEMS

The lump that keeps coming back in the same place

A 71-year-old woman has a third firm nodule at the same site in the left upper lid over 14 months. Two incision and curettage procedures, no specimen sent either time. Over the lesion the lid margin has lost its lashes and its posterior ridge, the tarsus is yellow and thickened, and the eye has stayed red through lid hygiene and 6 weeks of topical antibiotic. Left preauricular node 1 cm, firm and mobile.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Chronic staphylococcal blepharoconjunctivitis
- B Sebaceous carcinoma of the tarsal glands
- C Recurrent chalazion with granulation tissue
- D Nodular basal cell carcinoma of the lid
- E Squamous cell carcinoma of the lid margin

ANSWER _____ CONFIDENCE L • M • H

2 Which single feature argues most strongly against a benign chalazion?

DIAGNOSIS

- A Recurrence at the same site after curettage
- B Yellow colour of the thickened tarsal tissue
- C A red eye that has not settled on treatment
- D Firmness of the nodule on palpation
- E Lash loss with destruction of the posterior lid margin

ANSWER _____ CONFIDENCE L • M • H

3 Which procedure should be arranged to obtain the diagnosis?

TREATMENT

- A Repeat incision and curettage, curettings to histology
- B Shave biopsy of the lid margin lesion under anaesthesia
- C Excision with 2 mm margins as the first procedure
- D Fine needle aspiration of the nodule and the node
- E Full-thickness lid biopsy plus conjunctival map biopsies

ANSWER _____ CONFIDENCE L • M • H

4 Which three features of this tumour explain why it is missed?

DIAGNOSIS • SELECT 3

- A It metastasises before the lid lesion is visible
- B It is confined to the lower lid in most cases
- C Pagetoid spread within conjunctival epithelium
- D Multicentric foci with skip lesions in the tarsus
- E Lipid lost in processing unless fresh tissue is used
- F It arises from the eccrine glands of the lid

ANSWER _____ CONFIDENCE L • M • H

5 Histology confirms it. Which plan is appropriate?

TREATMENT

- A Cryotherapy to the lid margin with topical mitomycin
- B Primary radiotherapy to the lid and the orbit
- C Exenteration as the first definitive procedure
- D Excision alone, with review of the node in 3 months
- E Wide excision with margin control, plus node imaging

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-14 • LIDS, LACRIMAL, ADNEXA, ORBIT • 5 ITEMS

Watering since the first weeks of life

A 9-month-old boy has watered from the right eye since the age of 3 weeks, with crusted lashes each morning that clean away easily. No photophobia and no lid squeezing. Corneas clear, horizontal diameters 10.5 mm each. The right tear lake is high, and fluorescein clears from it in 8 minutes against 2 minutes on the left. Mucoid reflux on pressure over the sac. No sac swelling, no redness, no fever.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Congenital dacryocystocele with mucus reflux
- B Epiblepharon with lower lash contact
- C Congenital obstruction at the valve of Hasner
- D Primary congenital glaucoma with epiphora
- E Punctal agenesis of the lower canaliculus

ANSWER _____ CONFIDENCE L • M • H

2 Which finding most reliably excludes primary congenital glaucoma?

DIAGNOSIS

- A Corneal diameters of 10.5 mm, corneas clear
- B Absence of photophobia and of lid squeezing
- C Mucoid reflux on pressure over the sac
- D Dye disappearance delayed to 8 minutes
- E Symptoms present since the age of only 3 weeks

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate management now?

TREATMENT

- A Continuous topical antibiotic four times daily
- B Silicone intubation of the duct within 6 weeks
- C Refer for dacryocystorhinostomy before school age
- D Teach sac massage and hygiene, review at 12 months
- E Probing and irrigation under anaesthesia this month

ANSWER _____ CONFIDENCE L • M • H

4 Which three situations make earlier referral necessary?

TREATMENT • SELECT 3

- A Mucoid reflux on pressure over the lacrimal sac
- B Repeated episodes of acute dacryocystitis
- C A tense bluish swelling below the medial canthus
- D Symptoms unchanged beyond 12 to 15 months of age
- E Crusted lashes needing cleaning every morning
- F Watering that worsens in cold wind outdoors

ANSWER _____ CONFIDENCE L • M • H

5 At 13 months the symptoms are unchanged. What now?

TREATMENT

- A Topical antibiotic for a further 3 months
- B Refer for probing and irrigation under anaesthesia
- C Continue massage and review again at the age of 24 months
- D Balloon dacryoplasty as the first procedure
- E Dacryocystorhinostomy with silicone tubes

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-15 • EMERGENCIES AND TRAUMA • 5 ITEMS

The eye that will not look up after football

A 14-year-old took an elbow to the left orbit 5 hours ago at football, watched by his coach. Little bruising and no proptosis. Elevation OS stops 10° above the horizontal with pain on the attempt, vertical diplopia, and four episodes of vomiting. Pulse 48 and regular. Left cheek and upper gum numb. BVA 20/20 each eye, pupils equal, IOP 16 both, fundi normal. CT: small inferior wall defect with tissue drawn through it.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Inferior rectus contusion without entrapment
- B Retrobulbar haemorrhage limiting movement
- C Traumatic fourth nerve palsy on the left
- D Trapdoor floor fracture with entrapment
- E Large blowout fracture with enophthalmos

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate next step?

TREATMENT

- A Oral steroid and review in the fracture clinic next week
- B Wait 10 to 14 days for swelling to settle, then repair
- C Ice, nasal decongestant and review in 48 hours
- D Press-on prism for the diplopia, review in 2 weeks
- E Refer now for surgical release within 24 hours

ANSWER _____ CONFIDENCE L • M • H

3 Which action best addresses the pulse of 48 and the vomiting?

TREATMENT

- A ECG and a cardiology opinion before orbital repair
- B Reassure: bradycardia follows any orbital blow
- C Nil by mouth, urgent theatre, cardiac monitoring
- D Intravenous atropine, then review in 48 hours
- E Antiemetic and overnight observation on the ward

ANSWER _____ CONFIDENCE L • M • H

4 Which three duties apply before the child leaves the department?

LAW AND ETHICS • SELECT 3

- A Refer the same day by speaking to the operating team
- B Record the mechanism in the words given, and who attended
- C Note whether the injury fits the account, and escalate if not
- D Delay the referral until a parent signs a consent form
- E Notify safeguarding about every orbital injury in a minor
- F Send photographs of the face to the child's school

ANSWER _____ CONFIDENCE L • M • H

5 Which mechanism explains so little bruising with such tight restriction?

BASIC SCIENCE

- A Bone fragments transect the inferior rectus belly
- B Haemorrhage inside the muscle limits it mechanically
- C Orbital fat necrosis tethers the globe inferiorly
- D Infraorbital nerve injury weakens elevation
- E The elastic paediatric floor springs back on tissue

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-16 • CORNEA AND REFRACTIVE SURGERY • 5 ITEMS

Haze under the flap on the third day

A 34-year-old woman had bilateral LASIK 3 days ago and saw 20/20 with a clear interface on day 1. The right eye now has 24 hours of increasing pain, photophobia and blur, on prednisolone four times daily. BVA 20/60 OD. A 1.2 mm dense white infiltrate sits at the inferior flap edge and extends into the stroma beneath the flap, with an overlying epithelial defect, 2+ anterior chamber cells and dense sectoral injection.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Pressure-induced stromal keratopathy
- B Infection at the flap interface
- C Diffuse lamellar keratitis of grade 3
- D Central toxic keratopathy after LASIK
- E Epithelial ingrowth beneath the flap

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings separate this from diffuse lamellar keratitis?

DIAGNOSIS • SELECT 3

- A Topical steroid four times daily since surgery
- B A focal dense infiltrate with an epithelial defect
- C Pain and photophobia worsening on day 3
- D 2+ anterior chamber cells with sectoral injection
- E Haze visible at the lamellar interface
- F Best corrected acuity reduced to 20/60

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate next step?

TREATMENT

- A Oral doxycycline, lubricants, review in 48 hours
- B Lift the flap, scrape for culture, fortified drops
- C Increase prednisolone to hourly and review in 24 hours
- D Hourly moxifloxacin without lifting the flap
- E Amputate the flap and treat the bed with drops

ANSWER _____ CONFIDENCE L • M • H

4 Which empirical cover fits an onset on day 3, before cultures return?

TREATMENT

- A Topical natamycin with oral voriconazole
- B Hourly polyhexanide with propamidine drops
- C Moxifloxacin four times daily with lubricants
- D Fortified vancomycin with a fluoroquinolone
- E Topical amikacin with oral clarithromycin

ANSWER _____ CONFIDENCE L • M • H

5 Which property of the flap interface most affects treatment?

BASIC SCIENCE

- A Stromal lamellae there stay more hydrated
- B The flap is denervated, so surface healing is slower
- C Bowman layer blocks antibody entry to it
- D It is a space sealed from tears and limbal vessels
- E Keratocytes there do not express MHC class II

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-17 • EMERGENCIES AND TRAUMA • 5 ITEMS

Wet cement, and a cornea that has gone grey

A 41-year-old builder splashed wet cement into the right eye 30 minutes ago and rinsed with bottled water for about a minute on site. Severe pain, lids swollen, counting fingers OD. Conjunctiva blanched from 5 to 8 o'clock with perilimbal ischaemia over 4 clock hours, cornea hazy with the whole epithelium gone, iris detail faint. Grey particles in the upper fornix. Fornix pH 10.5.

1 What is the first action?

TREATMENT

- A Instil anaesthetic and antibiotic, then refer for irrigation
- B Neutralise with a dilute acid buffer, then irrigate
- C Irrigate with saline, sweep the fornices, recheck pH
- D Measure acuity and pressure, then start irrigation
- E Irrigate for 15 minutes by the clock, then examine

ANSWER _____ CONFIDENCE L • M • H

2 Which three belong in the first hour?

TREATMENT • SELECT 3

- A Topical anaesthetic so irrigation can be tolerated
- B Double eversion and sweeping of both fornices
- C Recheck of fornix pH 5 minutes after stopping
- D Applanation pressure before any irrigation
- E Pad the eye and transfer for irrigation there
- F Topical steroid before the surface is clean

ANSWER _____ CONFIDENCE L • M • H

3 Which finding most determines the visual prognosis?

DIAGNOSIS

- A Corneal haze obscuring iris detail
- B Perilimbal ischaemia over 4 clock hours
- C Loss of the whole corneal epithelium
- D Acuity of counting fingers on arrival
- E A fornix pH of 10.5 at presentation

ANSWER _____ CONFIDENCE L • M • H

4 Why does alkali keep damaging after the surface is rinsed?

BASIC SCIENCE

- A It is held in the epithelium and goes no deeper
- B It binds corneal collagen and stays on the surface
- C Hydroxyl ions saponify membranes and pass inwards
- D It coagulates surface protein into a barrier layer
- E Heat of hydration burns the deeper stroma

ANSWER _____ CONFIDENCE L • M • H

5 The pH is normal and the surface is clean. Which plan fits the next 2 weeks?

TREATMENT

- A Tears, steroid, cycloplegic, ascorbate and doxycycline
- B Topical steroid alone, tapered over the first 3 days only
- C Bandage lens with a topical antibiotic only
- D Antibiotic with pressure padding for a week
- E Amniotic membrane graft with no topical steroid

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-18 • CORNEA AND REFRACTIVE SURGERY • 5 ITEMS

A large epithelial defect that does not hurt

A 63-year-old man had herpes zoster ophthalmicus in the left V1 a year ago and uses preserved lubricants with a topical NSAID for discomfort. He reports 3 weeks of blur and no pain. BVA 20/80 OS. A 4 by 3 mm oval epithelial defect sits below centre with heaped grey edges, stromal thinning of about 30%, no infiltrate, no chamber cells. Corneal sensation absent in all quadrants. Closure complete, Bell present, Schirmer 12 mm.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Herpes simplex geographic ulcer
- B Exposure keratopathy from poor closure
- C Toxic keratopathy from preserved drops
- D Neurotrophic keratopathy after zoster
- E Microbial ulcer with a rolled edge

ANSWER _____ CONFIDENCE L • M • H

2 Which mechanism explains why the epithelium fails?

BASIC SCIENCE

- A Reduced aqueous tear volume from lacrimal scarring
- B Antibody deposition in the epithelial basement membrane
- C Limbal stem cell loss from the original vesicles
- D Persistent viral replication in the epithelium
- E Loss of sensory neuropeptides that drive repair

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate first change to treatment?

TREATMENT

- A Add a topical antibiotic four times daily
- B Start hourly preserved lubricant drops
- C Stop the NSAID and the preserved drops
- D Add a topical steroid to settle the heaped edge
- E Debride the edges and pad the eye for 48 hours

ANSWER _____ CONFIDENCE L • M • H

4 Which three measures are appropriate now?

TREATMENT • SELECT 3

- A Topical steroid to reduce the edge reaction
- B Repeated debridement of the heaped epithelium
- C Topical NSAID continued for the discomfort
- D Preservative-free lubricants through the day
- E A therapeutic contact lens with antibiotic cover
- F Autologous serum or recombinant nerve growth factor

ANSWER _____ CONFIDENCE L • M • H

5 Three weeks on, thinning is 70% with a clear central descemetocoele. What now?

TREATMENT

- A Amniotic membrane with temporary tarsorrhaphy, now
- B Continue lubricants and review in 4 weeks
- C Penetrating keratoplasty within the week
- D Topical steroid to reduce the melting
- E Bandage lens with hourly antibiotic alone

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-19 • LENS, CATARACT, IOL, PERIOPERATIVE • 5 ITEMS

Pain and a lost view on the third day after surgery

A 74-year-old with type 2 diabetes had uneventful phacoemulsification 3 days ago and read 20/25 on day 1. Pain and rapid loss began 12 hours ago. BVA hand movements. Lids swollen, conjunctiva intensely injected, wound Seidel negative, 2 mm hypopyon, fibrin across the intraocular lens, IOP 28. No fundus view. B-scan: dense vitreous opacities with the retina attached.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Acute bacterial endophthalmitis
- B Toxic anterior segment syndrome
- C Fibrinous uveitis after surgery
- D Retained lens matter with inflammation
- E Uveitis-glaucoma-hyphaema syndrome

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate next step?

TREATMENT

- A Vitreous tap and intravitreal vancomycin with ceftazidime
- B Hourly fortified topical antibiotics with review in 12 hours
- C Subconjunctival gentamicin and oral moxifloxacin
- D Intravitreal antibiotic after the cultures are reported
- E Increase the topical steroid and review tomorrow

ANSWER _____ CONFIDENCE L • M • H

3 Which three belong in the first hour?

TREATMENT • SELECT 3

- A Intravitreal amphotericin as routine cover
- B Systemic ciprofloxacin as the sole therapy
- C Removal of the intraocular lens at once
- D Aqueous and vitreous samples before injecting
- E Intravitreal vancomycin with ceftazidime
- F Vitrectomy today if vision is light perception

ANSWER _____ CONFIDENCE L • M • H

4 At 48 hours the eye is worse despite injection. What now?

TREATMENT

- A Add oral prednisolone and continue observation
- B Continue the current treatment and review in 48 hours
- C Change to intravitreal amphotericin for fungus
- D Proceed to enucleation to control the infection
- E Repeat intravitreal antibiotics and refer for vitrectomy

ANSWER _____ CONFIDENCE L • M • H

5 Which source and organism best fit this onset?

BASIC SCIENCE

- A Surface flora through the wound, mostly staphylococcal
- B Haematogenous streptococci from a distant focus
- C Propionibacterium sequestered in the capsular bag
- D Contaminated irrigating fluid causing sterile inflammation
- E Fungal spores carried in the viscoelastic device

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-20 • LENS, CATARACT, IOL, PERIOPERATIVE • 5 ITEMS

A cloudy cornea the morning after routine surgery

A 68-year-old woman is seen the morning after uneventful phacoemulsification. Painless blur, BVA 20/200. Limbus-to-limbus corneal oedema, 1.5 mm of fibrin in the chamber, a fixed mid-dilated pupil, IOP 32, lids barely swollen, no vitreous cells on examination or on ultrasound. Two other patients from the same operating list have similar findings this morning.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Retained lens fragment reaction
- B Corneal decompensation from surgical trauma
- C Toxic anterior segment syndrome
- D Acute bacterial endophthalmitis
- E Descemet membrane detachment

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings favour a toxic cause over infection?

DIAGNOSIS • SELECT 3

- A Fibrin layered in the anterior chamber
- B Intraocular pressure of 32 mmHg
- C Best acuity reduced to 20/200
- D Onset within 24 hours of surgery
- E Limbus-to-limbus corneal oedema
- F Two others from the same list affected

ANSWER _____ CONFIDENCE L • M • H

3 How should this eye be managed today?

TREATMENT

- A Urgent vitrectomy with intravitreal cover
- B Topical antibiotic hourly, review in 48 hours
- C Oral acetazolamide alone, review in a week
- D Intensive topical steroid with pressure control
- E Tap and intravitreal antibiotics within the hour

ANSWER _____ CONFIDENCE L • M • H

4 Which action addresses the cause for the rest of the list?

TREATMENT

- A Sequester the instruments and fluids, audit reprocessing
- B Change the antibiotic prophylaxis for future lists
- C Switch to a different intraocular lens model
- D Send the remaining viscoelastic for sterility testing only
- E Continue the list with extra topical antibiotic

ANSWER _____ CONFIDENCE L • M • H

5 Which exposure most commonly causes this syndrome?

BASIC SCIENCE

- A Preservative-free lidocaine given into the chamber
- B Povidone-iodine 5% applied to the lid skin
- C A hydrophobic acrylic lens in the capsular bag
- D Topical NSAID started on the first day
- E Residue left on reprocessed surgical instruments

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-21 • LENS, CATARACT, IOL, PERIOPERATIVE • 5 ITEMS

The eye that bleeds a little every few months

A 68-year-old man had phacoemulsification OD 14 months ago, complicated by a capsule tear, and a single-piece acrylic implant was placed in the sulcus. Four episodes since of ache and red haze OD. He takes aspirin 75 mg daily after a coronary stent. BVA 20/40 OD. A 1 mm layered hyphaema, 2+ cells, pigment on the endothelium, a superonasal iris transillumination defect, an open angle heavy with pigment. IOP 32 OD, 15 OS.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Uveitis-glaucoma-hyphaema syndrome from a sulcus implant
- B Herpetic anterior uveitis with trabeculitis and atrophy
- C Pigmentary glaucoma with spontaneous microhyphaema
- D Neovascular glaucoma with rubeosis and recurrent bleeding
- E Chronic pseudophakic endophthalmitis from *Cutibacterium acnes*

ANSWER _____ CONFIDENCE L • M • H

2 Which three steps belong in the initial management of this eye?

TREATMENT • SELECT 3

- A Laser trabeculoplasty to the pigmented angle
- B Intravitreal anti-VEGF for the recurrent bleeding
- C Ultrasound biomicroscopy to locate the haptics
- D Topical steroid with an aqueous suppressant
- E Referral for repositioning or exchange of the implant
- F A prostaglandin analogue as first-line pressure control

ANSWER _____ CONFIDENCE L • M • H

3 The pressure is 32 mmHg with a 1 mm hyphaema. Which pressure treatment fits today?

TREATMENT

- A Topical pilocarpine 2% with the steroid four times daily
- B Topical timolol with dorzolamide, review in 48 hours
- C Topical latanoprost at night, review in one week
- D Oral acetazolamide 250 mg four times daily for a week
- E Selective laser trabeculoplasty at this visit

ANSWER _____ CONFIDENCE L • M • H

4 Which mechanism best explains the raised pressure in this eye?

BASIC SCIENCE

- A A neovascular membrane contracting across the angle
- B Steroid response raising trabecular outflow resistance
- C Pigment and red cells loading the trabecular meshwork
- D Peripheral anterior synechiae closing the angle
- E Ciliary block with forward rotation of the ciliary body

ANSWER _____ CONFIDENCE L • M • H

5 He asks about his aspirin. Which plan is most appropriate?

TREATMENT

- A Ask cardiology for a shorter-acting substitute
- B Continue the aspirin and treat the chafing implant
- C Stop the aspirin until the hyphaema has cleared
- D Stop the aspirin and start tranexamic acid instead
- E Continue the aspirin and observe for six months

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-22 • LENS, CATARACT, IOL, PERIOPERATIVE • 5 ITEMS

A wobble in the vision nine years after surgery

A 79-year-old man had phacoemulsification OS nine years ago and read 20/20 afterwards. Six months of blur that changes with head position, with a monocular double image. BVA 20/70 OS, and 20/25 with +2.25 D added to his old prescription. Pseudoexfoliative material on both pupil margins. OS: the implant and a contracted capsule sit low and tilted, the optic edge crossing the pupil, moving on eye movement. IOP 22 OS, 19 OD.

1 What is the most likely cause of the change in vision?

DIAGNOSIS

- A Out-of-the-bag dislocation with a broken haptic
- B Capsular phimosis without loss of zonular support
- C Pseudophakic macular oedema causing fluctuation
- D Posterior capsule opacification behind a centred implant
- E Late in-the-bag dislocation of the implant and capsule

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings show that the zonule, not the capsule alone, has failed?

DIAGNOSIS • SELECT 3

- A Nine years since an uncomplicated operation
- B Phacodonesis of implant and capsule together
- C The complex sitting low and tilted in the pupil
- D A hyperopic shift of +2.25 D since surgery
- E Pseudoexfoliative material on both pupil margins
- F An intraocular pressure of 22 mmHg OS

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate next step?

TREATMENT

- A Nd:YAG laser to open the contracted anterior capsule
- B Observation three-monthly while acuity stays at 20/70
- C New spectacles with the +2.25 D change, review in six months
- D Referral for planned surgical refixation of the complex
- E Emergency same-day vitrectomy and implant removal

ANSWER _____ CONFIDENCE L • M • H

4 Which change would make this a same-day referral?

TREATMENT

- A The optic edge crossing the visual axis at rest
- B Free dislocation into the vitreous, pressure 42 mmHg
- C Acuity falling a further two lines over three months
- D Pressure rising to 26 mmHg on no treatment
- E The double image becoming constant in all positions

ANSWER _____ CONFIDENCE L • M • H

5 Which mechanism best explains the late loss of support here?

BASIC SCIENCE

- A Enzymatic digestion of the zonule by chronic uveitis
- B A capsule tear made during the original operation
- C Pseudoexfoliative fibrilopathy weakening the zonule
- D Residual lens epithelium contracting the capsule
- E Vitreous traction pulling on the implant haptics

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-23 • RETINA, CHOROID, VITREOUS • 5 ITEMS

A haze in the vision beside an old scar

A 28-year-old woman has 10 days of floaters and blur OD, without pain. Two years ago co-trimoxazole gave her a rash and fever. BVA 20/60 OD, 20/20 OS. OD: 2+ vitreous cells, a fluffy white retinal focus 1.5 disc diameters superotemporal to the fovea, its edge touching a pigmented atrophic scar, with haze in the vitreous above it. Fine keratic precipitates, 1+ cells, IOP 14. OS normal.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Acute retinal necrosis from varicella zoster virus
- B Toxoplasma retinochoroiditis at the edge of a scar
- C Cytomegalovirus retinitis in occult immunosuppression
- D Tuberculous focal choroiditis with vitreous spillover
- E Toxocara granuloma with dense localised vitritis

ANSWER _____ CONFIDENCE L • M • H

2 Which three belong in the initial plan?

TREATMENT • SELECT 3

- A Systemic antiparasitic treatment started this week
- B Oral steroid added 48 hours after the antiparasitic
- C Toxoplasma IgG and IgM to support the diagnosis
- D Oral steroid first, antiparasitic if it fails
- E Intravitreal foscarnet while serology is awaited
- F Topical steroid alone for the anterior reaction

ANSWER _____ CONFIDENCE L • M • H

3 Which regimen fits her drug history best?

TREATMENT

- A Oral aciclovir 800 mg five times daily
- B Oral prednisolone 40 mg daily on its own
- C Pyrimethamine, clindamycin and folinic acid
- D Pyrimethamine, sulfadiazine and folinic acid
- E Oral co-trimoxazole twice daily for six weeks

ANSWER _____ CONFIDENCE L • M • H

4 Why does the new focus sit at the edge of the old scar?

BASIC SCIENCE

- A Fresh infection is seeded there by the choroidal flow
- B Glial tissue obstructs flow in the border arterioles
- C Cysts at the scar margin rupture and release parasites
- D Immune complexes settle in previously damaged retina
- E New vessels grow from the scar and leak into retina

ANSWER _____ CONFIDENCE L • M • H

5 At four weeks, which finding most justifies continuing treatment?

TREATMENT

- A A fluffy lesion border with vitreous haze
- B A pigmented border with a clearing vitreous
- C Acuity 20/30 with a residual central floater
- D A high IgG titre on the repeat sample
- E A few fine keratic precipitates remaining

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-24 • CORNEA AND REFRACTIVE SURGERY • 5 ITEMS

A crescent of cornea melting at the edge

A 61-year-old woman with 12 years of seropositive rheumatoid arthritis, on methotrexate 20 mg weekly, has eight days of pain and watering OD. Her joints have been swollen for two months. ESR 68, CRP 42. BVA 20/50 OD. A crescentic peripheral stromal ulcer from 2 to 5 o'clock, 1.2 mm wide and 60% deep, with an epithelial defect, an infiltrated leading edge and limbal injection. No scleral oedema, no hypopyon.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Staphylococcal marginal hypersensitivity keratitis
- B Infectious peripheral keratitis from Staphylococcus aureus
- C Peripheral ulcerative keratitis from rheumatoid vasculitis
- D Mooren ulcer in an otherwise healthy cornea
- E Terrien marginal degeneration with peripheral thinning

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate systemic treatment?

TREATMENT

- A High-dose oral steroid with escalation of immunosuppression
- B Methotrexate increased to 25 mg weekly, review in a month
- C Hourly topical prednisolone with a bandage lens
- D Oral doxycycline with preservative-free lubricants
- E A full-dose oral NSAID with gastric protection

ANSWER _____ CONFIDENCE L • M • H

3 Which three local measures belong alongside systemic treatment?

TREATMENT • SELECT 3

- A Topical antibiotic cover over the epithelial defect
- B Hourly topical prednisolone to the ulcer bed
- C A topical NSAID four times daily for the pain
- D A bandage lens worn continuously for a month
- E Preservative-free lubricants through the day
- F Oral doxycycline as an anticollagenase

ANSWER _____ CONFIDENCE L • M • H

4 At day 10 a 0.8 mm perforation appears in the gutter. What now?

TREATMENT

- A Tissue adhesive or a patch graft, immunosuppression continued
- B A large penetrating keratoplasty as the first procedure
- C A bandage lens with hourly topical steroid alone
- D Aqueous suppression with a pressure patch for a week
- E Surgery deferred until the joint disease is quiet

ANSWER _____ CONFIDENCE L • M • H

5 Which mechanism explains the benefit of doxycycline here?

BASIC SCIENCE

- A It inhibits matrix metalloproteinases in the stroma
- B It suppresses the B cells making rheumatoid factor
- C It blocks prostaglandin synthesis in the epithelium
- D It clears commensal bacteria from the lid margin
- E It thickens the lipid layer of the tear film

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-25 • RETINA, CHOROID, VITREOUS • 5 ITEMS

The upper field that went and did not come back

A 44-year-old man has had type 1 diabetes for 27 years, HbA1c 96 mmol/mol, blood pressure 168/98, and has not attended for laser in three years. He drives a taxi. BVA 20/200 OD, 20/40 OS. OD: fibrovascular fronds at the disc and along the superior arcade, a taut hyaline membrane, a shallow concave detachment tented over the temporal macula with striae, no break seen, inferior vitreous blood.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Vitreous haemorrhage over a taut hyaline membrane
- B Tractional retinal detachment involving the macula
- C Rhegmatogenous detachment from a superior break
- D Combined tractional and rhegmatogenous detachment
- E Diabetic macular oedema with severe thickening

ANSWER _____ CONFIDENCE L • M • H

2 Which single feature separates this from a rhegmatogenous detachment?

DIAGNOSIS

- A Acuity of 20/200 in that eye at presentation
- B Retinal striae running across the macula
- C No afferent pupillary defect in that eye
- D A concave surface tented to fibrovascular fronds
- E Vitreous haemorrhage filling the inferior vitreous gel

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate management?

TREATMENT

- A Completion of panretinal photocoagulation, review in three months
- B Intravitreal anti-VEGF alone, review in four weeks
- C Glucose and blood pressure optimised before referral
- D Observation while the inferior vitreous blood clears
- E Vitrectomy with delamination of membranes and endolaser

ANSWER _____ CONFIDENCE L • M • H

4 What is the main risk of anti-VEGF injection alone in this eye?

TREATMENT

- A A sustained rise in intraocular pressure
- B Worsening of his systemic hypertension
- C Fibrovascular contraction that worsens the traction
- D Rapid progression of the inferior vitreous haemorrhage
- E Endophthalmitis from the injection itself

ANSWER _____ CONFIDENCE L • M • H

5 He drives a taxi. Which two duties follow from today's findings?

LAW AND ETHICS • SELECT 2

- A Asking his employer to confirm his fitness
- B Telling him the duty to notify the authority is his
- C Recording the acuities and the advice given
- D Notifying the licensing authority yourself this week
- E Telling him he may drive until his operation

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-26 • RETINA, CHOROID, VITREOUS • 5 ITEMS

An aching eye that lost its field in five days

A 57-year-old man on no medication has five days of floaters, ache and falling vision OS. He had shingles of the trunk two years ago. BVA 20/200 OS, 20/20 OD. OS: 3+ vitreous cells, keratic precipitates, IOP 30. Confluent creamy peripheral necrosis nasally and temporally, with scalloped borders spreading circumferentially, sheathed occluded arterioles, an oedematous disc, macula spared. OD normal.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Toxoplasma retinochoroiditis with dense vitritis
- B Behcet occlusive retinal vasculitis with infiltrate
- C Progressive outer retinal necrosis without vitritis
- D Cytomegalovirus retinitis with a mild vitreous reaction
- E Acute retinal necrosis from varicella zoster virus

ANSWER _____ CONFIDENCE L • M • H

2 Which three steps belong in the first 24 hours?

TREATMENT • SELECT 3

- A Oral steroid started before the antiviral
- B Barrier laser at the necrotic border today
- C Intravitreal ganciclovir instead of systemic cover
- D Intravenous aciclovir at induction dose
- E Aqueous or vitreous sample for viral PCR
- F Examination of the fellow eye with a warning

ANSWER _____ CONFIDENCE L • M • H

3 How should the pressure of 30 mmHg and the anterior cells be treated?

TREATMENT

- A Oral steroid 48 hours before the antiviral starts
- B A prostaglandin analogue as first-line control
- C Observation until the antiviral takes effect
- D Oral acetazolamide 250 mg four times daily alone
- E Topical steroid with an aqueous suppressant

ANSWER _____ CONFIDENCE L • M • H

4 Which mechanism explains the risk to the fellow eye?

BASIC SCIENCE

- A Viraemia seeding both choroids from the blood
- B Reactivation in both trigeminal ganglia together
- C Antibody against a shared retinal antigen
- D Spread of virus along the visual pathway
- E Immune complexes settling in both retinas at once

ANSWER _____ CONFIDENCE L • M • H

5 Which change at three weeks most needs surgical review?

TREATMENT

- A A necrotic border that consolidates and pigments
- B Vitritis persisting with acuity steady at 20/80
- C Pressure settling to 16 on one topical agent
- D PCR confirming varicella zoster virus in the tap
- E A new field defect with a peripheral break

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-27 • RETINA, CHOROID, VITREOUS • 5 ITEMS

The dark lesion that grew between two photographs

A 63-year-old woman with blue irides reports three months of a shadow in her nasal field OD. BVA 20/25 OD. A dome-shaped pigmented choroidal lesion temporal to the right disc, with orange pigment on its surface and subretinal fluid at its edge, no drusen and no halo. Ultrasound: 9.2 mm base, 3.6 mm thickness, low internal reflectivity, choroidal excavation. A photograph two years ago recorded a 7.1 mm base.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Circumscribed choroidal haemangioma
- B Hypertrophy of the retinal pigment epithelium
- C Choroidal naevus with surface drusen and a halo
- D Choroidal melanoma with subretinal fluid
- E Choroidal metastasis from an occult primary

ANSWER _____ CONFIDENCE L • M • H

2 Which three features carry the greatest risk of malignancy?

DIAGNOSIS • SELECT 3

- A A position temporal to the disc
- B Acuity of 20/25 in that eye
- C Thickness of 3.6 mm on ultrasound
- D Orange pigment on the surface
- E Subretinal fluid at the edge
- F Blue irides and fair skin

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate next step?

TREATMENT

- A Photographs repeated in six months for comparison
- B Laser to the area of subretinal fluid
- C Enucleation arranged at the local unit
- D Systemic staging first, then ophthalmic review
- E Referral to an ocular oncology service this week

ANSWER _____ CONFIDENCE L • M • H

4 After oncology assessment, which plan best fits a tumour of this size?

TREATMENT

- A Enucleation, because subretinal fluid is present
- B Transpupillary thermotherapy as sole treatment
- C Observation with three-monthly photographs
- D Systemic chemotherapy before local treatment
- E Plaque brachytherapy, with liver surveillance afterwards

ANSWER _____ CONFIDENCE L • M • H

5 Which route of spread explains where this tumour metastasises?

BASIC SCIENCE

- A Direct extension along the optic nerve sheath
- B Seeding of the vitreous and then the meshwork
- C Vortex veins draining to the regional nodes
- D Haematogenous, as the uvea has no lymphatics
- E Lymphatic, through the conjunctival channels

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-28 • OPTIC NERVE AND NEURO-OPHTHALMIC • 5 ITEMS

Two years of headache and a field she never noticed

A 38-year-old woman has 14 months of amenorrhoea with milky nipple discharge, and two years of dull headache. BVA 20/25 OD, 20/30 OS. Pupils equal and briskly reactive, with no afferent defect. Fields: OD has lost the upper and lower temporal quadrants, stopping at the vertical midline, and OS has lost the same temporal quadrants; the nasal half of each field is full. Mild temporal pallor of both discs.

1 Where must the lesion be?

DIAGNOSIS

- A The left optic nerve, close to the chiasm
- B The right optic tract, behind the chiasm
- C Both optic nerves, in the orbit
- D The left occipital cortex, below the calcarine
- E The optic chiasm, at the crossing fibres

ANSWER _____ CONFIDENCE L • M • H

2 Which finding would place the lesion in front of the chiasm instead?

DIAGNOSIS

- A Reduced red saturation in the worse eye only
- B Temporal pallor of both optic discs
- C Best acuity of 20/30 in the worse eye
- D Enlargement of both blind spots
- E An afferent pupillary defect on one side

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate next step?

TREATMENT

- A MRI of the pituitary and chiasm, with endocrine bloods
- B Urgent unenhanced CT of the head today
- C Prolactin measured alone, then a decision
- D Fields repeated in three months to confirm the pattern
- E Routine neurology referral in six weeks

ANSWER _____ CONFIDENCE L • M • H

4 Prolactin is 4200 mU/L on a macroadenoma. Which treatment comes first?

TREATMENT

- A Transsphenoidal decompression within the week
- B Fractionated radiotherapy to the sella
- C Observation with six-monthly field tests
- D Hydrocortisone cover, then delayed surgery
- E A dopamine agonist, with the fields monitored

ANSWER _____ CONFIDENCE L • M • H

5 Which two statements about the crossing fibres are correct?

BASIC SCIENCE • SELECT 2

- A Temporal retinal fibres cross to the other side
- B Macular fibres never reach the chiasm
- C Crossing fibres carry the nasal half of the field
- D Nasal retinal fibres cross in the chiasm
- E Inferonasal fibres cross low and anteriorly

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-29 • ACCOMMODATION, VERGENCE, OCULOMOTOR • 5 ITEMS

The eye that will not follow its partner across

A 31-year-old woman has six days of horizontal double vision, worse looking to the right, and two weeks of tingling in the right hand. Three years ago she lost vision OS over days with pain on eye movement, and recovered. BVA 20/20 both, mild temporal pallor OS. On right gaze the left eye will not adduct past the midline and the right eye shows coarse abducting nystagmus. Convergence is full. Pupils equal, lids level.

1 Where must the lesion be?

DIAGNOSIS

- A The left third nerve in its subarachnoid course
- B The left medial rectus or its insertion
- C The right abducens nucleus in the pons
- D The left medial longitudinal fasciculus in the pons
- E The right medial longitudinal fasciculus in the pons

ANSWER _____ CONFIDENCE L • M • H

2 Which finding shows that the medial rectus itself is working?

DIAGNOSIS

- A Full elevation and depression in both eyes
- B Equal pupils with symmetrical lids
- C Coarse abducting nystagmus on the right
- D Corrected acuity of 20/20 in each eye
- E Full convergence to a near target

ANSWER _____ CONFIDENCE L • M • H

3 Which three features point to demyelination rather than a stroke?

DIAGNOSIS • SELECT 3

- A Painful loss of vision OS three years ago
- B Temporal pallor of the left optic disc
- C Diplopia that is worse on right gaze
- D Coarse nystagmus in the abducting eye
- E Equal pupils and symmetrical lids
- F Age 31 with no vascular risk factors

ANSWER _____ CONFIDENCE L • M • H

4 What is the most appropriate next step?

TREATMENT

- A MRI of brain and cord, with neurology referral
- B CT of the head today with a vascular opinion
- C Prism correction and review in six weeks
- D Intravenous steroid before any imaging
- E Occlusion of one eye at home until the diplopia settles

ANSWER _____ CONFIDENCE L • M • H

5 Which pathway explains the failure of conjugate gaze here?

BASIC SCIENCE

- A Vestibular input through the medial vestibular nucleus
- B The rostral interstitial nucleus for vertical saccades
- C Direct supply of the medial rectus by the sixth nerve
- D Internuclear fibres from the abducens nucleus crossing over
- E Frontal eye field fibres descending to the pons

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-30 • OPTIC NERVE AND NEURO-OPHTHALMIC • 5 ITEMS

The same half of the world gone in both eyes

A 71-year-old man with atrial fibrillation, on no anticoagulant, noticed three days ago while gardening that he bumps into things on his left and misses the left of the page. Blood pressure 176/98. BVA 20/25 both. Fields: each eye has lost the whole left half, stopping at the vertical midline, with a spared island of about 4° around fixation. Pupils brisk, no afferent defect, discs normal.

1 Where must the lesion be?

DIAGNOSIS

- A The left occipital cortex, above the calcarine fissure
- B The right temporal lobe, in Meyer loop
- C The right parietal lobe, above the ventricle
- D The right occipital cortex, both calcarine banks
- E The right optic tract, just behind the chiasm

ANSWER _____ CONFIDENCE L • M • H

2 Which feature best separates a cortical lesion from an optic tract lesion?

DIAGNOSIS

- A Sudden onset in atrial fibrillation
- B Normal optic discs three days after onset
- C Congruous halves with a spared central island
- D Loss in each eye that respects the vertical midline
- E Normal pupil responses in both eyes

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate management today?

TREATMENT

- A Aspirin started, with clinic review in a month
- B Field retraining and prisms arranged first
- C Routine neurology referral in six weeks
- D Same-day referral to a stroke service for imaging
- E CT within a week, then a stroke clinic review

ANSWER _____ CONFIDENCE L • M • H

4 Which three belong in his care over the next month?

TREATMENT • SELECT 3

- A Occlusion of one eye to settle confusion
- B Weekly field tests to track recovery
- C Steroid to reduce cortical swelling
- D Anticoagulation decided by the stroke team
- E Blood pressure brought to target
- F Low vision assessment with reading strategies

ANSWER _____ CONFIDENCE L • M • H

5 He asks about driving. What is the correct advice?

LAW AND ETHICS

- A He may drive once the field stops changing
- B He may drive while acuity stays at 20/25
- C He may drive by day with a passenger beside him
- D No restriction follows loss on one side
- E He must stop driving and notify the authority

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-31 • SYSTEMIC HEALTH • 5 ITEMS

Vision that dropped like a blind for four minutes

A 71-year-old man has had three episodes of painless loss of vision OD in 10 days, each a grey curtain falling from above, lasting about four minutes, with full recovery. He smokes 20 a day. BP 168/94. No jaw claudication, no scalp tenderness, ESR 12 mm/h. BVA 20/25 both eyes, pupils equal with no RAPD. A bright refractile plaque lies at an arteriolar bifurcation OD. A right carotid bruit is audible.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Ocular ischaemic syndrome from carotid occlusion
- B Transient obscurations from raised intracranial pressure
- C Giant cell arteritis with transient visual loss
- D Retinal transient ischaemic attack from carotid embolism
- E Retinal migraine with recurrent monocular aura

ANSWER _____ CONFIDENCE L • M • H

2 Which three steps belong in the first 24 hours?

TREATMENT • SELECT 3

- A Same-day referral to a transient ischaemic attack clinic
- B Carotid duplex booked as an outpatient in four weeks
- C Oral prednisolone 60 mg daily while the ESR is repeated
- D A second antihypertensive today, with review in a month
- E Same-day carotid duplex ultrasound of both sides
- F Aspirin 300 mg today, then a maintenance antiplatelet

ANSWER _____ CONFIDENCE L • M • H

3 Duplex shows 78% stenosis of the right internal carotid. What is the most appropriate management?

TREATMENT

- A Endarterectomy within two weeks, with antiplatelet and a statin
- B Dual antiplatelet therapy for 90 days, then reassessment
- C Carotid stenting deferred until a further episode occurs
- D Warfarin to an INR of 2 to 3, with no surgical referral
- E Antiplatelet and statin alone, with duplex repeated in six months

ANSWER _____ CONFIDENCE L • M • H

4 Which source best explains the refractile plaque seen OD?

BASIC SCIENCE

- A Talc embolus from intravenous drug use
- B Lipid deposit from untreated hypertriglyceridaemia
- C Cholesterol embolus from an ulcerated carotid plaque
- D Calcific embolus from a stenosed aortic valve
- E Platelet-fibrin embolus from atrial fibrillation

ANSWER _____ CONFIDENCE L • M • H

5 He asks whether he may keep driving to work. What is the correct advice and record?

LAW AND ETHICS

- A Advise him to stop driving only once the surgery has been booked
- B Tell him the practice will notify the licensing authority on his behalf
- C Leave driving to the vascular clinic and make no entry now
- D No restriction, since acuity is 20/25 and the field is full
- E Advise him to stop driving now, notify the authority, and record it

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-32 • GLAUCOMA • 5 ITEMS

The eye that took a squash ball twenty years ago

A 44-year-old man was struck in the right eye by a squash ball at 24 and told nothing was broken. He has no symptoms now. IOP 30 OD, 15 OS, CCT 545 μm both eyes. Gonioscopy OD: a wide ciliary band over four clock hours, torn iris processes, focal meshwork pigment, angle open elsewhere. OS unremarkable. Sphincter tears and a focal lens opacity OD. C/D 0.8 with an inferior notch OD, 0.4 OS. Superior arcuate defect OD.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Ghost cell glaucoma from an old vitreous bleed
- B Primary open angle glaucoma, markedly asymmetrical
- C Pigmentary glaucoma from iris chafing on the lens
- D Angle recession glaucoma in the injured eye
- E Pseudoexfoliative glaucoma of unilateral onset

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings date the disease to the old injury rather than to primary glaucoma?

DIAGNOSIS • SELECT 3

- A Intraocular pressure of 30 mmHg in that eye
- B Cup-disc ratio 0.8 with an inferior notch OD
- C Central corneal thickness of 545 μm
- D A band of ciliary body visible over four clock hours
- E Torn iris processes at the recessed angle
- F Sphincter tears with a focal lens opacity OD

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate first-line treatment for the pressure OD?

TREATMENT

- A Selective laser trabeculoplasty to the recessed angle OD
- B Topical pilocarpine 2% four times daily to that eye
- C Trabeculectomy with mitomycin C in the right eye now
- D Oral acetazolamide 250 mg four times daily, long term
- E Topical prostaglandin analogue OD, review at four weeks

ANSWER _____ CONFIDENCE L • M • H

4 Which finding at review would most justify escalating treatment?

TREATMENT

- A IOP 24 mmHg with nerve fibre layer thinning progressing OD
- B IOP 24 mmHg with the disc and field unchanged over a year
- C IOP 19 mmHg with conjunctival hyperaemia and lash growth
- D IOP 30 mmHg recorded once on a day he missed the drop
- E A pressure difference of 9 mmHg between the two eyes

ANSWER _____ CONFIDENCE L • M • H

5 He drives to work. What is the correct position on his licence?

LAW AND ETHICS

- A Driving is his own affair, so nothing is advised or recorded
- B Assessment rests on the binocular field, so he drives on for now
- C Any glaucoma diagnosis must be notified to the authority at once
- D A monocular arcuate defect disqualifies him until pressure is controlled
- E He may drive while his treated pressure stays below 21 mmHg

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-33 • GLAUCOMA • 5 ITEMS

Both eyes blurred a week after a new tablet

A 34-year-old woman began topiramate for migraine 11 days ago, raised to 50 mg twice daily four days ago. Both eyes blurred over 36 hours, with brow ache. Refraction has moved from -1.00 to -4.50 D in each eye. BVA 20/40 each. IOP 42 OD, 40 OS. Corneas hazy, chambers shallow centrally and peripherally, iris plane flat with no bombe, angles closed on gonioscopy in both eyes. Lenses clear. B-scan shows annular choroidal elevation.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Topiramate effusion with secondary angle closure
- B Plateau iris syndrome presenting acutely in both eyes
- C Phacomorphic closure from an intumescent lens
- D Posner-Schlossman syndrome with bilateral spikes
- E Acute primary angle closure in both eyes at once

ANSWER _____ CONFIDENCE L • M • H

2 Which three steps belong in the first 24 hours?

TREATMENT • SELECT 3

- A Topical pilocarpine 2% hourly to both eyes
- B Argon laser peripheral iridoplasty to both eyes
- C Stop the topiramate today and tell the prescriber
- D Topical aqueous suppressants with atropine 1%
- E Topical prednisolone to settle the ciliary effusion
- F Bilateral laser peripheral iridotomy at this session

ANSWER _____ CONFIDENCE L • M • H

3 The pressure is 34 mmHg at six hours. What is the next step?

TREATMENT

- A Add pilocarpine 2% hourly to both eyes and review
- B Arrange urgent lens extraction for the eye with the higher pressure
- C Paracentesis of both anterior chambers at the slit lamp
- D Add oral acetazolamide and a hyperosmotic, keeping the cycloplegic
- E Perform laser peripheral iridotomy in both eyes now

ANSWER _____ CONFIDENCE L • M • H

4 Which mechanism explains the myopic shift?

BASIC SCIENCE

- A Ciliary muscle spasm driving an accommodative response
- B Axial elongation under a sustained high pressure
- C Anterior rotation of the ciliary body and lens
- D Osmotic swelling of the lens raising its refractive index
- E Corneal oedema steepening the anterior corneal curvature

ANSWER _____ CONFIDENCE L • M • H

5 Which single finding most reliably separates this closure from primary angle closure?

DIAGNOSIS

- A Brow ache with blurring over the last 36 hours
- B A central chamber as shallow as the periphery
- C Intraocular pressure above 40 mmHg in each eye
- D Corneal oedema with acuity reduced to 20/40
- E A closed angle on gonioscopy in both eyes

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-34 • GLAUCOMA • 5 ITEMS

Pressure that climbed again after a laser hole

A 42-year-old woman had bilateral laser iridotomies 14 months ago for appositional closure. Refraction +0.75 both eyes, BVA 20/20 each. IOP 15 OD, 30 OS. Both iridotomies are patent on retroillumination, the iris plane flat, central chamber depth normal. Indentation gonioscopy OS shows a double hump, angle closed to the meshwork over three quadrants. OD closes on dark provocation. Ciliary processes sit anteriorly on ultrasound.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Incomplete iridotomy with residual pupil block
- B Plateau iris syndrome persisting after iridotomy
- C Malignant glaucoma with aqueous misdirection
- D Chronic angle closure with peripheral synechiae
- E Pigment dispersion from chafing at the laser edge

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings show that pupil block is not the mechanism?

DIAGNOSIS • SELECT 3

- A Refraction of +0.75 in both eyes
- B Both iridotomies are patent on retroillumination
- C Normal central chamber depth with a flat iris
- D A double hump at the angle on indentation
- E Intraocular pressure of 30 mmHg in the left eye
- F The angle closes on dark provocation OD

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate treatment for the closure OS?

TREATMENT

- A Argon laser iridoplasty OS with a pressure-lowering drop
- B Repeat the iridotomy OS with a larger opening
- C Trabeculectomy with mitomycin C in the left eye
- D Clear lens extraction in the left eye this month
- E Observation with gonioscopy and imaging repeated in six months

ANSWER _____ CONFIDENCE L • M • H

4 How should the right eye be managed?

TREATMENT

- A Long-term topical pilocarpine 1% to the right eye
- B Repeat the iridotomy OD, since the angle still closes
- C Discharge to routine care, with no further gonioscopy
- D Regular gonioscopy, treating on a pressure rise or synechiae
- E Prophylactic iridoplasty OD at the same session as the left eye

ANSWER _____ CONFIDENCE L • M • H

5 She is offered a systemic anticholinergic for bladder symptoms. What is correct?

TREATMENT

- A Reassure her, since only eye drops dilate enough to matter
- B Tell the prescriber that dilation can close her angles
- C No precaution is needed now that the iridotomies are patent
- D Refuse the tablet until a trabeculectomy has been done
- E Add pilocarpine while she takes the tablet, then stop it

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

CASE S2-35 • GLAUCOMA • 5 ITEMS

The screening result he does not want to act on

A community programme screened 1000 adults over 60 for glaucoma, in a population where the prevalence is 4%. The instrument has a sensitivity of 90% and a specificity of 85%. A 63-year-old man screens positive: IOP 25 OD, 24 OS, cup-disc 0.7 both eyes, a brother with glaucoma. He has capacity, declines referral and further testing today, and asks for a copy of his record to take elsewhere.

1 What is the positive predictive value of a positive screen in this population?

BASIC SCIENCE

- A 4%, since that is the prevalence in the group
- B 20%, so four in five positive screens are false
- C 90%, the share of true cases the test detects
- D 85%, the share of healthy people it clears
- E About 99%, since very few cases are missed

ANSWER _____ CONFIDENCE L · M · H

2 The same test is used in a clinic where prevalence is 20%. Which two statements are correct?

BASIC SCIENCE • SELECT 2

- A The positive predictive value rises to about 60%
- B Sensitivity and specificity are unchanged
- C The negative predictive value rises to about 99%
- D False positives per 1000 screened rise to 200
- E Specificity rises because fewer positives are false
- F The positive predictive value stays at about 20%

ANSWER _____ CONFIDENCE L · M · H

3 He declines referral and further testing. What must be done today?

LAW AND ETHICS

- A Refer him anyway, since a positive screen obliges a referral
- B Involve his brother, who already understands the family risk
- C Record what was explained, the risks accepted, and his decision
- D Record that he refused and close the episode as declined
- E Ask him to sign a waiver releasing the practice from liability

ANSWER _____ CONFIDENCE L · M · H

4 He asks for a copy of his record to take to another practice. What is correct?

LAW AND ETHICS

- A Withhold the screening images as practice property
- B Charge for the copy and release it once it is paid
- C Refuse until he accepts the referral he has declined
- D Give him a copy within the statutory period
- E Send it to the other practice only, and not to him

ANSWER _____ CONFIDENCE L · M · H

5 What is the most appropriate clinical plan at this visit?

TREATMENT

- A Discharge him with advice to return if his vision changes
- B Write to his physician asking for the referral to be made
- C Offer fields and disc imaging here, with a written plan
- D Start a prostaglandin analogue today on the screening result
- E Repeat the screening test in twelve months and do nothing now

ANSWER _____ CONFIDENCE L · M · H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

PART 9

Teaching keys

Thirty-five keys, one per case, in the order you sat them. They start here and not on the back of each case, so the session can be answered cold.

READ THEM IN THIS ORDER

Wrong answers first. Then correct answers you were unsure of. Then the ones you were sure of and still missed, which are the expensive ones and the last thing anybody wants to look at.

THE DISTRACTOR TABLE IS THE POINT

Knowing why the answer is right is worth one item. Knowing why the trap is tempting is worth every item built the same way, and the exam builds a lot of them the same way.

WHAT EACH KEY CONTAINS

The letter, the answer in a few words, why the findings force it, and a table naming what makes each wrong option tempting and the specific finding that rules it out.

THEN WRITE THE RULE

Each case closes with one or two if-then sentences. Say them aloud. If you cannot, you have read the key rather than learned it.

Case S2-01 • why

1 • B Stub short answer The stem carries the host factor, the interval.

Stub explanation for pseudomyopia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

2 • C Stub short answer The stem carries the host factor, the interval.

Stub explanation for pseudomyopia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

3 • C Stub short answer The stem carries the host factor, the interval.

Stub explanation for pseudomyopia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

4 • A, D, E Stub short answer The stem carries the host factor, the interval.

Stub explanation for pseudomyopia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

5 • C Stub short answer The stem carries the host factor, the interval.

Stub explanation for pseudomyopia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

IF-THEN RULES FROM THIS CASE

Stub if-then rule The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and.

Case S2-02 • why

1 • B Stub short answer The stem carries the host factor, the interval.

Stub explanation for anisometropia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

2 • B Stub short answer The stem carries the host factor, the interval.

Stub explanation for anisometropia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

3 • A Stub short answer The stem carries the host factor, the interval.

Stub explanation for anisometropia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

4 • C, D, F Stub short answer The stem carries the host factor, the interval.

Stub explanation for anisometropia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

5 • D Stub short answer The stem carries the host factor, the interval.

Stub explanation for anisometropia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

IF-THEN RULES FROM THIS CASE

Stub if-then rule The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and.

Case S2-03 • why

1 • D Stub short answer The stem carries the host factor, the interval.

Stub explanation for vertical imbalance The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor,.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

2 • D Stub short answer The stem carries the host factor, the interval.

Stub explanation for vertical imbalance The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor,.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

3 • C Stub short answer The stem carries the host factor, the interval.

Stub explanation for vertical imbalance The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor,.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

4 • B, C, E Stub short answer The stem carries the host factor, the interval.

Stub explanation for vertical imbalance The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor,.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

5 • B Stub short answer The stem carries the host factor, the interval.

Stub explanation for vertical imbalance The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor,.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

IF-THEN RULES FROM THIS CASE

Stub if-then rule The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and.

Case S2-04 • why

1 • C Stub short answer The stem carries the host factor, the interval.

Stub explanation for toric rotation The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

2 • B Stub short answer The stem carries the host factor, the interval.

Stub explanation for toric rotation The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

3 • B Stub short answer The stem carries the host factor, the interval.

Stub explanation for toric rotation The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

4 • A, B, D Stub short answer The stem carries the host factor, the interval.

Stub explanation for toric rotation The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
F	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
F	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

5 • B Stub short answer The stem carries the host factor, the interval.

Stub explanation for toric rotation The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

IF-THEN RULES FROM THIS CASE

Stub if-then rule The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and.

Case S2-05 • why

1 • C Stub short answer The stem carries the host factor, the interval.

Stub explanation for flat GP fit The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

2 • B Stub short answer The stem carries the host factor, the interval.

Stub explanation for flat GP fit The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

3 • D Stub short answer The stem carries the host factor, the interval.

Stub explanation for flat GP fit The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

4 • A, C, F Stub short answer The stem carries the host factor, the interval.

Stub explanation for flat GP fit The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

5 • E Stub short answer The stem carries the host factor, the interval.

Stub explanation for flat GP fit The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

IF-THEN RULES FROM THIS CASE

Stub if-then rule The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and.

Case S2-06 • why

1 • B Stub short answer The stem carries the host factor, the interval.

Stub explanation for RP-spectrum low vision The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

2 • E Stub short answer The stem carries the host factor, the interval.

Stub explanation for RP-spectrum low vision The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

3 • C Stub short answer The stem carries the host factor, the interval.

Stub explanation for RP-spectrum low vision The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

4 • B, E, F Stub short answer The stem carries the host factor, the interval.

Stub explanation for RP-spectrum low vision The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

5 • A Stub short answer The stem carries the host factor, the interval.

Stub explanation for RP-spectrum low vision The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

IF-THEN RULES FROM THIS CASE

Stub if-then rule The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and.

Case S2-07 • why

1 • D Stub short answer The stem carries the host factor, the interval.

Stub explanation for accommodative insufficiency The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

2 • E Stub short answer The stem carries the host factor, the interval.

Stub explanation for accommodative insufficiency The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

3 • C Stub short answer The stem carries the host factor, the interval.

Stub explanation for accommodative insufficiency The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

4 • A, D, E Stub short answer The stem carries the host factor, the interval.

Stub explanation for accommodative insufficiency The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

5 • D Stub short answer The stem carries the host factor, the interval.

Stub explanation for accommodative insufficiency The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

IF-THEN RULES FROM THIS CASE

Stub if-then rule The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and.

Case S2-08 • why

1 • A Stub short answer The stem carries the host factor, the interval.

Stub explanation for high AC/A esophoria The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor,.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

2 • C Stub short answer The stem carries the host factor, the interval.

Stub explanation for high AC/A esophoria The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor,.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

3 • A Stub short answer The stem carries the host factor, the interval.

Stub explanation for high AC/A esophoria The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor,.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

4 • C, D, F Stub short answer The stem carries the host factor, the interval.

Stub explanation for high AC/A esophoria The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor,.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

5 • C Stub short answer The stem carries the host factor, the interval.

Stub explanation for high AC/A esophoria The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor,.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

IF-THEN RULES FROM THIS CASE

Stub if-then rule The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and.

Case S2-09 • why

1 • B Stub short answer The stem carries the host factor, the interval.

Stub explanation for divergence insufficiency The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

2 • D Stub short answer The stem carries the host factor, the interval.

Stub explanation for divergence insufficiency The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

3 • B Stub short answer The stem carries the host factor, the interval.

Stub explanation for divergence insufficiency The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
D	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

4 • B, C, E Stub short answer The stem carries the host factor, the interval.

Stub explanation for divergence insufficiency The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

5 • A Stub short answer The stem carries the host factor, the interval.

Stub explanation for divergence insufficiency The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

IF-THEN RULES FROM THIS CASE

Stub if-then rule The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and.

Case S2-10 • why

1 • A Stub short answer The stem carries the host factor, the interval.

Stub explanation for intermittent exotropia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

2 • A Stub short answer The stem carries the host factor, the interval.

Stub explanation for intermittent exotropia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
C	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

3 • E Stub short answer The stem carries the host factor, the interval.

Stub explanation for intermittent exotropia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

4 • A, B, D Stub short answer The stem carries the host factor, the interval.

Stub explanation for intermittent exotropia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
F	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
F	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

5 • E Stub short answer The stem carries the host factor, the interval.

Stub explanation for intermittent exotropia The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.
B	Stub label The stem carries.	Stub tempting The stem carries the host factor, the interval and the number that.	Stub ruled out The stem carries the host factor, the interval and the number that.

IF-THEN RULES FROM THIS CASE

Stub if-then rule The stem carries the host factor, the interval and the number that decides between two otherwise identical options, at length. The stem carries the host factor, the interval and.

Case S2-11 • why

1 • A Ethambutol toxic optic neuropathy

Symmetric painless loss over weeks, dense red-green failure and matched central scotomas, on 23 mg/kg with an eGFR of 42, is dose-related toxicity. Flat discs place it behind the globe.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Bilateral NAION	Age and vascular risk make ischaemia the reflex answer in an older man.	Ischaemia is sudden, usually sequential, and swells the disc it damages.
C	Chiasmal compression	Bilateral field loss with normal discs does point towards a chiasm.	Compression gives temporal loss and asymmetry, not matched central scotomas.

2 • A, B, C Colour loss, matched scotomas, no RAPD

Dyschromatopsia, symmetric central scotomas and an absent RAPD in eyes that both see 20/60 or worse localise to the optic nerves and say the two nerves are equally affected.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D, E	Drug and dose	They name the cause and are the reason the drug has to stop.	Exposure and dose explain why the nerves are failing, not where.
F	Acuity down in both eyes	It is the abnormality the patient actually came with.	Reduced acuity grades severity and localises nothing at all.

3 • A Stop ethambutol today, through the TB team

Toxicity is dose and duration dependent, and recovery depends on how early the drug stops. The TB service substitutes cover the same day so the regimen is not left short.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Dose reduction	15 mg/kg is the low-risk dose, so it reads as the safe compromise.	Risk is low at that dose in intact nerves. This nerve is already failing.
C	Stop all four drugs	It removes every possible culprit in one decision.	Only ethambutol needs to go. Losing cover invites resistant disease.

4 • B Warn of further decline, review at 6 weeks

Loss from ethambutol often progresses for 1 to 3 months after withdrawal, and recovery, when it comes, takes 6 months or longer. Early review records the floor and catches other causes.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Restart at a lower dose	The drug is useful and the dose was high, so a smaller dose feels safe.	Rechallenge after toxicity risks permanent loss. Substitute instead.
C	Normal within 2 weeks	Removing a cause usually stops a disease, so quick recovery sounds right.	Decline for weeks after stopping is typical, and recovery is slow.

5 • D Mitochondrial failure in papillomacular axons

Ethambutol chelates copper and zinc and impairs complex IV. The small, heavily loaded axons of the papillomacular bundle fail first, which produces central scotomas and red-green loss.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Ciliary artery infarction	It is the mechanism behind the ischaemic neuropathy on this list.	Infarction is abrupt, usually unilateral, and swells the disc.
C	Magnocellular loss	Naming a ganglion cell population sounds precise and mechanistic.	Toxicity hits small parvocellular axons, which is why colour goes first.

IF-THEN RULES FROM THIS CASE

If colour and central vision fall in both eyes on ethambutol, stop the drug the same day. If the loss is asymmetric or a disc swells, look for a second cause.

Case S2-12 • why

1 • E Retinoblastoma, right eye

Leukocoria with an elevated chalky retinal mass and vitreous seeds in a quiet, clear-media eye at 14 months is tumour. Strabismus is the second commonest way it is noticed.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Coats disease	It also gives leukocoria in a young child with a normal front of the eye.	Coats shows yellow exudate and telangiectasia, not a chalky mass with seeds.
D	Stage 5 ROP	A total detachment behind the lens does look white through the pupil.	Born at term with no oxygen therapy, so the risk period never existed.

2 • C Same-day referral to a specialist centre

Delay costs the eye and, once tumour leaves the globe, the child. Diagnosis and staging happen at the specialist centre, and no needle enters an eye suspected of holding retinoblastoma.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Needle biopsy	Histology is how most other paediatric tumours are confirmed first.	Perforating the globe seeds tumour outside it and changes survival.
D	Scan, review in 2 weeks	Imaging first looks orderly, and 2 weeks feels prompt in clinic terms.	Growth over 2 weeks can turn an eye-sparing case into an orbital one.

3 • A, E, F EUA of both eyes, MRI, RB1 testing

Both eyes are examined under anaesthesia because disease is often bilateral. MRI stages the nerve and pineal region without radiation, and RB1 testing drives screening of siblings.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	CT of the head	Calcification is the classic sign and CT shows it best.	It irradiates a child who may carry a germline RB1 mutation.
C, D	Biopsy, marrow	Tissue and marrow are how paediatric tumours are usually staged.	Biopsy seeds the orbit, and marrow is for proven extraocular disease.

4 • A Refer today; if she is not brought, escalate

The duty is a same-day referral made by the clinician who holds the suspicion, with that suspicion named in writing. A child not brought to the appointment is a safeguarding matter.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Letter via the parents	It keeps the family informed and involves their usual doctor.	It hands the timing of a sight and life threatening referral to others.
E	Safeguarding first	Involving safeguarding early looks like the careful, thorough route.	Nothing delays the ophthalmic referral. Safeguarding follows non-attendance.

5 • B Germline RB1 is possible in unilateral disease

Laterality does not settle germline status, so testing is offered in unilateral disease too. A germline result changes screening of this child, of siblings and of any future children.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Unilateral rules it out	Bilateral disease is the taught marker of heritable retinoblastoma.	A germline mutation can still produce a single unilateral tumour.
D	Recessive inheritance	Two hits at one gene sounds like recessive transmission.	Transmission is dominant with high penetrance. The second hit is somatic.

IF-THEN RULES FROM THIS CASE

If a child has leukocoria or new strabismus with no red reflex, refer the same day to a retinoblastoma service. If the child is not brought, escalate rather than discharge.

Case S2-13 • why

1 • B Sebaceous carcinoma of the tarsus

Recurrence at one site, lash loss with destruction of the posterior margin, yellow tarsal thickening and a unilateral chronic conjunctivitis are destructive disease. The node raises the stakes.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Recurrent chalazion	Two surgeons treated it as one, and chalazia genuinely do recur.	A chalazion does not destroy lash follicles or the posterior lid margin.
A	Chronic blepharitis	Unilateral chronic conjunctivitis with lid disease reads as blepharitis.	Blepharitis is bilateral and builds no firm nodule inside the tarsus.

2 • E Madarosis with loss of margin architecture

Loss of lash follicles and of the posterior margin means the lesion is replacing tissue rather than distending it. Recurrence, colour and chronic redness each have benign explanations.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Recurrence after surgery	Three episodes in 14 months is what prompted the referral.	Incompletely drained chalazia recur too. Recurrence is not destruction.
B, D	Yellow, firm nodule	Yellow lipid and firmness feel like the signature of this tumour.	Old granulation tissue is firm and yellow as well.

3 • E Full-thickness biopsy with map biopsies

The tumour sits within the tarsus and spreads inside conjunctival epithelium, so a superficial sample misses it. Mapping defines pagetoid extent, which decides how much tissue must go.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Curettage with histology	It samples the lesion and repeats a familiar, quick procedure.	Curettings miss tarsal depth and pagetoid spread, and blur the margins.
C	Excision with 2 mm margins	Excising it looks decisive and treats the lesion in one visit.	Margins are planned after histology and mapping, not before them.

4 • C, D, E Pagetoid spread, skip lesions, lipid lost

Intraepithelial spread imitates chronic blepharoconjunctivitis, skip lesions make one negative sample falsely reassuring, and lipid dissolves in routine processing so the cells look bland.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	Eccrine origin	Naming a gland of the lid sounds like the pathological detail.	It arises in meibomian and Zeis glands, which are sebaceous.
A, B	Early spread, lower lid	Both sound like facts that would explain a missed tumour.	The upper lid is the commoner site, and nodes follow visible disease.

5 • E Wide excision with margins, node assessed

Clearance is judged on controlled margins, and pagetoid disease needs conjunctival treatment as well. A palpable node is imaged and sampled because this tumour spreads through lymphatics.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Cryotherapy, mitomycin	Topical treatment spares the lid and is used for conjunctival spread.	They are adjuncts to excision, not treatment for an invasive tarsal mass.
D	Excision, node later	The node is small and mobile, so watching it seems reasonable.	A 1 cm node with an invasive tumour is imaged and sampled now.

IF-THEN RULES FROM THIS CASE

If a chalazion recurs at one site, or takes lashes or the posterior lid margin, send full-thickness tissue. If a lid carcinoma is confirmed, examine and image the nodes.

Case S2-14 • why

1 • C Congenital nasolacrimal duct obstruction

Watering from the first weeks with morning crusting, a high tear lake, delayed dye disappearance and mucoid reflux from the sac places the block low in a patent, comfortable eye.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Congenital glaucoma	Epiphora since infancy is the presentation the two conditions share.	Corneal diameters of 10.5 mm, clear corneas, no photophobia or squeezing.
A	Dacryocystocele	Mucoid reflux comes from a sac that is holding its contents.	A dacryocystocele presents in the neonate as a tense bluish swelling.

2 • A Normal corneal diameter with a clear cornea

Raised pressure in infancy stretches and clouds the cornea, so normal diameters with clear corneas are hard evidence against it. The other findings show a blocked duct but exclude nothing.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	No photophobia	Photophobia and squeezing are the taught triad with epiphora.	Early congenital glaucoma can water without obvious light aversion.
C	Mucoid reflux	Reflux proves the duct is blocked, which feels like enough.	A blocked duct and a raised pressure can sit in the same eye.

3 • D Massage and hygiene, review at 12 months

Most obstructions at Hasner open during the first year, so conservative care with a review date is the default at 9 months. Antibiotic is for purulent episodes, not as a standing course.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Probing now	It cures the problem in one visit after 8 months of watering.	At 9 months most open without anaesthesia. Probing can wait.
A	Continuous antibiotic	Morning crusting looks infective and drops settle it briefly.	It treats discharge, not obstruction, and selects resistant organisms.

4 • B, C, D Dacryocystitis, tense sac, persistence

Infection of the sac, a distended sac in a neonate, which can also obstruct the nose, and persistence past the age at which spontaneous opening stops all move the child to surgical care.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E, F	Crusting, wind	They are the symptoms the parents most want stopped.	Both are typical of simple obstruction that still resolves on its own.
A	Mucoid reflux	Reflux from the sac sounds like a sac that needs an operation.	It is expected in ordinary obstruction and does not change the timing.

5 • B Probing and irrigation under anaesthesia

Spontaneous opening becomes unlikely after the first year, and probing at this age succeeds as a single procedure in most children. Intubation and bypass surgery are for failures.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Wait until 24 months	Conservative care was the right answer for the whole first year.	The resolution curve flattens after 12 months. Waiting buys nothing.
E	Primary bypass surgery	It is the definitive operation, so it looks like the thorough answer.	It is reserved for failed probing and intubation at this age.

IF-THEN RULES FROM THIS CASE

If an infant waters with clear corneas of normal diameter, treat as duct obstruction and review at 12 months. If the sac becomes infected or tensely distended, refer at once.

Case S2-15 • why

1 • D Trapdoor floor fracture, tissue entrapped

A white eye with severe painful limitation of elevation, infraorbital numbness, vomiting and bradycardia after blunt injury in a teenager is tissue gripped by a floor that has sprung shut.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Large blowout fracture	It is the adult pattern after this mechanism and also gives diplopia.	A large defect gives bruising and enophthalmos, not tight painful tethering.
A	Rectus contusion	Contusion explains pain and limitation without invoking surgery.	Contusion does not drive repeated vomiting with a pulse of 48.

2 • E Surgical release today, within 24 hours

Entrapped muscle is ischaemic, and in children fibrosis with permanent restriction follows within days. The interval that suits an adult blowout is the interval in which this eye loses movement.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Wait 10 to 14 days	It is the correct rule for an adult blowout fracture with diplopia.	Paediatric trapdoor entrapment is released within a day or two.
A	Steroid, clinic next week	Steroid does reduce orbital oedema, and the eye still sees 20/20.	The problem is mechanical trapping, and a week of it scars the muscle.

3 • C Nil by mouth and urgent theatre

Bradycardia with vomiting is the oculocardiac reflex from traction on trapped tissue, and it settles when the tissue is freed. It marks urgency and warns the anaesthetist.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Atropine, review later	Atropine does treat reflex bradycardia during orbital handling.	It masks the sign while the muscle stays trapped and ischaemic.
A	Cardiology first	A pulse of 48 in a teenager looks like a cardiac problem.	The cause is orbital. Waiting for cardiology delays the release.

4 • A, B, C Same-day referral, record account, check fit

The duties are a same-day spoken referral, a contemporaneous record of the account and of who gave it, and a judgement on whether the injury fits that account. Escalation follows a discrepancy.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Wait for consent	Surgery on a minor needs consent, so waiting looks correct.	Consent is arranged alongside the referral. It does not delay it.
E, F	Blanket notify, photographs	Both feel like caution about a child with a facial injury.	Notification follows concern, and school sharing has no lawful basis.

5 • E Elastic bone snaps back and grips tissue

A child's floor is greenstick and returns to position, so the defect stays small, blood is contained and muscle or its sheath is held. That is why the eye looks well and moves badly.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Muscle transected	Total loss of elevation suggests the muscle has been divided.	A divided muscle is slack, not tight. Forced duction would be free.
D	Nerve injury	Cheek numbness proves the infraorbital nerve is injured.	That nerve is sensory. It has no part in elevating the eye.

IF-THEN RULES FROM THIS CASE

If a child cannot elevate after blunt orbital injury, treat entrapment as a same-day operation, and more urgently still with nausea or bradycardia. The adult 2-week rule does not apply.

Case S2-16 • why

1 • B Infectious keratitis at the interface

A focal dense infiltrate with an epithelial defect, pain worsening on day 3, chamber cells and sectoral injection is infection. Sterile interface disease is diffuse, comfortable and quiet.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Diffuse lamellar keratitis	Interface haze after LASIK on steroid is the pattern taught first.	It is diffuse granular haze, peaks on day 1 or 2, epithelium intact.
D	Central toxic keratopathy	It also appears in the first week and drops acuity sharply.	It is central and non-inflammatory, with striae, no infiltrate, no cells.

2 • B, C, D Focal infiltrate, rising pain, chamber cells

Focality with a break in the epithelium, pain that worsens after day 2, and an anterior chamber reaction are infective. Interface haze, reduced acuity and steroid use belong to both.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E, F	Haze, acuity	They are the findings that make the eye look abnormal at the lamp.	Both occur in sterile lamellar keratitis, so neither separates them.
A	Steroid since surgery	Steroid is the treatment for DLK, so its use hints at that diagnosis.	Every LASIK eye is on steroid. It is routine, not a discriminator.

3 • B Lift the flap, culture, fortified antibiotics

The organism sits in a space that drops reach poorly, so the flap is lifted to sample, irrigate and treat directly. Steroid comes down while the infection is uncontrolled.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Hourly steroid	It is the correct treatment for the sterile disease this mimics.	Hourly steroid over an untreated infection is how interfaces melt.
D	Drops, flap left down	Fluoroquinolone cover looks adequate and avoids a procedure.	Penetration under the flap is poor and the culture is never taken.

4 • D Fortified vancomycin and a fluoroquinolone

Infection in the first week after LASIK is usually staphylococcal or streptococcal, so gram-positive cover with broad support starts at once. Atypical and fungal cover belongs to late onset.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Amikacin, clarithromycin	It is the classic regimen for the atypical mycobacteria of this surgery.	Those organisms present at 2 to 8 weeks, not on the third day.
C	Drops four times daily	Moxifloxacin covers the likely organisms and is already prescribed.	Four times daily does not treat an established stromal infiltrate.

5 • D A sealed space with poor drug access

The interface has no vessels and is shielded from the tear film, so drug levels there stay low and host defence arrives late. Access is why the flap is lifted rather than treated through.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	MHC expression	An immune privilege explanation sounds like the mechanistic answer.	Drug and cell access, not antigen presentation, is what fails here.
B	Denervation	LASIK does cut nerves, and that does slow epithelial healing.	Denervation does not explain why an interface infiltrate resists drops.

IF-THEN RULES FROM THIS CASE

If a post-LASIK interface is focally infiltrated, or the epithelium is broken, or the chamber reacts, lift and culture. If haze is diffuse with intact epithelium, treat as sterile with steroid.

Case S2-17 • why

1 • C Irrigate to a normal pH, fornices swept

Alkali burns for as long as it remains, and retained cement keeps releasing it, so irrigation runs to a pH endpoint with the particles removed. Measurement and referral wait their turn.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Acuity and IOP first	Recording vision before treatment is the habit of every clinic.	Every minute of contact deepens the burn. Numbers can be taken later.
E	Fifteen minutes, then stop	A fixed interval is easy to remember and feels thorough.	Time is not the endpoint. Cement holds the pH up far beyond 15 minutes.

2 • A, B, C Anaesthetic, fornix sweep, delayed pH recheck

Anaesthesia makes irrigation effective, sweeping removes the reservoir of cement, and a delayed recheck catches the pH climbing again as trapped particles keep dissolving.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D, E	Pressure, transfer	Both are standard trauma care and both feel responsible.	Neither removes alkali, and both spend the minutes that matter.
F	Early steroid	Inflammation is intense and steroid is used in these eyes.	It belongs after decontamination, not while alkali is still present.

3 • B Extent of perilimbal ischaemia

Limbal vessels mark the stem cell reserve, and their loss predicts failure to re-epithelialise, conjunctivalisation and scarring. Epithelial loss and haze at 30 minutes say little.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C, D	Epithelium, acuity	They are the most dramatic findings in front of you.	Both recover fully if the limbus survives, and both mislead early.
E	A pH of 10.5	It grades the insult and drives the irrigation endpoint.	It measures the surface now, not the reserve the eye heals from.

4 • C Hydroxyl ions saponify lipid and penetrate

Alkali dissolves cell membranes and travels through stroma into the chamber, so injury continues after the surface looks clean. Acid coagulates protein and is more often held at the surface.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B, D	Barrier formation	Coagulation limiting depth is true, but it is true of acid.	Alkali forms no barrier. It saponifies its way inwards.
E	Heat of hydration	Cement does generate heat, and the eye is intensely painful.	Depth here comes from hydroxyl ions, not from temperature.

5 • A Tears, steroid, cycloplegic, ascorbate, doxycycline

Steroid controls inflammation while the epithelium is watched, cycloplegia settles the iris, and ascorbate with doxycycline supports collagen and limits melt. Lubrication carries the surface.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Steroid, 3-day taper	Steroid is the mainstay, and a short course avoids stromal melt.	Three days is too short for this burn and misses the melting phase.
E	Membrane, no steroid	Amniotic membrane does help severe burns re-epithelialise.	It is an adjunct. Withholding steroid leaves the inflammation untreated.

IF-THEN RULES FROM THIS CASE

If the injury is alkali, irrigate to a pH endpoint with the fornices swept before anything is measured. If the limbus is ischaemic, the prognosis follows the clock hours lost.

Case S2-18 • why

1 • D Neurotrophic keratopathy

A painless persistent defect with heaped edges and thinning, in a cornea with no sensation a year after zoster, is failure of trophic supply. No infiltrate and no cells argue against infection.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Microbial ulcer	Any persistent defect with thinning raises infection first.	There is no infiltrate, no chamber reaction, and no pain in 3 weeks.
B	Exposure keratopathy	Post-zoster lids scar, and exposure also gives a lower defect.	Closure is complete with a Bell phenomenon, and tear volume is adequate.

2 • E Loss of trophic sensory neuropeptides

Sensory nerves release substance P and CGRP, which support epithelial mitosis, migration and adhesion. Without them the surface neither closes nor sticks, and blinking is not prompted.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Aqueous deficiency	Dryness is the usual reason a surface fails after zoster.	Schirmer 12 mm is adequate, and dryness gives punctate staining.
D	Live virus	The trigger was viral, so ongoing infection sounds plausible.	This is a year on, with no dendrites and no infiltrate.

3 • C Stop the NSAID and preserved drops

Topical NSAIDs impair epithelial healing and can drive melt in an anaesthetic cornea, and preservative adds surface toxicity. Removing both costs nothing and may close the defect.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Add steroid	The heaped edge looks inflamed and steroid flattens it.	Steroid slows healing and thins a cornea already down by 30%.
A	Add antibiotic	An open surface invites infection, so cover feels prudent.	Preserved cover adds toxicity without treating why it stays open.

4 • D, E, F Unpreserved tears, bandage lens, serum or NGF

Treatment replaces the missing trophic support and protects the surface: unpreserved lubrication, a lens that shields the epithelium under cover, and serum or growth factor to restore healing.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A, C	Steroid, NSAID	Both quieten an eye that looks irritated at the slit lamp.	Both slow epithelial closure and both invite melt in this cornea.
B	Repeated debridement	Heaped edges do not adhere, so removing them feels constructive.	Repeated debridement of an anaesthetic cornea enlarges the defect.

5 • A Amniotic membrane and tarsorrhaphy today

A descemetocoele is hours from perforation, so the surface is covered and the lids are closed to take the mechanical load off it. Grafting an anaesthetic cornea repeats the same failure.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Keratoplasty this week	The cornea is nearly perforated, and a graft replaces it.	A graft on an anaesthetic surface melts. Cover and close first.
B	Lubricants, 4 weeks	Conservative care has been the theme of the whole illness.	At 70% thinning, the next month is when the cornea perforates.

IF-THEN RULES FROM THIS CASE

If a corneal defect is painless in an anaesthetic cornea, stop every toxic drop and support the surface. If thinning passes half depth, cover the cornea and close the lids.

Case S2-19 • why

1 • A Acute bacterial endophthalmitis

Pain with loss to hand movements on day 3, a hypopyon, fibrin and dense vitreous opacities is infection inside the eye. Vitreous involvement is what separates it from anterior causes.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Toxic anterior segment	It also gives fibrin and a hypopyon after routine surgery.	It starts within 24 hours, spares the vitreous, and is not painful.
D	Retained lens matter	Retained cortex does inflame an eye during the first week.	It does not produce a hypopyon with dense vitreous opacity in hours.

2 • A Tap and inject vancomycin with ceftazidime

Bacteria multiply in vitreous within hours and only intravitreal drug reaches it, so the sample and the injection happen at one sitting, today. Topical and oral routes do not reach vitreous.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Topical, review in 12 hours	Intensive drops treat corneal infection and feel proportionate.	Drops do not reach the vitreous, and 12 hours is the retina's margin.
D	Wait for cultures	Targeted therapy after a result is sound antimicrobial practice.	Therapy here is empirical. Sampling and injection go together.

3 • D, E, F Sample, inject, vitrectomy if light perception

Samples are taken before drug is given, cover is broad and intravitreal, and presenting vision decides surgery: light perception only is the threshold at which vitrectomy does better.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Amphotericin routinely	Fungal endophthalmitis is devastating, so cover looks safer.	Fungus is rare on day 3 and amphotericin is toxic to the retina.
B, C	Oral only, lens out	Both look aggressive in roughly the right direction.	Oral drug underdoses the vitreous, and the lens stays for now.

4 • E Reinject and refer for vitrectomy

Failure to improve means the organism load has not been cleared, so drug is given again and the vitreous is removed with it. Steroid alone and further delay both cost retina.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Oral steroid	Inflammation is doing the visible damage inside the eye.	Steroid with uncontrolled infection accelerates the loss.
C	Switch to an antifungal	Failure of antibacterial therapy suggests the wrong organism.	Reinjection and vitrectomy come first. Fungal onset is later and slower.

5 • A Ocular surface flora through the wound

Most acute cases come from the patient's own lid and conjunctival flora entering at surgery, which is why antisepsis matters more than the choice of antibiotic. Coagulase-negative staphylococci lead.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Propionibacterium	It is the organism every postoperative list carries.	It presents weeks to months later, indolently, with a capsular plaque.
D	Sterile, from fluid	Contaminated fluid does cause clusters after cataract surgery.	Sterile syndromes appear on day 1 and spare the vitreous.

IF-THEN RULES FROM THIS CASE

If an eye loses vision with pain in the first week after surgery and the vitreous is involved, tap and inject the same hour.
If vision is light perception, arrange vitrectomy.

Case S2-20 • why

1 • C Toxic anterior segment syndrome

Onset within 24 hours, limbus-to-limbus oedema with a fixed dilated pupil, no vitreous involvement and three eyes from one list is a toxic insult delivered into the anterior chamber.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Endophthalmitis	Fibrin with a raised pressure after surgery reads as infection.	Infection is painful, starts later, and takes the vitreous with it.
B	Surgical decompensation	A hard cataract can leave a swollen cornea on day 1.	Trauma gives focal oedema, not a fixed pupil in three patients.

2 • D, E, F Day 1 onset, whole-cornea oedema, a cluster

Toxic injury is immediate, spread across the whole cornea because the insult was inside the chamber, and shared by whoever shared the exposure. Fibrin, pressure and acuity occur in both.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Fibrin in the chamber	A fibrinous chamber is the sign that dominates the examination.	Both diagnoses produce fibrin, so it never decides the question.
B, C	Pressure, acuity	Both are abnormal and both look like markers of severity.	Severity is shared. Timing and distribution are not.

3 • D Intensive steroid, pressure control, same-day review

The injury is a sterile chemical one, so treatment is anti-inflammatory with pressure control and a recheck the same day. If vitreous cells appear or pain begins, treat the eye as infected.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Tap and inject	Missing endophthalmitis is the costlier error, so cover it.	Day 1 onset, clear vitreous, no pain, and three eyes on one list.
B	Antibiotic, 48 hours	It hedges towards infection and avoids an injection.	It treats nothing here and leaves the eye unseen for 2 days.

4 • A Quarantine instruments and fluids, audit reprocessing

A cluster points at the process rather than the patient, so shared instruments, solutions and reprocessing steps are held and examined. Nothing else stops the next eye being injured.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Change prophylaxis	Adjusting antibiotics feels like the standard response to a cluster.	The insult is toxic, not infective. Prophylaxis changes nothing.
D	Test viscoelastic only	Viscoelastic is a real culprit, so testing it looks targeted.	Testing one item leaves detergent, autoclave and solution errors in use.

5 • E Residue on reprocessed instruments

Poor cleaning of phaco tips and cannulas leaves detergent, denatured viscoelastic and endotoxin to enter the chamber. Preserved intracameral drugs and solutions of wrong pH do the same.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Preservative-free lidocaine	Intracameral drug is an established cause of this syndrome.	It is preserved drug that injures. Unpreserved lidocaine is standard.
B, C	Iodine, lens	Both touch the eye, and iodine is toxic to endothelium at strength.	Antisepsis at 5% on skin and an acrylic lens give no diffuse oedema.

IF-THEN RULES FROM THIS CASE

If the anterior segment inflames within 24 hours with limbus-to-limbus oedema and a clear vitreous, treat as toxic and examine the list. If the vitreous is involved, treat as infection.

Case S2-21 • why

1 • A UGH syndrome from a sulcus implant

A single-piece haptic in the sulcus chafes the back of the iris: a transillumination defect, dispersed pigment, recurrent bleeding, cells and 32 mmHg in that eye alone, with a normal fellow eye.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Cutibacterium endophthalmitis	It is also a late, low-grade, relapsing uveitis in a pseudophakic eye.	No capsular plaque, and blood with an iris defect is mechanical.
C	Pigmentary glaucoma	Heavy angle pigment and iris transillumination are its signature.	That is bilateral, in young myopes, and it does not bleed.

2 • C, D, E Image the haptics, settle the eye, refer

The eye is settled medically while the cause is confirmed and the implant dealt with. Chafing is mechanical, so no drop cures it, and laser and anti-VEGF treat mechanisms not at work here.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	Prostaglandin first line	It is the strongest single agent for a pressure of 32 mmHg.	It is avoided while the chamber is inflamed and bleeding.
A, B	Trabeculoplasty, anti-VEGF	One targets the pigmented angle, the other the bleeding vessels.	The angle is loaded by chafe, and there are no new vessels.

3 • B Aqueous suppression with a beta-blocker and a CAI

Aqueous suppression lowers pressure without stirring the eye, and 48 hours is short enough to catch a further rise. Prostaglandins and miotics worsen inflammation, and laser adds pigment load.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Latanoprost at night	It is the usual first choice and the most effective single drop.	It aggravates active uveitis, which is what is driving this eye.
A	Pilocarpine with steroid	Miosis pulls the iris off the haptic and outflow improves.	Miotics increase inflammation and drag iris across the optic.

4 • C Pigment and blood clog the meshwork

Gonioscopy shows an open angle heavy with pigment, and the chamber holds red cells. Both load the meshwork, which is why pressure rises with each bleed and falls between them.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Angle closure by synechiae	Chronic uveitis does scar the angle shut and raise pressure.	Gonioscopy shows the angle open along its whole circumference.
B	Steroid response	He is likely to be given steroid, and steroid does raise pressure.	The pressure was already 32 mmHg before any steroid was given.

5 • B Keep the aspirin, deal with the implant

The bleeding is mechanical, from a haptic rubbing iris vessels, so withdrawing antiplatelet cover risks stent thrombosis without stopping the bleed. The implant is the treatable part.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stop the aspirin	Antiplatelet drugs do prolong bleeding, and this eye keeps bleeding.	A coronary stent makes that a cardiac risk for no ocular gain.
E	Aspirin, then observe	Bleeds have settled by themselves four times already.	Pressure is 32 with each episode. Waiting costs optic nerve.

IF-THEN RULES FROM THIS CASE

If a pseudophakic eye bleeds again with pigment dispersion and an iris transillumination defect, image the haptics. If pressure and bleeding recur, move the implant.

Case S2-22 • why

1 • E Late in-the-bag dislocation

Implant and contracted capsule move together, sit low and tilted, and the eye has gained +2.25 D, so the whole complex has fallen back on a failing zonule. Pseudoexfoliation explains why.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Capsule opacification	It is the usual reason vision fades years after cataract surgery.	A membrane does not tilt the optic, wobble, or shift refraction.
A	Out-of-the-bag dislocation	Both leave a displaced implant in an eye operated years ago.	The capsule has come with it, so the haptics are still in the bag.

2 • B, C, D Wobble, tilt and a hyperopic shift

Movement of implant and capsule as one unit, a tilted low position and a change in refraction all measure lost support. A capsule that has only contracted stays where the zonule holds it.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Pseudoexfoliation	It is the reason the zonule failed, so it reads as the evidence.	It predicts failure. It does not show that anything has moved.
A, F	IOP 22, nine years	One is abnormal, the other the interval that invites suspicion.	Pressure reflects the meshwork, and time alone measures nothing.

3 • D Planned surgical refixation

The complex is mobile and the optic edge crosses the pupil, so acuity will not hold and full dislocation is likely. It is a planned list operation, not an emergency and not a laser.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Emergency vitrectomy	A mobile implant in a nine-year-old eye feels like an emergency.	It is still in place with a flat retina. The list can be planned.
A	YAG to the phimosis	The capsule has contracted, and laser does open a tight rim.	Cutting a supporting rim on a loose complex hastens the drop.

4 • B Free dislocation with a pressure of 42

An implant loose in the vitreous with a pressure of 42 threatens optic nerve and retina today. The other changes are reasons to bring the planned operation forward, not to admit him.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Two lines over months	Falling acuity is what the operation is for in the first place.	A slow drop moves him up the list. It is not an emergency.
A	Edge across the axis	It explains the double image and the poor acuity he reports.	It is the same disease at the same speed, with the eye intact.

5 • C Pseudoexfoliative zonular weakness

Pseudoexfoliative material on both pupil margins marks a fibrilopathy that degrades the zonule over years, and it is the commonest reason a bagged implant descends late.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Capsule contraction	The capsule here has plainly contracted around the optic.	Contraction pulls the rim in. It does not detach the zonule.
B	Old capsule tear	A tear at surgery is the classic route to a displaced implant.	That implant falls early, and his operation was uneventful.

IF-THEN RULES FROM THIS CASE

If pseudophakic vision changes with head position, look for movement of the whole complex rather than a membrane. If it is mobile but still in place, plan surgery rather than admit.

Case S2-23 · why

1 · B Toxoplasma reactivation at a scar

A fluffy white focus of retinitis touching an old pigmented scar, with vitritis over it and a mild anterior reaction in a healthy young woman, is reactivation at the border of an old lesion.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Acute retinal necrosis	Both give white retinitis with heavy vitritis and an aching eye.	That spreads circumferentially in days and occludes arterioles.
C	CMV retinitis	It also produces a white retinal lesion with haemorrhage in it.	It needs deep immunosuppression and provokes little vitritis.

2 · A, B, C Antiparasitic, then steroid, with serology

A lesion 1.5 disc diameters from the fovea with vitritis is treated systemically. Steroid follows the antiparasitic by about two days, because steroid alone lets the organism multiply.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Steroid first	Vision is being lost to inflammation, which steroid settles fast.	Unopposed steroid enlarges the lesion. Cover the organism first.
E, F	Foscarnet, topical only	One covers a virus, the other treats the reaction you can see.	The organism is not viral, and drops do not reach the retina.

3 · C Pyrimethamine with clindamycin and folinic acid

She reacted to co-trimoxazole with rash and fever, so sulfonamides are out. Clindamycin substitutes for sulfadiazine, and folinic acid protects the marrow from pyrimethamine.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Sulfadiazine regimen	It is the standard combination and the one most often quoted.	Sulfadiazine is a sulfonamide, and she reacted to one already.
E	Co-trimoxazole	It is a convenient oral alternative with real evidence behind it.	It contains the sulfonamide that gave her rash and fever.

4 · C Cysts at the scar margin rupture

Bradyzoites persist in cysts in and around the healed lesion. When a cyst ruptures, parasites invade the adjacent retina, which is why recurrence is satellite to the scar.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Immune complexes	Recurrence looks immunological, and steroid does help the eye.	The lesion is an infection with live parasites, not a deposit.
A	Fresh infection	New exposure from cat litter or raw meat feels like the trigger.	Reactivation is endogenous, from cysts already in that retina.

5 · A A fluffy border with active vitritis

Activity is judged at the lesion border and in the vitreous. A soft edge with haze means live parasites, whereas pigmentation and a clearing vitreous are the signs treatment can stop.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Pigmented border	Any remaining abnormality feels like a reason to keep treating.	Pigment at the edge is the healed state, and the endpoint.
D	High IgG titre	A high titre reads like a measure of how active the disease is.	IgG marks past exposure and stays raised for life.

IF-THEN RULES FROM THIS CASE

If a focal white retinitis touches a pigmented scar with vitritis over it, treat toxoplasma reactivation. If steroid is used, it follows the antiparasitic and never leads it.

Case S2-24 • why

1 • C Rheumatoid peripheral ulcerative keratitis

A crescentic peripheral ulcer with an epithelial defect and an infiltrated advancing edge, in seropositive disease with swollen joints and an ESR of 68, is corneal vasculitis.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Mooren ulcer	It looks identical at the slit lamp and melts the periphery too.	Mooren is idiopathic, diagnosed once vasculitis is excluded.
E	Terrien degeneration	It thins the periphery and is easy to reach for in an older eye.	Terrien is painless, epithelialised, and slow over years.

2 • A Oral steroid plus stronger immunosuppression

Corneal melt in rheumatoid disease is active systemic vasculitis and it destroys tissue in days. Steroid buys control while a stronger agent is started with the rheumatologists.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	More methotrexate	She is on it already, so dose escalation looks like the next step.	It takes weeks to act, and it has failed her at 20 mg weekly.
C	Topical steroid only	Steroid is the anti-inflammatory reflex for a melting cornea.	Topical steroid thins the stroma further and misses the cause.

3 • A, E, F Lubricants, doxycycline, antibiotic cover

The local aims are surface protection, less proteolysis and no secondary infection over a bare bed. Steroid drops and NSAID drops both thin a gutter that is already 60% deep.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Hourly prednisolone	It settles the visible inflammation at the leading edge.	It suppresses stromal repair, and the ulcer deepens instead.
C, D	NSAID drops, bandage lens	One eases severe pain, the other covers a bare stromal bed.	NSAIDs are linked with melt, and a lens hides progression.

4 • A Seal it, keep the immunosuppression running

A small perforation is sealed with the least tissue that will hold, because a large graft in vasculitic cornea melts at its own edge. Systemic control continues through the repair.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Large keratoplasty	A definitive graft seems to solve cornea and perforation at once.	The host bed is inflamed, so the new rim melts in the same way.
E	Defer surgery	Operating on active vasculitis is genuinely a poor bet.	The eye is open now. A leaking cornea cannot wait for joints.

5 • A Metalloproteinase inhibition

Tetracyclines inhibit the matrix metalloproteinases released by neutrophils and keratocytes, which is the enzymatic step digesting collagen at the base of the gutter.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Lid margin bacteria	Doxycycline is prescribed for blepharitis for exactly that.	This is a sterile vasculitic melt, not a lid-driven ulcer.
B	B cell suppression	Rheumatoid factor is central, and B cell therapy does help.	That is rituximab. Doxycycline does not touch B cells.

IF-THEN RULES FROM THIS CASE

If rheumatoid disease brings a peripheral gutter with an epithelial defect, treat it as systemic vasculitis and escalate systemic therapy. If it perforates, seal it and keep treating.

Case S2-25 · why

1 · B Tractional detachment, macula involved

A shallow concave elevation tented to fibrovascular fronds, with striae across the macula and no break, is traction. Twenty-seven years of poor control and absent laser explain the fronds.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Combined detachment	Long traction does tear retina, and that would change surgery.	No break is seen, and the surface stays concave, not convex.
C	Rhegmatogenous detachment	Blood and floaters with a field defect is the classic route in.	A break-driven detachment is convex, mobile, and reaches the ora.

2 · D A concave surface pulled to fronds

Traction pulls retina towards the vitreous, so the surface is concave and tethered where the fronds insert. Blood, poor acuity, striae and normal pupils occur in either kind.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Vitreous haemorrhage	Blood accompanies both a retinal tear and a bleeding frond.	It says the vitreous is full of blood, not why retina is off.
B, C	Striae, normal pupils	Both are findings in front of you and both feel diagnostic.	Striae follow any macular distortion, and pupils are often normal.

3 · E Vitrectomy with delamination and endolaser

The macula is detached by traction, and only surgery relieves it. Laser, injection and metabolic control all matter, but none of them lifts a taut membrane off the fovea.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Finish the laser	Laser is overdue and does treat the ischaemia driving all this.	Laser cannot relieve traction, and three months loses the macula.
C	Optimise glucose first	An HbA1c of 96 and 168/98 are genuinely part of his problem.	They change his ten-year risk, not the fovea detached today.

4 · C Fibrous contraction, more traction

Blocking VEGF makes fronds involute and fibrose, and fibrous tissue shortens. On a macula already tented, that tightens the pull. It is given days before surgery, not instead of it.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Endophthalmitis	It is the risk quoted at every injection consent.	It is rare, and unrelated to the traction question asked here.
D	More haemorrhage	The eye already has blood, and injections do enter the vitreous.	Anti-VEGF reduces bleeding from fronds rather than provoking it.

5 · B, C Tell him he must notify, and record it

Vocational standards are stricter, and this eye has lost acuity and field, so fitness is decided by the licensing authority. The duty to declare is his, and the advice is documented.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Notify them yourself	A taxi carries passengers, so acting directly feels protective.	The duty is his. You disclose only if he drives on despite advice.
A	Tell his employer	The employer controls whether he works, so it seems practical.	It breaches confidence, and the employer has no assessing role.

IF-THEN RULES FROM THIS CASE

If a detachment is concave and tented to fibrovascular tissue with no break, it is traction and it needs vitrectomy. Incomplete laser does not change that order.

Case S2-26 • why

1 • E Acute retinal necrosis, zoster

Confluent peripheral necrosis with scalloped borders spreading circumferentially, dense vitritis and occluded arterioles over five days, in a well man with previous shingles, is zoster.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	CMV retinitis	Both give white retina with vessel change in an older adult.	It needs immunosuppression, spreads slowly, and barely inflames.
A	Toxoplasma retinitis	Heavy vitritis over a white focus is exactly its picture.	Its lesion is single, satellite to a scar, and not arteritic.

2 • D, E, F IV antiviral, PCR sample, check the other eye

Necrosis advances hour by hour, so systemic antiviral starts today, and the tap is taken as it begins. The fellow eye is at real risk and has to be examined and the risk explained.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Steroid first	The eye is full of cells, and steroid settles that quickly.	Steroid ahead of antiviral cover lets the necrosis spread faster.
B, C	Laser now, intravitreal only	One walls off detachment, the other hits the virus hard.	Laser waits for a quieter eye, and this virus needs systemic cover.

3 • E Topical steroid with aqueous suppression

Once antiviral cover is running, topical steroid is safe and treats the trabeculitis raising the pressure, with an aqueous suppressant while it settles. Prostaglandins worsen uveitis.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Steroid before antiviral	Steroid does control the inflammation that is blurring vision.	In a necrotising retina it must follow cover, never precede it.
C	Observe the pressure	The pressure may fall as the inflammation is treated.	30 mmHg with an oedematous disc is not a pressure to watch.

4 • D Virus travels the visual pathway

Untreated, about a third of these eyes are followed by the fellow eye, because virus passes through the chiasm and central connections. Systemic antiviral is what lowers that risk.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Immune complexes	Bilateral disease sounds autoimmune rather than infective.	This is lytic infection, and antiviral therapy prevents it.
B	Both ganglia at once	Latent zoster does sit in sensory ganglia on both sides.	Ganglia supply skin. Retinal spread follows the visual pathway.

5 • E A break in necrotic retina with a field defect

Necrotic retina tears as the vitreous contracts, and detachment follows in most untreated eyes. A new field defect with a break is a vitreoretinal problem, not a medical one.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Border pigmentation	Any change at the border reads as progression at first glance.	Pigment at the border is healing, and the point of treatment.
D	PCR result	It confirms the diagnosis, which feels like a decision point.	It validates the drug already running. Nothing needs surgery.

IF-THEN RULES FROM THIS CASE

If peripheral necrosis spreads circumferentially with dense vitritis and arteritis, start systemic antiviral the same day and examine the fellow eye. If retina tears, refer for surgery.

Case S2-27 • why

1 • D Choroidal melanoma

A dome-shaped pigmented lesion 3.6 mm thick with orange pigment, subretinal fluid, low internal reflectivity and excavation, whose base has grown from 7.1 to 9.2 mm, is malignant.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Choroidal naevus	Most pigmented choroidal lesions are naevi and stay harmless.	A naevus is flat, carries drusen and a halo, and does not grow.
A	Choroidal haemangioma	It is dome-shaped and leaks fluid under the retina as well.	It is orange-red rather than pigmented, and reflects sound highly.

2 • C, D, E Thickness, orange pigment, fluid

Thickness above two millimetres, orange lipofuscin and subretinal fluid each predict growth in a small pigmented lesion. Iris colour is a risk for the disease, not for this lesion.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	Blue irides, fair skin	Light pigmentation is the population risk factor for melanoma.	It says who develops one, not whether this lesion is growing.
A, B	Position, good acuity	Both feel relevant: near the disc, and vision still preserved.	Neither predicts growth. Good acuity is usual in thick tumours.

3 • E Ocular oncology referral this week

Diagnosis, treatment choice and eye-conserving options belong to a specialist service, and growth is already documented. Staging is arranged there, alongside the local treatment plan.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Repeat photographs	Serial imaging is the correct approach to a suspicious naevus.	Growth is documented already. The interval decision is behind her.
C	Enucleation locally	It removes the tumour definitively in a single operation.	Most tumours this size keep the eye with plaque radiotherapy.

4 • E Plaque brachytherapy with liver follow-up

A medium tumour of 3.6 mm thickness is treated with plaque radiotherapy, which keeps the eye and matches enucleation for survival. Surveillance afterwards is aimed at the liver.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Enucleation for fluid	Fluid signals activity, which sounds like a reason to remove it.	Fluid is common and settles after radiotherapy. Size decides.
B	Thermotherapy alone	It is eye-sparing and avoids a radiation plaque altogether.	Recurrence is high in a 3.6 mm tumour treated by heat alone.

5 • D Haematogenous, the uvea has no lymphatics

The uveal tract has no lymphatic drainage, so cells leave in the blood and lodge first in the liver. That is why surveillance is hepatic and why nodes are not sampled.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Lymphatic spread	Most solid tumours reach nodes first, so it is the default.	The uvea has no lymphatics, and nodes are not the first site.
A	Optic nerve sheath	Direct extension occurs and looks anatomically plausible.	It is rare, and it does not explain hepatic metastasis.

IF-THEN RULES FROM THIS CASE

If a pigmented choroidal lesion is over two millimetres thick, carries orange pigment or leaks fluid, refer it as a melanoma. Documented growth removes the option of watching.

Case S2-28 • why

1 • E The optic chiasm

Loss of both temporal fields, stopping at the vertical midline, can only come from nasal retinal fibres, and they cross in one place. Amenorrhoea with galactorrhoea names what sits below.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Left optic nerve	One eye is a line worse, which invites a lesion on that side.	A nerve lesion takes the whole field of one eye, not two halves.
B	Right optic tract	A tract lesion is retrochiasmal and splits at the midline too.	That gives a left homonymous defect, not both temporal halves.

2 • E A one-sided afferent pupillary defect

Asymmetry of pupillary input means one optic nerve conducts worse than the other, which is prechiasmal. Colour, acuity and disc pallor all fall with chiasmal compression as well.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Bilateral temporal pallor	Pallor points at the optic nerve, so it reads as prechiasmal.	Chronic chiasmal compression atrophies crossing fibres in both.
A, C	Colour, acuity	Both measure afferent function and both are abnormal here.	Chiasmal compression reduces both. Neither localises in front.

3 • A MRI of the sella with endocrine bloods

MRI shows the mass, its extent and the chiasm, and hormones decide the treatment, so the two are ordered together. Compression that has grown for two years is imaged within days.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Unenhanced CT	It is quick, available, and excludes a large mass or a bleed.	It underestimates soft tissue at the sella and misses the chiasm.
D	Repeat the fields	Confirming an unfamiliar field defect feels careful.	The pattern is unambiguous, and delay costs crossing fibres.

4 • E A dopamine agonist first

A prolactin of 4200 mU/L makes this a prolactinoma, and these shrink on a dopamine agonist, often with fields recovering in weeks. Surgery is for failure or sudden visual collapse.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Surgery first	The chiasm is compressed, and decompression sounds mechanical.	A prolactinoma shrinks medically. Surgery is the second line.
C	Observation	Acuity is nearly normal, so watching looks proportionate.	Both temporal fields have gone already. It is progressing.

5 • D, E Nasal fibres cross, inferonasal ones lowest

Nasal retina, which sees the temporal field, crosses in the chiasm, and the inferonasal bundle crosses lowest and furthest forward. That is why superior temporal loss often comes first.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Temporal fibres cross	Half the fibres cross, so which half is easy to invert.	Temporal retina stays uncrossed. Nasal retina is what crosses.
C	Nasal half of the field	Nasal fibres and nasal field sound like the same thing.	Nasal retina sees the temporal field. Retina and field are flipped.

IF-THEN RULES FROM THIS CASE

If both temporal fields are lost at the vertical midline, localise to the chiasm and image within days. If one pupil shows an afferent defect, the disease is in front of it.

Case S2-29 • why

1 • D The left medial longitudinal fasciculus

Fibres run from the right abducens nucleus across to the left third nerve nucleus in the left fasciculus. Left adduction fails on right gaze while convergence, which bypasses it, is full.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Right fasciculus	The nystagmus is in the right eye, which drags the side across.	A right lesion would stop the right eye adducting on left gaze.
C	Right abducens nucleus	The fibres do start there, and rightward gaze is what fails.	That lesion abolishes all gaze to the right, abduction included.

2 • E Convergence is full

Convergence reaches the medial rectus by a supranuclear route that does not use the fasciculus. A muscle that adducts fully for a near target cannot be paretic for gaze.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Abducting nystagmus	It is the other half of the syndrome and impossible to miss.	It reports the yoked eye overshooting, not what the muscle can do.
A, B	Vertical gaze, pupils	Both are normal, which feels like proof the muscles are intact.	Neither tests adduction, and both can be normal in a palsy.

3 • A, B, F Young, previous optic neuritis, disc pallor

Youth without vascular risk, an earlier painful visual loss and pallor of that disc give lesions separated in time and place. Two weeks of hand tingling adds a third site.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Diplopia on right gaze	It is the symptom that brought her in, so it feels central.	It localises the lesion. It says nothing about the cause.
D, E	Nystagmus, pupils	Both belong to the examination that made the diagnosis.	Both are identical whether the cause is a plaque or an infarct.

4 • A MRI of brain and cord, neurology

The question is how many lesions there are and where, which needs MRI of brain and cord with contrast. Treatment and prognosis follow the scan, so imaging comes before steroid.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	CT and vascular review	Brainstem signs do raise stroke, and CT is the quickest scan.	CT cannot see a pontine plaque, and she has no vascular risk.
D	Steroid first	Steroid shortens a relapse, and she is symptomatic now.	Steroid before imaging blurs the diagnosis it means to treat.

5 • D Internuclear fibres from the abducens nucleus

Interneurons in the abducens nucleus cross the midline and ascend in the opposite fasciculus to the medial rectus subnucleus. Cutting that link uncouples the two eyes on gaze.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Frontal eye field	It initiates the saccade, so it sits upstream of the movement.	A frontal lesion moves both eyes together, without a split.
C	Sixth nerve to medial rectus	The abducens nucleus is involved, so its nerve looks implicated.	The sixth nerve supplies lateral rectus. Medial rectus is third.

IF-THEN RULES FROM THIS CASE

If adduction fails on gaze to one side while convergence stays full, the lesion is internuclear and in the pons. In a young adult, image brain and cord for demyelination.

Case S2-30 • why

1 • D The right occipital cortex

The left half of both fields has gone at the vertical midline, so the lesion is retrochiasmal on the right. A complete, congruous defect with a spared central island is occipital.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Meyer loop	A temporal lesion is retrochiasmal and gives homonymous loss.	Meyer loop takes the upper quadrant, not a whole half field.
A	Left occipital cortex	Occipital is the right place, and sides are easy to invert.	The left cortex serves the right half of both fields.

2 • C Congruity with a spared central island

Fibres from the two eyes lie together in the cortex, so cortical defects match closely and often spare a central island. Tract lesions are incongruous and may give a pupil defect.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Respects the midline	The midline rule is the first thing taught about these fields.	Everything behind the chiasm respects it, the tract included.
E	Normal pupils	Normal pupils feel like evidence the lesion is far back.	Pupils are often normal in tract lesions too. It rarely helps.

3 • D Same-day stroke service referral

Three days from onset, in atrial fibrillation with no anticoagulation, he is at high risk of a second embolus. The pathway that investigates and prevents that starts at the stroke service.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Aspirin and a month	Antiplatelet therapy is part of secondary prevention.	Fibrillation needs anticoagulation, decided after imaging.
B	Prisms and retraining	Rehabilitation is what will help the field he has lost.	It treats the deficit and ignores the artery about to close.

4 • D, E, F Anticoagulation, blood pressure, low vision help

The month belongs to preventing the next infarct, so rhythm and a pressure of 176/98 are treated, while reading strategies restore what a left-sided field defect takes away.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Occlude one eye	Occlusion helps double vision, so it sounds like a comfort measure.	Both eyes lose the same half. Covering one removes nothing.
B, C	Weekly fields, steroid	One tracks recovery, the other treats swelling around an infarct.	Fields are repeated at months, and steroid has no role here.

5 • E Stop driving, notify the authority

A homonymous hemianopia fails the visual field standard whatever the acuity. He stops driving now, the duty to declare is his, and any return is decided on a formal field test.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Acuity is enough	The acuity standard is the one most drivers know about.	Field and acuity are separate standards. This one fails on field.
A	Wait until it is stable	Some cortical field loss does recover in the first months.	Recovery is judged by the authority, not by how he feels.

IF-THEN RULES FROM THIS CASE

If both eyes lose the same half of the field at the vertical midline, the lesion is behind the chiasm and it is a stroke referral. Congruity with a spared centre means occipital.

Case S2-31 • why

1 • D Retinal TIA, carotid embolic source

Painless monocular loss lasting minutes with full recovery, repeated three times, with a refractile plaque at a bifurcation and an ipsilateral bruit, is embolic retinal ischaemia from the carotid.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Giant cell arteritis	It is the transient loss that must never be missed, and he is the right age for it.	ESR 12 with no jaw claudication, scalp tenderness or systemic symptoms.
E	Retinal migraine	Brief stereotyped episodes with complete recovery fit its description well.	First episodes at 71, with a visible plaque and a bruit, are embolic.

2 • A, E, F Same-day duplex, aspirin, TIA clinic

Stroke risk peaks in the days after a retinal TIA, so imaging, antiplatelet cover and specialist assessment all happen today. The benefit of surgery falls sharply after the first two weeks.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Oral prednisolone	Covering for arteritis feels like the cautious move in a man of 71.	ESR 12 with no cranial symptoms, and steroid does nothing to an embolus.
B, D	Duplex in four weeks, BP	Both are correct actions at the tempo of a routine vascular review.	The window for endarterectomy is measured in days. Four weeks wastes it.

3 • A Endarterectomy within two weeks

Symptomatic stenosis of 70 to 99% carries a high early stroke risk, and the gain from surgery is largest inside two weeks of the last event. Antiplatelet and statin run alongside, not instead.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Medical therapy alone	Drugs carry no operative risk and the eye looks normal today.	At 78% and symptomatic, the stroke risk exceeds the risk of operating.
D	Warfarin	Anticoagulation sounds stronger than an antiplatelet for stopping emboli.	This is platelet debris from plaque, not thrombus from a fibrillating atrium.

4 • C Cholesterol embolus from the carotid plaque

A bright yellow refractile fleck lodged at a bifurcation is cholesterol shed from an ulcerated plaque. The bruit places the source in the ipsilateral carotid rather than the heart.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Calcific, valvular	Cardiac emboli are the other classic cause of transient monocular loss.	Calcific emboli are chalky white and dull, and they sit more proximally.
E	Platelet-fibrin embolus	Fibrillation is common at 71 and does throw emboli to the retina.	Platelet emboli are dull grey and pass within minutes. This one persists.

5 • E Stop driving now, he notifies, advice recorded

Three events in 10 days make him unfit to drive from today, and the duty to notify the licensing authority rests with the licence holder. The advice and his response are entered at this visit.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Practice notifies for him	It feels like the protective act, and the risk to other road users is real.	Notification is his duty. Disclosure follows only if he drives on regardless.
D	No restriction	He sees 20/25 and the eye is entirely normal between the attacks.	The hazard is a sudden loss at the wheel, not his acuity in the chair.

IF-THEN RULES FROM THIS CASE

If monocular loss lasted minutes and recovered fully, call it a retinal TIA and image the carotids the same day. If symptomatic stenosis reaches 70%, refer for surgery within two weeks.

Case S2-32 • why

1 • D Angle recession glaucoma OD

A wide band of ciliary body over four clock hours with torn iris processes is a recessed angle, and the sphincter tears and focal lens opacity date it to the injury. The fellow eye is normal.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Asymmetrical POAG	Open angles with a notched disc and arcuate loss is its usual picture.	Primary disease does not recess the ciliary band or tear iris processes.
E	Pseudoexfoliation	It is the commonest cause of a genuinely unilateral high pressure.	No flakes on the pupil margin or capsule, and the pigment is focal.

2 • D, E, F Recessed band, torn processes, sphincter tears

Only the structural scars carry a date: a recessed ciliary band, torn processes, and a torn sphincter beside a focal lens opacity. The other findings measure severity, not mechanism.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A, B	Pressure and cupping	They are the abnormal numbers, and they are what drives the treatment.	Both occur in primary disease, so neither dates the damage to 20 years ago.
C	Corneal thickness	Thickness corrects the pressure reading, which makes it feel diagnostic.	545 μ m is average, and it says nothing about why this angle is damaged.

3 • E Topical prostaglandin OD, review at four weeks

A notched disc with a matching arcuate defect needs the pressure down, and a prostaglandin gives the largest single-agent reduction. Medical therapy comes before laser or a filtering operation.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Laser trabeculoplasty	It is a standard first line where the meshwork is open and visible.	A traumatised meshwork answers laser poorly and unpredictably.
C	Trabeculectomy now	Established field loss at 30 mmHg looks like a surgical problem.	He has had no medical trial. Surgery is not the opening move here.

4 • A Progressive thinning at 24 mmHg

Escalation follows progression rather than a number. Structural loss continuing at 24 mmHg means the target is too high for this nerve, while a stable disc at the same pressure means it is not.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Stable at 24 mmHg	24 sits above the usual target, so treating the number feels safer.	A nerve holding steady for a year says the current target is working.
D	Spike after a missed dose	A reading of 30 on treatment reads as outright failure of the drop.	An unmedicated pressure measures adherence, not the regimen.

5 • B Binocular field decides, advice recorded

Fitness turns on the binocular field, and a full fellow field covers this arcuate defect. He is told the position, formally field tested if the defect spreads, and the advice is recorded.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Notify at once	The word glaucoma sounds notifiable in and of itself.	One eye affected with a full fellow field does not meet the test.
E	Pressure under 21 mmHg	21 is the number patients remember, so it feels like the threshold.	Licensing is decided on field and acuity, never on a pressure reading.

IF-THEN RULES FROM THIS CASE

If one eye alone runs a high pressure, gonioscope both angles before naming the disease. If a wide ciliary band is seen, expect a poor answer to trabeculoplasty and monitor for life.

Case S2-33 • why

1 • A Drug-induced ciliochoroidal effusion

Simultaneous bilateral closure, a 3.50 D myopic shift, shallowing at the centre as well as the periphery, and annular choroidal elevation place the block behind the iris. The drug started 11 days ago.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Primary angle closure	The pressures, the haze and the closed angles are identical to it.	Primary closure takes one eye at a time and shifts no refraction.
C	Phacomorphic closure	A lens moving forward does shallow a chamber and close an angle.	Both lenses are clear, and both eyes closed together in 36 hours.

2 • C, D, E Stop the drug, aqueous suppression, atropine, steroid

The block sits behind the iris, so the drug comes out, the ciliary body is relaxed and aqueous is suppressed while the effusion settles. The diaphragm falls back over days and the angles reopen.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	Laser iridotomy	Closed angles at 40 mmHg is the textbook indication for a hole.	There is no pupil block to relieve, so the angle stays shut afterwards.
A, B	Pilocarpine, iridoplasty	Both pull peripheral iris out of the angle in other closures.	Miotics rotate the ciliary body further forward and worsen this one.

3 • D Acetazolamide and a hyperosmotic, atropine kept

Medical control buys the days the effusion needs once the drug is withdrawn. Nothing here calls for a hole in the iris or the loss of a clear lens, and both act on a mechanism that is absent.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Iridotomy now	A pressure of 34 on maximal drops reads as failure needing laser.	The angle is shut by forward rotation, which an iridotomy leaves intact.
B	Lens extraction	Removing the lens does deepen a crowded anterior segment for good.	These lenses are clear and the mechanism resolves in days regardless.

4 • C Forward lens movement on a rotated ciliary body

Fluid in the supraciliary space rotates the ciliary body forward and slackens the zonule, so the lens moves anteriorly and thickens. The eye turns myopic and the chamber shallows across its width.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Lens swelling	Lens hydration is a real cause of acute myopia in hyperglycaemia.	These lenses are clear, and swelling would not lift the choroid.
A	Ciliary spasm	Spasm gives myopia and brow ache, and both are present here.	Spasm closes no angle and produces no annular choroidal elevation.

5 • B Central as well as peripheral shallowing

Pupil block bows the peripheral iris and leaves central depth near normal. Uniform shallowing with a myopic shift means the whole lens-iris diaphragm has come forward, which is a different disease.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Pressure above 40	The height of the pressure feels like a measure of the mechanism.	Both mechanisms reach the forties, so the number separates nothing.
D, E	Oedema, closed angle	These are the findings that name an angle closure in the first place.	Shared by both, and neither says what pushed the iris forward.

IF-THEN RULES FROM THIS CASE

If both angles close at once with a myopic shift, stop the new drug and look behind the iris. If the chamber is shallow centrally as well as peripherally, an iridotomy will not open it.

Case S2-34 • why

1 • B Plateau iris syndrome after iridotomy

Appositional closure with a rising pressure despite two patent iridotomies, a flat iris with normal central depth, a double hump on indentation, and anteriorly placed ciliary processes define plateau iris.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Residual pupil block	Closure after a laser suggests the hole is too small or has sealed.	Both holes transilluminate and the central chamber depth is normal.
C	Malignant glaucoma	It too closes an angle in an eye that already has a patent iridotomy.	It follows filtering surgery and shallows the chamber centrally.

2 • B, C, D Patent holes, flat iris, double hump

A patent hole removes block, and its signature, a bowed iris over a shallow centre, is absent. The double hump shows ciliary processes holding the peripheral iris against the meshwork instead.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A, E	Pressure and refraction	One is the abnormal number, the other the classic closure risk factor.	Neither locates the block, and +0.75 argues against a crowded eye.
F	Closure in the dark	It proves the angle does close, which is the worry in front of you.	It shows that closure happens, not what pushes the iris forward.

3 • A Iridoplasty OS with a pressure-lowering drop

Iridoplasty contracts peripheral iris stroma and pulls it out of the angle, which is the only laser that treats this configuration. A drop holds the pressure while the angle is reopened.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	A second iridotomy	Another hole feels logical when the first one did not hold the angle.	The hole is patent, so a second changes nothing about the ciliary body.
C	Trabeculectomy	30 mmHg after a laser that failed looks like a surgical eye.	Filtration is for uncontrolled damage, not an angle that can be opened.

4 • D Gonioscopy on a schedule, treat on change

This eye holds 15 mmHg with no synechiae, so it is followed with the test that shows apposition. Treatment is triggered by a pressure rise or new adhesions, not by the configuration alone.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Iridoplasty both eyes	The configuration is bilateral, so treating both looks consistent.	No pressure rise and no synechiae here. There is nothing yet to treat.
C	Discharge	15 mmHg with 20/20 and a patent iridotomy reads as a normal eye.	An angle that closes in the dark needs gonioscopy, not discharge.

5 • B Warn the prescriber, dilation can close the angle

A patent iridotomy removes pupil block but not plateau configuration, so a dilated pupil still crowds the angle. The prescriber needs that before the first tablet, not after the attack.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	No precaution needed	The iridotomy is the treatment everyone associates with angle closure.	It prevents block alone, and this angle closes without any block.
A	Only drops matter	Topical drugs reach the iris directly, which sounds like the greater risk.	Systemic anticholinergics dilate the pupil as effectively as a drop.

IF-THEN RULES FROM THIS CASE

If an angle still closes with a patent iridotomy, a flat iris and normal central depth, the block sits at the iris root. Treat the configuration with iridoplasty, and warn about dilation.

Case S2-35 • why

1 • B 20%

Of 1000 screened, 40 have disease and 36 are detected, while 144 of the 960 healthy screen positive. That makes 180 positives, of which 36 are true, so 36 divided by 180 gives 20%.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	90%, the sensitivity	Sensitivity is the figure quoted for the test, so it reads as its accuracy.	Sensitivity looks forward from disease, not back from a positive result.
A	4%, the prevalence	It is the other number in the stem, and the disease is genuinely rare.	A positive result moves his probability from 4% up to 20%.

2 • A, B PPV rises to 60%, the test itself unchanged

At 20% prevalence, 200 of 1000 have disease and 180 are detected, while 120 of the 800 healthy screen positive. So 180 of 300 positives are true. The instrument has not changed at all.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	NPV rises to 99%	A better positive value at higher prevalence sounds like it lifts both.	More disease means more missed cases, so the negative value falls to 97%.
D, E	200 false positives	Both follow from treating prevalence as a property of the instrument.	With 800 healthy people, false positives fall to 120, and 85% is fixed.

3 • C Document the explanation, the risk and the refusal

A refusal is valid when a person with capacity has been told what the test is for and what untreated disease costs. What makes it informed is the record of that exchange, and an open door back.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Signed waiver	A signature looks like the strongest protection the practice can hold.	Liability cannot be signed away, and a form is not a conversation.
A	Refer him anyway	The clinical risk is real, and referral feels like the safe default.	Acting against the refusal of a competent adult is the greater wrong.

4 • D A copy to him, inside the statutory period

The file belongs to the practice and the information in it belongs to him, so a person with capacity may have a copy on request. The original, including the refusal entry, stays on file.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Clinician to clinician	A transfer between professionals feels tidier and more secure.	He is entitled to his own record, and he decides where it goes next.
C	Refuse until he agrees	It reads as leverage that might yet protect his sight.	Access to a record is not conditional on accepting treatment.

5 • C Definitive testing offered here, then review

A positive screen is not a diagnosis, and four in five are false, so fields and imaging come before any drop. Offering the tests he will accept keeps him in view and dates the next decision.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Start a prostaglandin	25 mmHg with a 0.7 cup looks like enough to justify treating now.	Treating on a screen commits him to therapy for an unmade diagnosis.
E	Rescreen in a year	A year is a reasonable interval for following a glaucoma suspect.	He holds an untested positive result and a family history, not a clearance.

IF-THEN RULES FROM THIS CASE

If a screen is positive, work out the positive predictive value before acting on it. If a person with capacity refuses, record what was explained, what was refused, and the review offered.

PART 10

Reference

The answer key, an eighteen-day schedule keyed to real case numbers, a cut-line recall deck, and the sources with their link status recorded.

ANSWER KEY

Letters were generated to a distribution, not chosen by an author. The audit that proves it is on page 4, printed from the same build.

RECALL DECK

Cut on the dashed lines. Ten seconds per prompt, answered aloud. Misses go back into tomorrow's pile.

EIGHTEEN DAYS

New cases on the left, recall on the right. If you are short of time, cut new cases and keep the recall column.

SOURCES

Every claim carries a state: verified, unconfirmed, or removed. Nothing in this manual is certified by its own author.

Answer key

Letters are generated to the distribution audited on page 4, not chosen by whoever wrote the item. Guessing one letter throughout scores close to a fifth of the paper, which is what it should score. Multiple letters in a cell mean an all-or-none item: every one, or no mark.

Session 2

CASE	1	2	3	4	5	6
S2-01	B	C	C	A, D, E	C	
S2-02	B	B	A	C, D, F	D	
S2-03	D	D	C	B, C, E	B	
S2-04	C	B	B	A, B, D	B	
S2-05	C	B	D	A, C, F	E	
S2-06	B	E	C	B, E, F	A	
S2-07	D	E	C	A, D, E	D	
S2-08	A	C	A	C, D, F	C	
S2-09	B	D	B	B, C, E	A	
S2-10	A	A	E	A, B, D	E	
S2-11	A	A, B, C	A	B	D	
S2-12	E	C	A, E, F	A	B	
S2-13	B	E	E	C, D, E	E	
S2-14	C	A	D	B, C, D	B	
S2-15	D	E	C	A, B, C	E	
S2-16	B	B, C, D	B	D	D	
S2-17	C	A, B, C	B	C	A	
S2-18	D	E	C	D, E, F	A	
S2-19	A	A	D, E, F	E	A	
S2-20	C	D, E, F	D	A	E	
S2-21	A	C, D, E	B	C	B	
S2-22	E	B, C, D	D	B	C	
S2-23	B	A, B, C	C	C	A	
S2-24	C	A	A, E, F	A	A	
S2-25	B	D	E	C	B, C	
S2-26	E	D, E, F	E	D	E	
S2-27	D	C, D, E	E	E	D	
S2-28	E	E	A	E	D, E	
S2-29	D	E	A, B, F	A	D	
S2-30	D	C	D	D, E, F	E	
S2-31	D	A, E, F	A	C	E	
S2-32	D	D, E, F	E	A	B	
S2-33	A	C, D, E	D	C	B	
S2-34	B	B, C, D	A	D	B	
S2-35	B	A, B	C	D	C	

Eleven days

One session, eleven days. Cases are answered cold, once. Everything after that is recall, not rereading. The right-hand column is the part people skip and the part that moves the score.

DAY	NEW	WORK	RECALL
Day 1	S2-01 to S2-05	Cases cold, keys the same evening	-
Day 2	S2-06 to S2-10	Cases cold, keys the same evening	S2-01 to S2-05, recall cards
Day 3	S2-11 to S2-15	Cases cold, keys the same evening	S2-01 to S2-10, recall cards
Day 4	S2-16 to S2-20	Cases cold, keys the same evening	S2-06 to S2-15, recall cards
Day 5	S2-21 to S2-25	Cases cold, keys the same evening	S2-11 to S2-20, recall cards
Day 6	S2-26 to S2-30	Cases cold, keys the same evening	S2-16 to S2-25, recall cards
Day 7	S2-31 to S2-35	Cases cold, keys the same evening	S2-21 to S2-30, recall cards
Day 8	Repair	The two weakest item-type columns	Condition cards and pharmacology
Day 9	Repair	Every item missed with high confidence	Rewrite each as one if-then rule
Day 10	Timed run	All 175 items in one 3.5 hour block	Score by item type, compare to the first pass
Day 11	Consolidate	Emergency gate and competing pairs, aloud	Formulas drilled, then Volume 1

BOTH VOLUMES, THREE WEEKS

Run this volume, then Volume 1 on the same eleven-day shape. Compare the two score sheets by item-type column, not by total. A total that rose while the treatment column fell is a warning, not progress.

IF YOU HAVE FEWER THAN ELEVEN DAYS

Keep the recall column and cut new cases, not the reverse. Ten cases reviewed to the point of recall beat thirty-five read once. With four days: the emergency gate, the competing pairs, and one timed half session.

Cut-line recall deck

Cut on the dashed lines. Ten seconds per prompt, answered aloud.

Misses go back into tomorrow's pile.

Painful red eye in a contact lens wearer Microbial keratitis until the slit lamp says otherwise. Measure it, culture if central, treat intensively, review inside 24 hours. Never patch.	Pain out of proportion, water exposure, poor antibiotic response Acanthamoeba. Culture, PCR, or confocal, repeated if suspicion persists. Ring infiltrate is late, not required. Steroid makes it worse.
Lid swelling plus painful restricted motility Orbital cellulitis. Emergency department, contrast CT of orbits and sinuses, intravenous antibiotics, ENT. Colour and motility are the gate.	Over 50, profound loss, chalky pallid disc oedema Arteritic AION. ESR, CRP, platelets, and immediate high-dose systemic steroid. A normal ESR does not clear it.
Sudden painless profound monocular loss Retinal artery occlusion, an acute stroke syndrome. Stroke pathway now, embolic source workup, GCA screen by age. No office massage.	Flashes, shower of floaters, pigment in the vitreous Retinal break until the periphery has been examined with indentation. A normal central OCT proves nothing.

Deck, continued

New distortion with subretinal fluid in an older patient

Neovascular AMD until proven otherwise. Urgent retina referral for anti-VEGF.

Good acuity is a reason to treat quickly, not to wait.

Bilateral myopic shift with uniformly shallow chambers on a new drug

Ciliochoroidal effusion, commonly topiramate. Stop the drug with the prescriber, suppress aqueous, cycloplege.

Not pupillary block: no iridotomy, no miotics.

Acute ptosis and miosis after neck pain

Painful Horner: carotid dissection until imaged. Emergency head and neck vascular imaging.

Do not wait to finish localisation drops.

Recurrent chalazion at the same site with lash loss

Sebaceous carcinoma. Full-thickness biopsy with conjunctival mapping, examine nodes.

Not a third curettage.

Multiple-response item

Count the requested number, judge each option alone, choose patient-specific over generally true.

Partial sets score zero.

High-confidence miss

Top of the error log. Name the false rule, write the replacement as one if-then sentence, retest tomorrow from a changed stem.

FINAL 48 HOURS

Red flags, drug contraindications, formulas, image algorithms, and logistics only. No new content. Between the two sessions, do not autopsy Session 1: eat, hydrate, reset, and start the second session on pacing.

Sources and status

Official NBEO documents control scope and weighting. Everything below was checked on 15 August 2026, and link status is recorded so a reader can tell what was actually opened.

Official

STATUS	DOCUMENT
READ	NBEO. Part II PAM/TMOD exam information. optometry.org/exams/part-ii-pam-tmod . Source for item count, case size, session structure, timing, TMOD breakout, and delivery.
RESOLVES	NBEO. Part II Content Matrix, PDF dated May 2026. Link returns 200; content not machine-read for this edition, so domain ranges remain unconfirmed.
RESOLVES	NBEO. Part II PAM Content Outline and TMOD Content Outline, PDFs dated May 2026.
RESOLVES	NBEO. PAM Sample Cases and Test Day Instructions, PDFs dated May 2026.
DEAD	Pearson VUE tutorial deep link. Did not resolve; NBEO states the tutorial is being updated. Reach it from the exam page.

Clinical anchors

Mac Grory B, et al. Management of central retinal artery occlusion. *Stroke* 2021;52:e282-e294. · Marmor MF, et al. Recommendations on screening for chloroquine and hydroxychloroquine retinopathy, 2016 revision. *Ophthalmology* 2016;123:1386-1394. · Herpetic Eye Disease Study Group. *N Engl J Med* 1998;339:300-306. · Beck RW, et al. Corticosteroids in acute optic neuritis. *N Engl J Med* 1992;326:581-588. · AREDS2 Research Group. *JAMA* 2013;309:2005-2015. · CDC. Sexually transmitted infections treatment guidelines, 2021. · American Diabetes Association. Standards of care in diabetes.

Learning method

Dunlosky J, et al. Improving students' learning with effective learning techniques. *Psychol Sci Public Interest* 2013;14:4-58. · Roediger HL, Karpicke JD. Test-enhanced learning. *Psychol Sci* 2006;17:249-255.

KNOWN LIMITS OF THIS EDITION

Domain ranges and item-type bands are transcribed, not confirmed against the current matrix. Both sessions are complete: 70 cases and 350 items, every one with a generated answer letter and a distractor table. Clinical anchors are exam-oriented simplifications: verify labelling, dose, interactions, and local law before treating a patient.

ORIGINALITY AND INDEPENDENCE

Every case, option, rationale, table, and diagram here is original. Nothing reproduces NBEO secure items or copyrighted case images. This is an independent resource, not endorsed by NBEO, and no resource can guarantee a passing score.

How this was made

Recorded because the previous edition's defects were invisible to its reader, and because a claim you cannot check is worth less than a method you can.

Build

Content	Cases are structured data. Each item carries its options with the correct one flagged in place, an explanation, and a distractor table naming what makes each wrong option tempting and the finding that rules it out.
Answer letters	Generated after authoring, to an even distribution with no run longer than three and every letter live in both sessions. Options are rotated rather than shuffled, so ordered options stay ordered.
Layout	Every page is a fixed letter-sized box. A cost model estimates the height of each block and packs content into sheets, and a headless browser then measures the real render and fails the build if any page is clipped.
Numbering	Page numbers, the table of contents, both case indexes and the printed answer key are all generated after layout, so they cannot drift away from the content.
Print	Printed through the browser's own engine at letter size with the document owning its page geometry. The exported file is then checked for page count, page box and embedded fonts.

Typography

Display	Instrument Serif, for the page titles only.
Text	Spectral at 10.5 point, down to 8 point in the distractor tables and nowhere smaller. Chosen because it has static weights: the print engine silently substitutes a fallback for variable-axis fonts, which loses the body face in the exported file without any warning on screen.
Labels and numbers	IBM Plex Mono, for anything that has to be scanned rather than read.
One deliberate exception	The Greek delta in prism notation is set in a fallback face, in both weights, because Spectral has no Greek coverage. That was established by exporting the file with the character removed and comparing the embedded fonts. The alternative was to write prism notation incorrectly.
Colour	Petrol keys structure and navigation, red keys danger and prohibition, ochre keys a caution or a rule to memorise. Red is never used decoratively, so a red strip on a page always means the same thing.

Known limits of this edition

Unconfirmed figures	Per-domain item ranges and item-type percentage bands are transcribed from the prior edition. The official Content Matrix resolves but was not machine-read for this build. Both are marked unconfirmed wherever they appear.
Clinical simplification	Drug figures are exam anchors. Verify labelling, dose, interactions and local law before treating anybody.
Independence	Every case, option, rationale, table and diagram is original. Nothing reproduces secure NBEO items or copyrighted case images. This is an independent resource, not endorsed by NBEO, and no resource can guarantee a passing score.

Built 15 August 2026. 104 pages, 70 cases, 350 items.